Engineering Mathematics

Kee-Youn Yoo

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Lecture Notes (pdf)

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1 Introduction to Differential Equations

1.1 Definitions and Terminology

- A **differential equation (DE)** is an equation containing the **derivatives** of **one or more dependent variables** with respect to **one or more independent variables**
- Notation
  - Leibniz notation $\frac{d^2y}{dx^2}$
  - Prime notation y''
  - Newton's dot notation $\ddot{y}$
  - Subscript notation $u_{xx} + u_{yy} = 0$

1.1.1 Classification by Type

- **Ordinary differential equation (ODE):** Derivatives are with respect to a single independent variable

$$\frac{dx}{dt} + \frac{dy}{dt} = 3x + 2y$$

- **Partial differential equation (PDE):** Derivatives are with respect to two or more independent variables

$$\frac{\partial u}{\partial y} = -\frac{\partial u}{\partial x}$$

1.1.2 Classification by Order

- The order of an ODE or PDE is the order of the highest derivatives in the equation

$$\underbrace{\frac{d^2y}{dx^2}}_{\text{highest order}} + 5\left(\frac{dy}{dx}\right)^3 - 4y = e^x$$

$$2\underbrace{\frac{\partial^4 u}{\partial x^4}}_{\text{highest order}} + \frac{\partial^2 u}{\partial t^2} = 0$$

1.1.3 Classification by Linearity

- An n-th order ODE, $F\left(x, y, y', \cdots, y^{(n)}\right) = 0,$ is **linear** in the variable y if F is linear in $y, y', \cdots, y^{(n)}$
- An ODE is **nonlinear** if:
  - The coefficients of $y, y', \cdots, y^{(n)}$ contain the dependent variable y or its derivatives
  - Powers of $y, y', \cdots, y^{(n)}$ appear in the equation *or*
  - Nonlinear functions of the dependent variable or its derivatives (e.g., $\sin y$ or $e^{y'}$) appear in the equation

1.1.4 Solution of an ODE

- Any function ϕ, defined on an interval I and possessing at least n derivatives that are continuous on I, which when substituted into an ODE reduces the equation to an identity, is **a solution**
- Interval I can be an open interval (a, b), a closed interval $[a, b]$, an infinite interval (a, ∞), etc.
- A solution of a differential equation that is identically zero on an interval I is a **trivial solution**
- The graph of a solution ϕ of an ODE is a **solution curve** and it is continuous on its interval I while the domain of ϕ may differ from the interval I
- An **explicit solution** is one in which the dependent variable is expressed solely in terms of the independent variable and constants
- $G(x, y) = 0$ is an **implicit solution**, if at least one function ϕ exists, that satisfies the relation G and the ODE on I

1.1.5 Families of Solutions

- Similar to integration, we obtain a solution to a first-order differential equation containing an arbitrary constant c
- A solution with a constant c represents a set $G(x, y, c) = 0$ of solutions, called a **one-parameter family of solutions**
- An n-parameter family of solutions

$$G(x, y, c_1, c_2, \cdots, c_n) = 0$$

  solves an n-th order differential equation

1.1.6 Systems of Differential Equations

- Two or more equations involving the derivatives of two or more unknown functions of a single independent variable are called the **system of differential equations**

$$\begin{aligned} \frac{dx}{dt} &= f(t, x, y) \\ \frac{dy}{dt} &= g(t, x, y) \end{aligned}$$

- A solution of a system is a set of functions defined on a common interval I that satisfies each equation of the system on this interval

1.2 Initial-Value Problems - IVP

In an **IVP**, we seek a solution $y(x)$ of a differential equation so that $y(x)$ satisfies **initial conditions** at x_0

- n-th order differential equation and Initial conditions

$$\begin{aligned} \frac{d^n y}{dx^n} &= f\left(x, y, y', \cdots, y^{(n-1)}\right) \\ y(x_0) = y_0,\ y'(x_0) &= y_1, \cdots,\ y^{(n-1)}(x_0) = y_{n-1} \end{aligned}$$

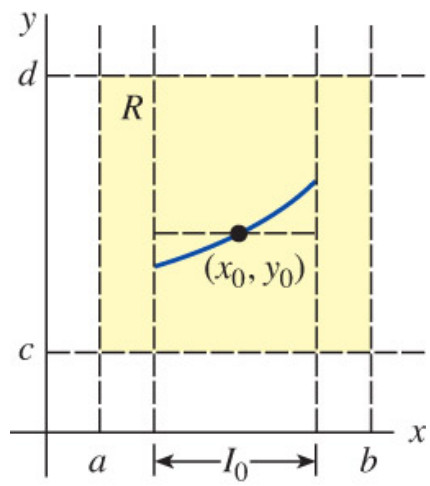
$$\Downarrow$$

$$\frac{d\mathbf{y}}{dx} = \mathbf{f}(x, \mathbf{y}), \quad \mathbf{y} = \mathbf{y}_0$$

When solving an IVP, consider whether a solution exists and whether the solution is unique

- **Existence:** Does the differential equation possess solutions and do any of the solution curves pass through the initial point (x_0, y_0)?
- **Uniqueness:** When can we be certain there is precisely one solution curve passing through the initial point (x_0, y_0)?

Theorem 1.1. *gives conditions that are sufficient to guarantee the existence and uniqueness of a solution $y(x)$ to a first-order IVP*



- $f(x, y)$ *and* $\dfrac{\partial f}{\partial y}$ *are continuous on the region* R *for the interval* I_0

Example First-order IVP

- $y(x) = ce^x$ is a one-parameter family of solutions of the first-order DE $y' = y$
- Find a solution of the first-order IVP with initial condition $y(0) = 3$

Solution

- From the initial condition, we obtain $3 = ce^0$
- Solving, we find $c = 3$
- The solution of the IVP is $y = 3e^x$

Example Second-order IVP

- $x(t) = c_1 \cos 4t + c_2 \sin 4t$ is a two-parameter family of solutions of the second-order DE $x'' + 16x = 0$

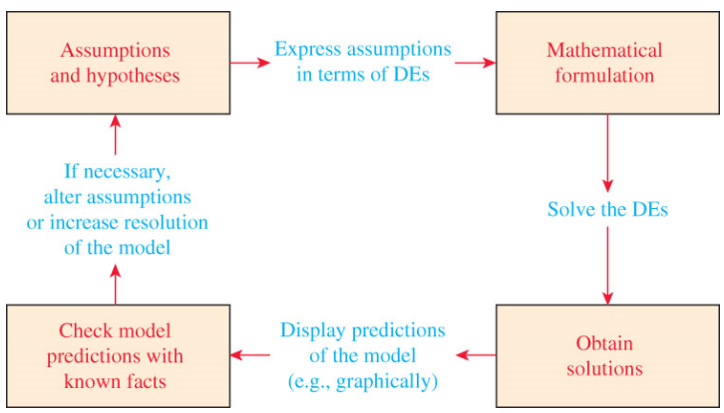
- Find a solution of the second-order IVP with initial conditions $x\left(\frac{\pi}{2}\right) = -2,\ x'\left(\frac{\pi}{2}\right) = 1$

Solution

- Substituting for initial conditions and solving for constants, we find $c_1 = -2$ and $c_2 = 1/4$
- The solution of the IVP is $x(t) = -2\cos 4t + \frac{1}{4}\sin 4t$

1.3 Differential Equations as Mathematical Models

- A **mathematical model** is a description of a system or a phenomenon
- Differential equation models are used to describe behavior in various fields
  - Biology
  - Chemistry
  - Physics
- Steps of the modeling process



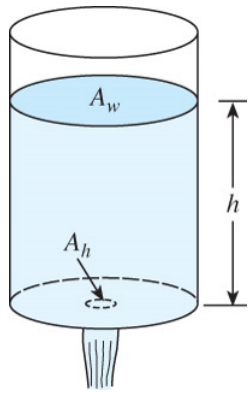
Example Radioactive Decay

- In modeling the phenomenon, it is assumed that the rate $\dfrac{dA}{dt}$ at which the nuclei of a substance decays is proportional to the amount $A(t)$ remaining at time t given an initial amount of radioactive substance on hand A_0

$$\frac{dA}{dt} = -kA,\ A(0) = A_0$$

- This differential equation also describes a first-order chemical reaction

Example Draining Tank



- Toricelli's law states that the exit speed v of water through a sharp-edged hole at the bottom of a tank filled to a depth h is the same as the speed it would acquire falling from a height h, $v = \sqrt{2gh}$
- Volume V leaving the tank per second is proportional to the area of the hole A_h

$$\frac{dV}{dt} = -A_h\sqrt{2gh}$$

- The volume in the tank at t is $V(t) = A_w h$, in which A_w is the *constant* area of the upper water surface
- Combining expressions gives the differential equation for the height of water at time t

$$\frac{dh}{dt} = -\frac{A_h}{A_w}\sqrt{2gh}$$

Worked Exercises

1.2: 1. $y = 1/(1 + c_1 e^{-x})$ is a one-parameter family of solutions of the first-order DE $y' = y - y^2$. Find a solution of the first-order IVP consisting of this differential equation and the given initial condition:

$$y(0) = -\frac{1}{3}$$

Solution

$$
\begin{aligned}
y &= \frac{1}{1+c_1 e^{-x}} \\
&\Downarrow \text{ at } x{=}0 \\
-\frac{1}{3} &= \frac{1}{1+c_1} \\
&\Downarrow \\
c_1 &= -4 \\
&\Downarrow \\
y &= \frac{1}{1-4e^{-x}}
\end{aligned}
$$

1.2: 3. $y = 1/(x^2+c)$ is a one-parameter family of solutions of the first-order DE $y'+2xy^2 = 0$. Find a solution of the first-order IVP consisting of this differential equation and the given initial condition. Give the largest interval I over which the solution is defined

$$
y(2) = \frac{1}{3}
$$

Solution

$$
\begin{aligned}
y &= \frac{1}{x^2+c} \\
&\Downarrow \text{ at } x{=}2 \\
\frac{1}{3} &= \frac{1}{4+c} \\
&\Downarrow \\
c = -1\,, &\quad I = (1, \infty) \\
&\Downarrow \\
y &= \frac{1}{x^2-1}
\end{aligned}
$$

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-5, 5, 200)
y = 1.0 /(x**2 -1)

threshold = 20
y[y > threshold] = np.inf
y[y <-threshold] = np.inf
```

```
plt.plot(x,y)
plt.scatter(2, 1/3, color='red')

plt.title(r'$y=\frac{1}{x^2 -1}$')
plt.xlabel('$x$')
plt.ylabel('$y$')
plt.grid()

plt.show()
```

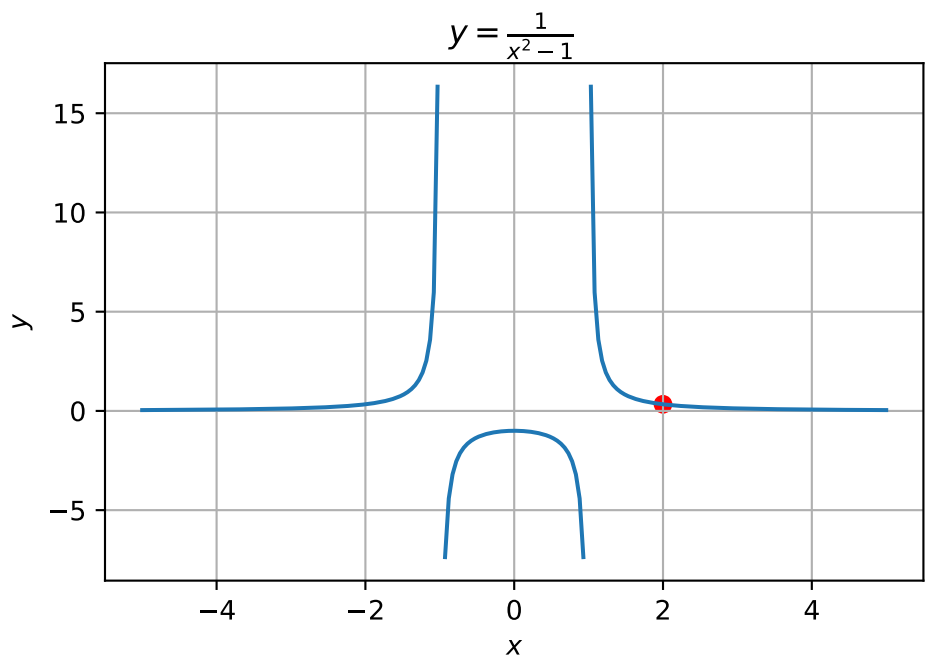


Figure 1.1: Exercises 1.2: 3

1.2: 7. $x(t) = c_1 \cos t + c_2 \sin t$ is a two-parameter family of solutions of the second-order DE $x'' + x = 0$. Find a solution of the second-order IVP consisting of this differential equation and the given initial conditions

$$x(0) = -1,\ x'(0) = 8$$

Solution

$$
\begin{aligned}
x(t) &= c_1 \cos t + c_2 \sin t \\
&\Downarrow \ {\scriptstyle x(0)=-1,\ x'(0)=8} \\
-1 &= c_1 \\
8 &= c_2 \\
&\Downarrow \\
x(t) &= -\cos t + 8 \sin t
\end{aligned}
$$

1.2: 17. Determine a region of the xy-plane for which the given differential equation would have a unique solution whose graph passes through a point (x_0, y_0) in the region.

$$\frac{dy}{dx} = y^{2/3}$$

Solution

1.2: 29.

1. By inspection, find a one-parameter family of solutions of the differential equation $xy' = y$. Verify that each member of the family is a solution of the initial value problem $xy' = y, y(0) = 0$
2. Explain part 1. by determining a region R in the xy-plane for which the differential equation $xy' = y$ would have a unique solution through a point (x_0, y_0) in R
3. Verify that the piecewise-defined function

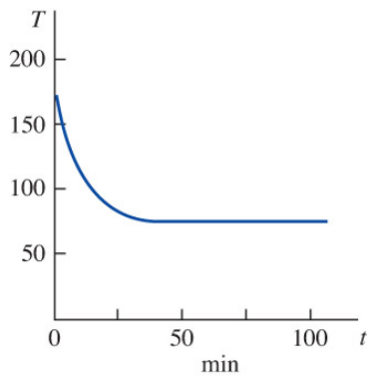
$$y = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}$$

satisfies the condition $y(0) = 0$. Determine whether this function is also a solution of the initial-value problem in part 1

Solution

1.3: 5. A cup of coffee cools according to Newton's law of cooling. Use data from the graph of the temperature $T(t)$ in figure to estimate the constants T_m, T_0, and k in a model of the form of the first-order initial-value problem

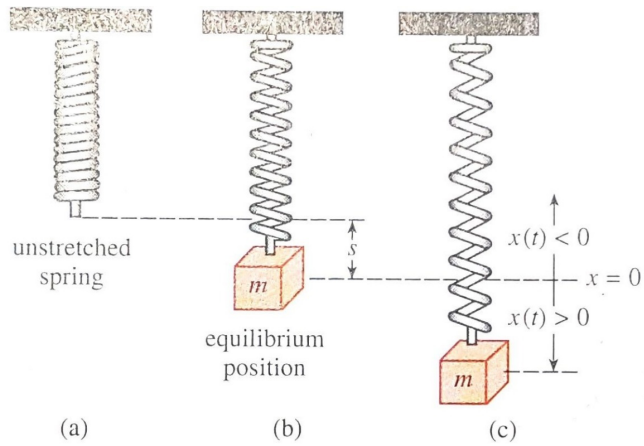
$$\frac{dT}{dt} = k(T - T_m), \ \ T(0) = T_0$$



Solution

$$\frac{dT}{dt} = k(T - T_m), \quad T(0) = T_0$$
$$\Downarrow$$
$$T(0) = T_0$$
$$T(\infty) = T_m$$
$$\ln\left(\frac{T - T_m}{T_0 - T_m}\right) = kt$$
$$\Downarrow$$
$$k \text{ is the slope of the graph of } t \text{ vs. } \ln\left(\frac{T - T_m}{T_0 - T_m}\right)$$

1.3: 19. After a mass m is attached to a spring, it stretches s units and then hangs at rest in the equilibrium position as shown in Figure (b). After the spring/mass system has been set in motion, let $x(t)$ denote the directed distance of the mass beyond the equilibrium position. As indicated in Figure (c), assume that the downward direction is positive, that the motion takes place in a vertical straight line through the center of gravitiy of the mass, and that the only forces acting on the system are the weight of the mass and the restoring force of the stretched spring. Use Hooke's law: The restoring force of a spring is proportional to its total elongation. Determine a differential equation for the displacement $x(t)$ a time t


unstretched
spring
s
m
equilibrium
position
x(t) < 0
x = 0
x(t) > 0
m
(a)
(b)
(c)

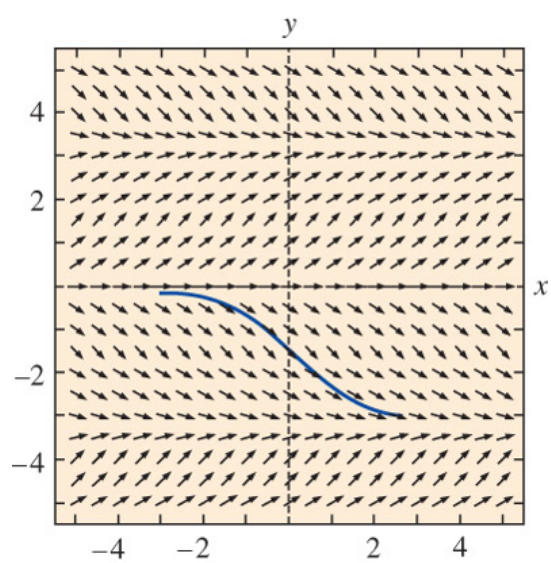
Solution

2 First-Order Differential Equations

2.1 Solution Curves Without a Solution

- DEs can be analyzed qualitatively, allowing us to approximate a solution curve without solving the problem
- Two approaches are:
  - Direction fields
  - Autonomous first-order DEs
- Direction fields:
  - Slope of the **lineal element** at $(x, y(x))$ on a solution curve is the value of $\frac{dy}{dx}$ at this point
  - **Direction/slope fields** of $\frac{dy}{dx} = f(x, y)$ are collections of lineal slope elements that visually suggest the shape of a family of solution curves

For example, $\dfrac{dy}{dx} = \sin y$

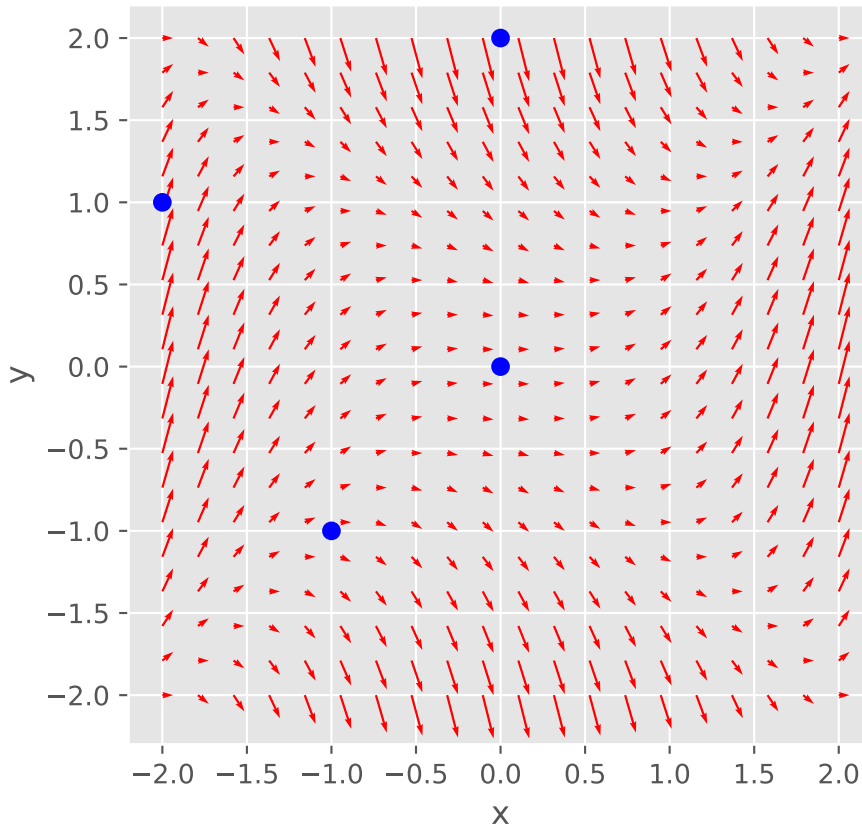


- Using the given computer-generated direction field, sketch, by hand, an approximate solution curve that passes through each of the indicated points:

$$\frac{df}{dx} = x^2 - y^2$$

$$\begin{aligned}
&(a)\ \ y(-2) = 1\\
&(b)\ \ y(-1) = -1\\
&(c)\ \ y(0) = 2\\
&(d)\ \ y(0) = 0
\end{aligned}$$

```
import numpy as np
from scipy import integrate

import matplotlib.pyplot as plt
plt.style.use('ggplot')

x = np.linspace(-2, 2, 20)
y = np.linspace(-2, 2, 20)
X, Y = np.meshgrid(x, y)

dy = (X*X -Y*Y)
dx = np.ones(dy.shape)

plt.figure(figsize=(5, 5))
plt.quiver(X, Y, dx, dy, color='red')
plt.plot([-2, -1, 0, 0], [1, -1, 2, 0], 'o', color="blue")
plt.xlim([-2.2, 2.2])
plt.ylim([-2.3, 2.2])
plt.xticks(np.arange(-2, 2.5, 0.5))
plt.yticks(np.arange(-2, 2.5, 0.5))
plt.xlabel('x')
plt.ylabel('y')

plt.show()
```



- **Autonomous** first-order DEs, $\dfrac{dy}{dx} = f(y)$

  An ODE in which the independent variable does not appear explicitly

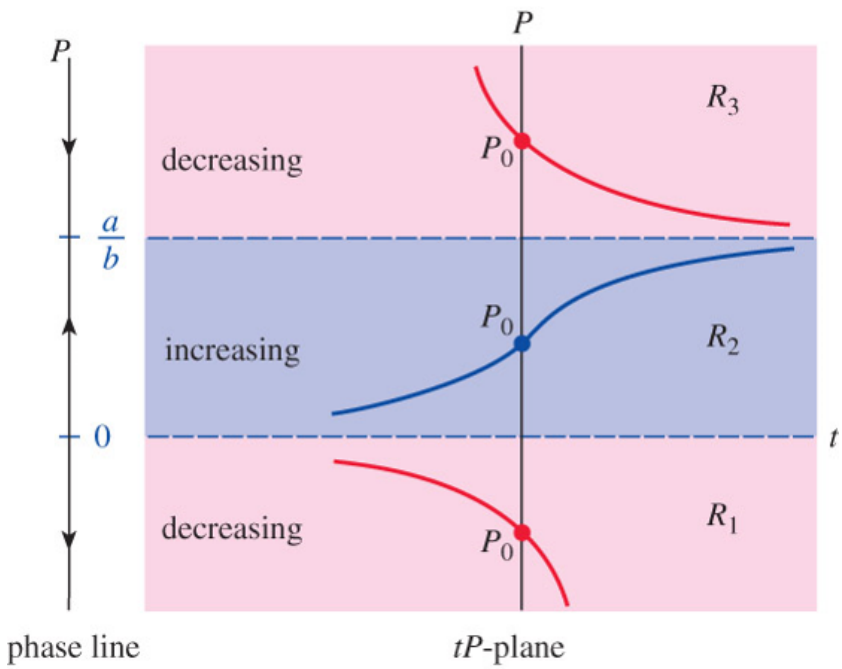
$$\frac{dy}{dx} = 1 + y^2 \quad \text{autonomous}$$
$$\frac{dy}{dx} = 0.2\,xy \quad \text{nonautonomous}$$

- **Critical points**, $f(c) = 0$, are constant (or equilibrium) solutions of autonomous DEs
- A **phase portrait** is made by putting critical points on a vertical line with phase lines pointing up or down, depending on the sign of the function over intervals between the points
- Some conclusions can be drawn about nonconstant solution curves to autonomous DEs
  - If a solution $y(x)$ passes through (x_0, y_0) in sub-region R_i, then $y(x)$ remains in R_i
  - By continuity, $f(y)$ cannot change signs in a sub-region R_i

– Since $f(y)$ is either positive or negative in R_i, a solution is either increasing or decreasing and has no relative extremum

Example Phase portrait and solution curves

$$\frac{dP}{dt} = P(a - bP)$$



Example Consider the autonomous first-order differential equation

$$\frac{dy}{dx} = y^2 - y^4$$

and the initial condition $y(0) = y_0$. Sketch the graph of a typical solution $y(x)$ when y_0 has the given values

$$\begin{array}{ll} (a) & y_0 < -1 \\ (b) & -1 < y_0 < 0 \\ (c) & 0 < y_0 < 1 \\ (d) & 1 < y_0 \end{array}$$

2.2 Separable Equations

- Consider $\dfrac{dy}{dx} = f(x)$
  - When f does not depend on y, $\dfrac{dy}{dx} = f(x)$, which can be solved by integration
  - The solution $y = \displaystyle\int f(x)dx = F(x) + c$, where $F(x)$ is an antiderivative (indefinite integral)
  - Some functions, termed **nonelementary**, do not possess an antiderivative that is an elementary function

- A first-order DE of the form $\dfrac{dy}{dx} = g(x)h(y)$ is said to be **separable**, or have **separable variables**
  - A separable equation can be rewritten in the form

$$\frac{1}{h(y)}dy = g(x)dx$$

    which is solved by integrating both sides

Example Solve a separable equation $\dfrac{dy}{dx} = y^2 - 9$, $y(0) = 0$

- Separating and using partial fractions

$$\frac{dy}{(y-3)(y+3)} = dx$$
$$\Downarrow$$
$$\frac{1}{6}\left[\frac{1}{y-3} - \frac{1}{y+3}\right]dy = dx$$

- Integrating and solving for y yields

$$\frac{1}{6}\ln\left|\frac{y-3}{y+3}\right| = x + c_1$$
$$\Downarrow c = e^{6c_1}$$
$$y = 3\frac{1 - ce^{6x}}{1 + ce^{6x}}$$

- Finally, applying $y(0) = 0$ gives $c = 1$

Example Solve the given differential equation by separation of variables

- $\dfrac{dy}{dx} = \sin 5x$
- $dx + e^{3x}dy = 0$
- $\dfrac{dy}{dx} = x\sqrt{1-y^2}$

Example Find an implicit and an explicit solution of the given initial-value problem

- $x^2\dfrac{dy}{dx} = y - xy, \quad y(-1) = -1$
- $\dfrac{dx}{dt} = 4(x^2+1), \quad x(\pi/4) = 1$

2.3 Linear Equations

A first-order DE of the form $a_1(x)\dfrac{dy}{dx} + a_0(x)y = g(x)$ is a **linear equation** in the dependent variable y

- The DE is **homogeneous** when $g(x) = 0$; otherwise, it is **nonhomogeneous**
- The standard form of a linear DE is obtained by dividing both sides by the lead coefficient

$$\frac{dy}{dx} + P(x)y = f(x)$$

The **standard form** equation has the property that its solution y is the sum of the solution of the associated homogeneous equation y_h and the particular solution of the nonhomogeneous equation y_p :

$$y = y_h + y_p$$

- The homogeneous equation $\dfrac{dy_h}{dx} + P(x)y_h = 0$ is separable, allowing us to solve for y_h

$$\frac{dy_h}{y_h} = -P(x)dx$$
$$\Downarrow$$
$$\ln|y_h| = -\int P(x)\,dx + \bar{c} \Rightarrow y_h = c\exp\left(-\int P(x)\,dx\right)$$

- **Variation of parameters** $y_p = u(x)y_h$ can be used to solve the nonhomogeneous equation of y_p

$$y_h\frac{du}{dx} + \underbrace{\left(\frac{dy_h}{dx} + P(x)y_h\right)}_{=0} u = f(x)$$
$$du = \frac{f(x)}{y_h}dx \Rightarrow u(x) = \int \frac{f(x)}{y_h(x)}dx$$
$$\Downarrow$$
$$y_p = y_p\int \frac{f(x)}{y_h(x)}dx$$

Example Find the general solution of the given differential equation:

- $\dfrac{dy}{dx} + 2y = 0$
- $y' + 2xy = x^3$
- $x\dfrac{dy}{dx} + 2y = 3$
- $xy' + (1+x)y = e^{-x}\sin 2x$

Example Solve the given initial-value problem. Give the largest interval I over which the solution is defined

- $y\dfrac{dx}{dy} - x = 2y^2, \;\; y(1) = 5$
- $(x+1)\dfrac{dy}{dx} + y = \ln x, \;\; y(1) = 10$

Example The given differential equation is not linear in y. Nevertheless, find a general solution of the equation

- $dx = (x + y^2)dy$
- $ydx + (2x + xy - 3)dy = 0$

Example The sine integral function is defined as

$$\mathrm{Si}(x) = \int_0^x \frac{\sin t}{t}\, dt$$

where the integrand is defined to be 1 at $x = 0$. Express the solution of the initial value problem

$$x^3 \frac{dy}{dx} + 2x^2 y = 10 \sin x, \quad y(1) = 0$$

in terms of $\mathrm{Si}(x)$

2.4 Exact Equations

- A differential expression $M(x,y)dx + N(x,y)dy$ is an **exact differential** in a region R of the xy-plane if it corresponds to the differential of some function $f(x,y)$:

$$df(x,y) = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

- and a condition of exact differentials is:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

- $M(x,y)\, dx + N(x,y)\, dy = 0$ is an **exact equation** if the left side is an exact differential

Example Solving an exact DE, $2xy\, dx + (x^2 - 1)\, dy = 0$

- **Integrating Factor** of the first-order linear DE

$$\frac{dy}{dx} + P(x)y = f(x)$$
$$\Downarrow$$
$$(P(x)y - f(x))\,dx + dy = 0$$
$$\Downarrow \times I(x): \text{ Integrating Factor}$$
$$I(x)\,(P(x)y - f(x))\,dx + I(x)dy = 0$$

To be an exact equation $\Downarrow$

$$\frac{\partial}{\partial y}\left\{I(x)\,(P(x)y - f(x))\right\} = I(x)P(x) = \frac{d}{dx}I(x)$$
$$\Downarrow\ I(x) = \exp\left(\int P(x)dx\right)$$

Then

$$I(x)\frac{dy}{dx} + I(x)P(x)y = I(x)f(x) \ \Rightarrow\ \frac{d}{dx}\left\{I(x)y\right\} = I(x)f(x)$$
$$\Downarrow$$
$$y(x) = I(x)^{-1}y(x_0)I(x_0) + I(x)^{-1}\int_{x_0}^{x} I(x)f(x)dx$$

Example Solve $\dfrac{dy}{dx} - 2xy = 2,\ y(0) = 1$

$$\frac{dy}{dx} - 2xy = 2$$
$$\Downarrow \times e^{-x^2}$$
$$\frac{d}{dx}[e^{-x^2}y] = 2e^{-x^2}$$
$$y = ce^{x^2} + 2e^{x^2}\int_0^x e^{-t^2}dt$$
$$\Downarrow y(0) = 1 \to c = 1$$
$$y = e^{x^2}\left[1 + \sqrt{\pi}\,\underbrace{\left(\frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt\right)}_{\mathrm{erf}(x)}\right]$$
$$= e^{x^2}\left[1 + \sqrt{\pi}\,\mathrm{erf}(x)\right]$$

Example Determine whether the given differential equation is exact. If it is exact, solve it

- $(2x-1)dx + (3y+7)dy = 0$
- $(5x+4y)dx + (4x-8y^2)dy = 0$
- $(2xy^2-3)dx + (2x^2y+4)dy = 0$
- $(x^2-y^2)dx + (x^2-2xy)dy = 0$

Example Solve the given initial-value problem

- $(x+y)^2dx + (2xy+x^2-1)dy = 0, \;\; y(1) = 1$
- $(4y+2t-5)dt + (6y+4t-1)dy = 0, \;\; y(-1) = 2$

Example Solve the given differential equation by finding an appropriate integrating factor

- $y(x+y+1)dx + (x+2y)dy = 0$

2.5 Solutions by Substitutions

Substitution is often used to get a DE in a form that a known procedure can be used to find a solution

- Reduction to separation of variables can be facilitated in the DE

$$\frac{dy}{dx} = f(Ax+By+C)$$

by substituting $u = Ax+By+C, \; B \neq 0$

Example Solve the IVP $\dfrac{dy}{dx} = (-2x+y)^2 - 7, \; y(0) = 0$

Let $u = -2x+y$,

then $\dfrac{du}{dx} = -2 + \dfrac{dy}{dx}$ giving $\dfrac{du}{dx} = u^2 - 9$

- **Homogeneous** first-order DE

  A first-order ordinary DE in the form:

$$\frac{dy}{dx} = f(x,y) = -\frac{M(x,y)}{N(x,y)}$$

  is a homogeneous type if both function $M(x,y)$ and $N(x,y)$ are homogeneous functions of the same degree n:

$$M(\lambda x, \lambda y) = \lambda^n M(x,y), \quad N(\lambda x, \lambda y) = \lambda^n N(x,y)$$

  Thus, we can let $t = 1/x$ to simplify this quotient to a function f of the single variable y/x:

$$\frac{M(x,y)}{N(x,y)} = \frac{M(tx,ty)}{N(tx,ty)} = \frac{M\left(1, \frac{y}{x}\right)}{N\left(1, \frac{y}{x}\right)} = -f\left(\frac{y}{x}\right) = -\frac{dy}{dx}$$

  The change of variables $y = ux$ transforms the original differential equation into the separable form:

$$\frac{du}{f(u) - u} = \frac{dx}{x}$$

- **Bernoulli** DE:

$$y' + P(x)y = Q(x)y^n \quad \text{where} \quad n \neq 0 \quad \text{and } n \neq 1$$

  - Bernoulli equations are special because they are nonlinear differential equations with known exact solutions
  - The substitution $u = y^{1-n}$ reduces any Bernoulli equation to a linear differential equation

Example Solve $y' - \dfrac{2}{x}y = -x^2y^2$

Changing variables $u = \frac{1}{y}$, $u' = -\frac{1}{y^2}y'$ gives the equation

$$u' + \frac{2}{x}u = x^2$$

- **Riccati** DE
  - Riccati equation is any first-order ordinary differential equation that is quadratic in the unknown function. In other words, it is an equation of the form

    $$y' = q_0(x) + q_1(x)y + q_2(x)y^2$$

    where $q_0(x) \neq 0$ and $q_2(x) \neq 0$. If $q_0(x) = 0$, the equation is Bernoulli one, and if $q_2(x) = 0$, the equation is linear one
  - The new variable $v = yq_2$ satisfies an equation

    $$v' = v^2 + R(x)v + S(x)$$

    where $S(x) = q_2 q_0$ and $R(x) = q_1 + \left(\frac{q_2'}{q_2}\right)$
  - Substituting $v = -\frac{u'}{u}$, u satisfies the linear 2nd order ODE

    $$u'' - R(x)u' + S(x)u = 0$$

    A solution of this equation will lead to a solution $y = -\frac{u'}{q_2 u}$ of the original Riccati equation

Example Solve the given differential equation by using an appropriate substitution

- $(x - y)dx + xdy = 0$
- $xdx + (y - 2x)dy = 0$
- $\dfrac{dy}{dx} = \dfrac{y - x}{y + x}$
- $x\dfrac{dy}{dx} = y + \sqrt{x^2 - y^2}, \quad x > 0$

Example Solve the given initial-value problem

- $xy^2\dfrac{dy}{dx} = y^3 - x^3, \quad y(1) = 2$
- $(x^2 + 2y^2)\dfrac{dx}{dy} = xy, \quad y(-1) = 1$

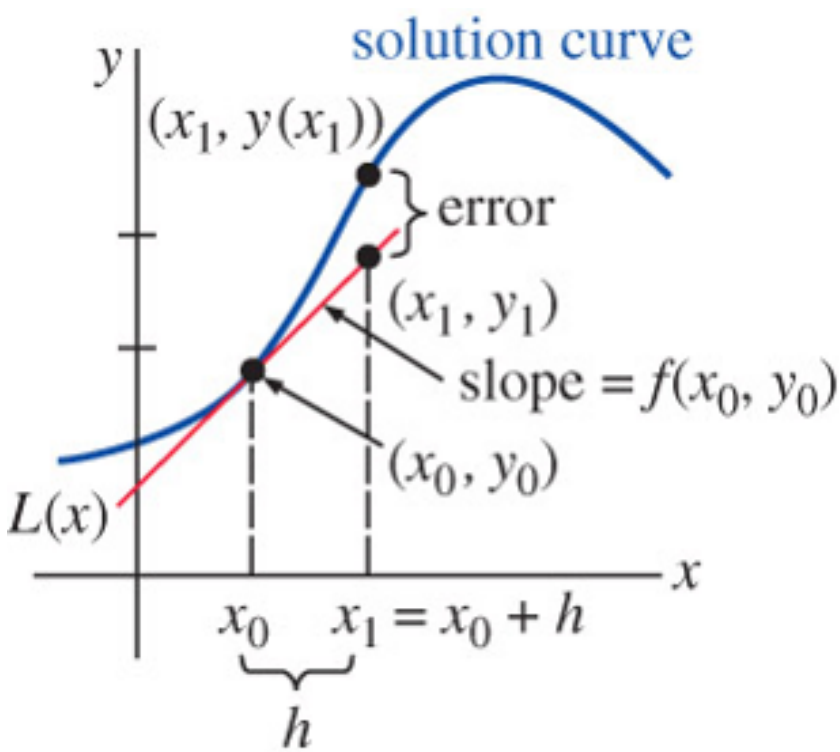
Example Solve the given differential equation by using an appropriate substitution

- $x\dfrac{dy}{dx} + y = \dfrac{1}{y^2}$
- $3(1+t^2)\dfrac{dy}{dt} = 2ty(y^3 - 1)$
- $\dfrac{dy}{dx} = \cos(x+y), \;\; y(0) = \dfrac{\pi}{4}$

2.6 A Numerical Method

Numerical method: an algorithm for approximating the unknown solution to a DE

- **Linearization** approximates solutions within a small area around lineal elements in direction fields
- The procedure of evaluating successive tangent lines is **Euler's method**



Example Consider $y' = 0.1\sqrt{y} + 0.4x^2,\; y(2) = 4$ and approximate $y(2.5)$ using $h = 0.1$

- Substituting into the general formula for Euler's method gives

$$y_{n+1} = y_n + h\left(0.1\sqrt{y_n} + 0.4x_n^2\right)$$

- Considering the initial condition and $n = 0$

$$\begin{aligned}
y_1 &= 4 + 0.1\left(0.1\sqrt{4} + 0.4 \cdot 2^2\right) = 4.1800 \\
y_2 &= 4.18 + 0.1\left(0.1\sqrt{4.18} + 0.4 \cdot 2.1^2\right) = 4.3768 \\
&\vdots \\
y_5 &= 5.0768
\end{aligned}$$

```python
plt.style.use('ggplot')

x0 = 2.0
y0 = 4.0
xf = 2.5

h = 0.1
n = int((xf - x0)/h) + 1

x = np.linspace(x0, xf, n)
y = np.zeros(n)

y[0] = y0
for i in range(1, n):
    y[i] = y[i -1] + h*(0.1 * np.sqrt(y[i -1]) + 0.4 * x[i -1]**2)
    print(f'x = {x[i] : 3.1f}, y = {y[i] : 5.4f}')

plt.figure(figsize=(4, 4))
plt.plot(x, y, 'o')
plt.xlabel("$x$")
plt.ylabel("$y$")
plt.title("$h=0.1$")

plt.show()
```

```
x =  2.1, y =  4.1800
x =  2.2, y =  4.3768
x =  2.3, y =  4.5914
x =  2.4, y =  4.8244
x =  2.5, y =  5.0768
```

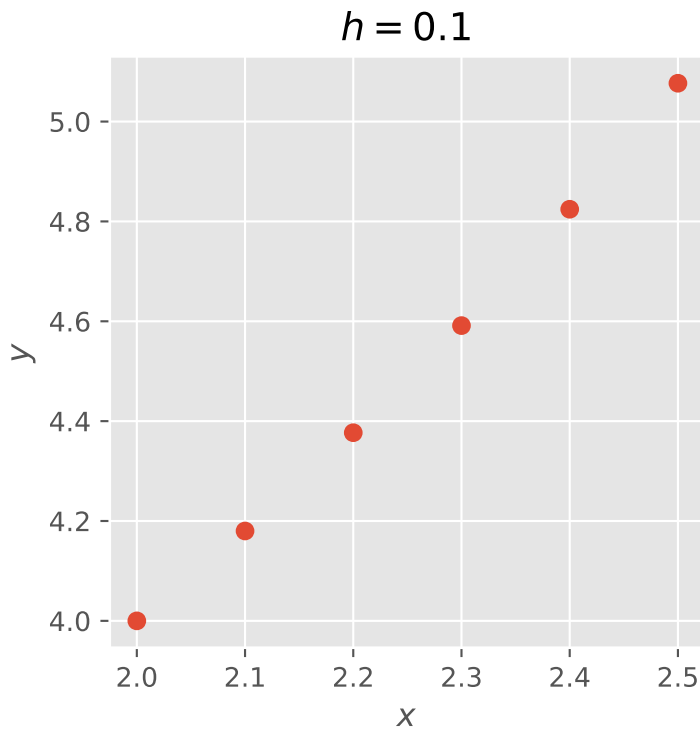


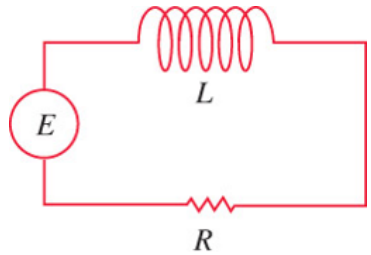
Figure 2.1: Euler's Method

Example Use Euler's method to obtain a four-decimal approximation of the indicated value. Carry out the recursion, first using $h = 0.1$ and then using $h = 0.05$

$$y' = 2x - 3y + 1, \;\; y(1) = 5; \;\; y(1.2)$$

2.7 Linear Models

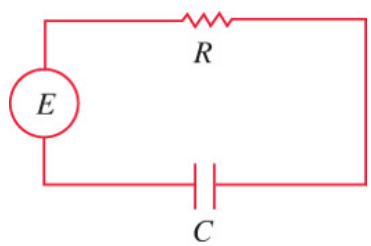
Example Series Circuits - For a series circuit containing a resister R and an inductor L,



Kirchoff's second law gives

$$L\frac{di}{dt} + Ri = E(t)$$

For a series circuit containing a resister and a capacitor,



Kirchoff's second law gives

$$Ri + \frac{1}{C}q = E(t)$$

where $i = \dfrac{dq}{dt}$. Then $R\dfrac{dq}{dt} + \dfrac{1}{C}q = E(t)$

Example The population of a community is known to increase at a rate proportional to the number of people present at time t. If an initial population P_0 has doubled in 5 years, how long will it take to triple? To quadruple?

2.8 Nonlinear Models

Example The logistic model

- Verhulst proposed a model, called the logistic model, for population growth in 1838. It does not assume unlimited resources. Instead, it assumes there is a carrying capacity K for the population
- This carrying capacity is the stable population level. If the population is above K, then the population will decrease, but if below, then it will increase
- For this model, it is assumed that ther rate of change $\frac{dy}{dt}$ of the population y is proportional to the product of the current population y and $K - y$

$$\frac{dy}{dt} = \alpha y(K - y)$$

2.9 Modeling with Systems of First-Order DEs

Example Radioactive Series:

$$X \xrightarrow{k_1} Y \xrightarrow{k_2} Z$$

- The decay of the initial element X is described by

$$\frac{dx}{dt} = -k_1 x$$

- The second element Y is produced by the decay of X and loses from its own spontaneous decay

$$\frac{dy}{dt} = k_1 x - k_2 y$$

- The stable element z is generated from the decay of Y

$$\frac{dz}{dt} = k_2 y$$

Example Consider the Lotka-Volterra predator-prey model defined by

$$\frac{dx}{dt} = -0.1x + 0.02xy$$
$$\frac{dy}{dt} = 0.2y - 0.025xy$$

where the populations $x(t)$ (predator) and $y(t)$ (prey) are measured in the thousands and $x(0) = 6$ and $y(0) = 6$

- Use a numerical solver to graph $x(t)$ and $y(t)$

```
# Lotka-Volterra predator-prey model
a, b, c, d = -0.1, 0.02, 0.2, -0.025

def f(t, xy):
    x, y = xy
    return [a *x +b *x *y, c *y +d *x *y]

# Initial condition and Time span
xy0 = [6, 6]
tf = 400
t_eval = np.linspace(0, tf, 5*tf)

# Numerical Solver
sol = integrate.solve_ivp(f, [0, tf], xy0, t_eval=t_eval)

t = sol.t
xy_t = sol.y.T
```

- Use the graphs to approximate the period of each population

```
fig, ax = plt.subplots(1, 1, figsize=(6, 5))

ax.plot(t, xy_t[:, 0], 'r', label="Predator")
ax.plot(t, xy_t[:, 1], 'b', label="Prey")

ax.set_xlabel("Time")
ax.set_ylabel("Number of animals")
ax.set_xlim(0, tf)
ax.set_ylim(0, 12)
ax.legend()

plt.show()
```

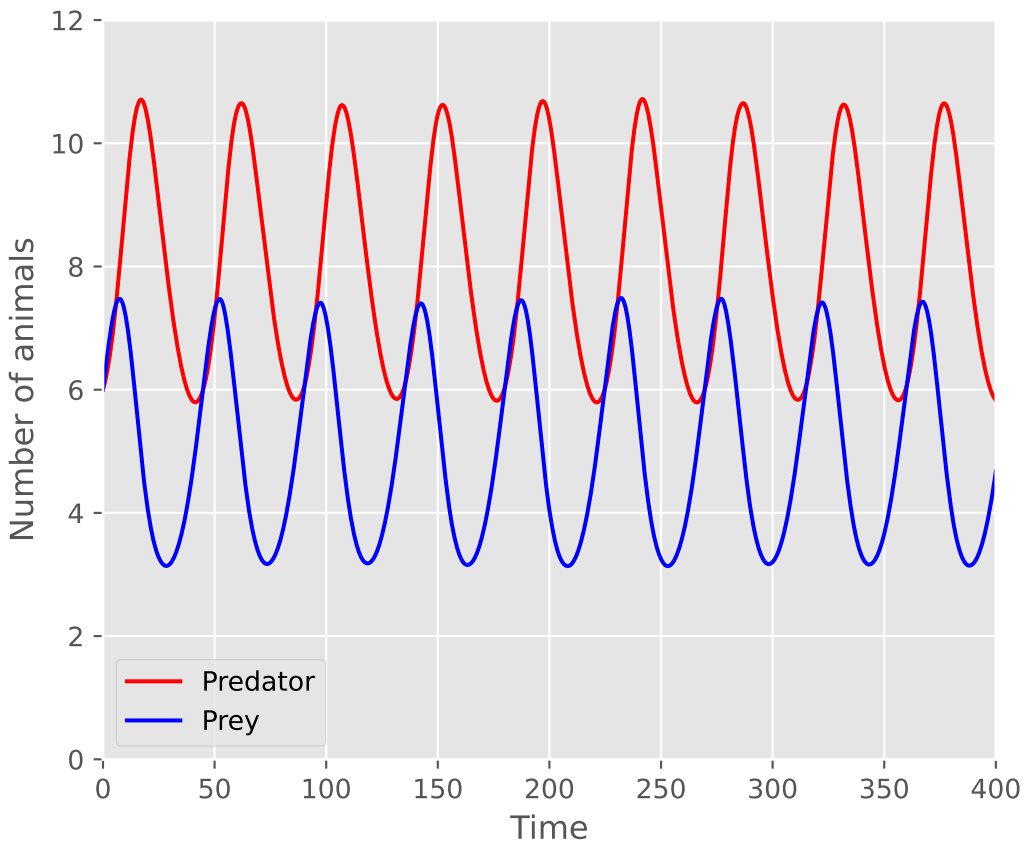


Figure 2.2: Lotka-Volterra predator-prey model

Worked Exercises

Solve the given differential equation by separation of variables

2.2: 1. $\dfrac{dy}{dx} = \sin 5x$

Solution

$$\begin{aligned} \frac{dy}{dx} &= \sin 5x \\ &\Downarrow \\ dy &= \sin 5x\, dx \\ &\Downarrow \\ y &= -\frac{1}{5}\cos 5x + c \end{aligned}$$

2.2: 3. $dx + e^{3x}dy = 0$

Solution

$$\begin{aligned} dy &= -e^{-3x}\, dx \\ &\Downarrow \\ y &= \frac{1}{3}e^{-3x} + c \end{aligned}$$

2.2: 21. $\dfrac{dy}{dx} = x\sqrt{1-y^2}$

Solution

$$\begin{aligned} \frac{1}{\sqrt{1-y^2}}dy &= x\, dx \\ &\Downarrow \\ \arcsin y &= \frac{1}{2}x^2 + c \\ &\Downarrow \\ y &= \sin\left(\frac{1}{2}x^2 + c\right) \end{aligned}$$

2.2: 25. Find an implicit or an explicit solution of the given initial-value problem

$$x^2\frac{dy}{dx} = y - xy, \quad y(-1) = -1$$

Solution

$$\frac{1}{y}dy = \frac{1-x}{x^2}\,dx$$
$$\Downarrow$$
$$\ln|y| = -\left(\frac{1}{x} + \ln|x|\right) + c$$
$$\Downarrow y(-1) = -1 \;\Rightarrow\; c = -1$$
$$\ln|y| = -\left(1 + \frac{1}{x}\right) - \ln|x|$$
$$\Downarrow$$
$$y = \frac{1}{x}e^{-(1+\frac{1}{x})}$$

2.7: 41. Evaporating Raindrop As a raindrop falls, it evaporates while retaining its spherical shape. If we make the further assumptions that the rate at which the raindrop evaporates is proportional to its surface area and that air resistance is negligible, then a model for the velocity $v(t)$ of the raindrop is

$$\frac{dv}{dt} + \frac{3(k/\rho)}{(k/\rho)t + r_0}v = g$$

Here ρ is the density of water, r_0 is the radius of the raindrop at $t = 0$, $k < 0$ is the constant of proportionality $\frac{dm}{dt} = 4\pi r^2 k$, and the downward direction is taken to be the positive direction

(a) Solve for $v(t)$ if the raindrop falls from the rest

(b) Show that the radius of the raindrop at time t is $r(t) = (k/\rho)t + r_0$

(c) If $r_0 = 0.01$ ft and $r = 0.007$ ft at time $t = 10$ sec after the raindrop falls from a cloud, determine the time at which the raindrop has evaporated completely

Solution

(a)

$$\frac{dv}{dt} + \frac{3(k/\rho)}{(k/\rho)t + r_0} v = g$$

$$\Downarrow \text{ multiply by the integral factor}$$

$$\frac{d}{dt}\left[\left(\frac{k}{\rho}t + r_0\right)^3 v\right] = \left(\frac{k}{\rho}t + r_0\right)^3 g$$

$$\Downarrow$$

$$v(t) = \frac{g\rho}{4k}\left(\frac{k}{\rho}t + r_0\right) + c\left(\frac{k}{\rho}t + r_0\right)^{-3}$$

$$\Downarrow \; v(0) = 0 \;\rightarrow\; c = -\frac{g\rho}{4k} r_0^4$$

$$v(t) = \frac{g\rho}{4k}\left(\frac{k}{\rho}t + r_0\right)\left[1 - \left(\frac{r_0}{\frac{k}{\rho}t + r_0}\right)^4\right]$$

(b)

$$m = \frac{4}{3}\pi r^3 \rho$$

$$\Downarrow$$

$$\frac{dm}{dt} = 4\pi r^2 \rho \frac{dr}{dt}$$

$$\Downarrow \; \tfrac{dm}{dt} = 4\pi r^2 k$$

$$\frac{dr}{dt} = \frac{k}{\rho}$$

$$\Downarrow \; r = r_0 \text{ at } t = 0$$

$$r(t) = \frac{k}{\rho}t + r_0$$

(c)

$$r(10) = \frac{k}{\rho} \times 10 + 0.01 = 0.007 \;\Rightarrow\; \frac{k}{\rho} = -0.0003$$

$$\Downarrow$$

$$r(t) = -0.0003t + 0.01 = 0 \;\Rightarrow\; t = \frac{0.01}{0.0003} \approx 33.3 \text{ sec}$$

3 Higher-Order Differential Equations

3.1 Theory of Linear Equations

- In an **initial-value problem (IVP)**, we seek a solution $y(x)$ of a nth order linear DE so that $y(x)$ satisfies **initial conditions** at x_0
  - n^{th} order linear DE: $a_n(x) \neq 0$

$$\underbrace{a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y}_{L(y)\text{: Linear Operator}} = g(x)$$

  - Initial conditions

$$y(x_0) = y_0,\ y'(x_0) = y_1, \cdots,\ y^{(n-1)}(x_0) = y_{n-1}$$

- **Boundary-value problem (BVP)** consists of solving a linear DE of order 2 or greater, in which the dependent variable y or its derivatives are specified at **different points**
  - For example,

$$a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

  - Boundary conditions

$$y(x_0) = y_0,\ y(x_1) = y_1$$

- The sum, or **superposition** , of two or more solutions of a homogeneous linear DE is also a solution
- Any set of n linearly *independent* solutions $y_1, y_2, \cdots, y_n$ of the nth order homogeneous linear DE on interval I is **a fundamental set of solutions**
- If two functions are linearly dependent, then one is a constant multiple of the other (otherwise, they are linearly independent)

- If $\{y_1, y_2, \cdots, y_n\}$ are a set of linearly independent functions, the **Wronskian** function is not singular:

$$W(y_1, y_2, \cdots, y_n) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix} \neq 0$$

- General solution of nth order *homogeneous* linear DE is

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x)$$

  where $y_1, y_2, \cdots, y_n$ is a fundamental set of solutions and $c_i,\ i = 1, 2, \cdots, n$ are arbitrary constants

- General solution of nth order *nonhomogeneous* linear DE is

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x) + y_p(x)$$

  where $y_1, y_2, \cdots, y_n$ is a fundamental set of solutions, y_p is a particular solution, and $c_i,\ i = 1, 2, \cdots, n$ are arbitrary constants

Example Given that $x(t) = c_1 \cos \omega t + c_2 \sin \omega t$ is the general solution of $x'' + \omega^2 x = 0$ on the interval $(-\infty, \infty)$, show that a solution satisfying the initial conditions $x(0) = x_0$, $x'(0) = x_1$, is given by

$$x(t) = x_0 \cos \omega t + \frac{x_1}{\omega} \sin \omega t$$

Example Determine whether the given set of functions is linearly dependent or linearly independent on the interval $(-\infty, \infty)$

$$f_1(x) = x,\ f_2(x) = x^2,\ f_3(x) = 4x - 3x^2$$

$$f_1(x) = 5,\ f_2(x) = \cos^2 x,\ f_3(x) = \sin^2 x$$

Example Verify that the given functions form a fundamental set of solutions of the differential equation on the indicated interval. Form the general solution of the equation

$$y'' - y' - 12y = 0; \quad e^{-3x}, \; e^{4x}, \quad (-\infty, \infty)$$

$$y'' - 2y' + 5y = 0; \quad e^{x}\cos 2x, \quad e^{x}\sin 2x, \quad (-\infty, \infty)$$

Example Verify that the given two-parameter family of functions is the general solution of the nonhomogeneous differential equation on the indicated interval

$$y'' - 7y' + 10y = 24e^{x}; \quad y = c_1 e^{2x} + c_2 e^{5x} + 6e^{x}; \quad (-\infty, \infty)$$

$$y'' - 4y' + 4y = 2e^{2x} + 4x - 12; \quad y = c_1 e^{2x} + c_2 x e^{2x} + x^2 e^{2x} + x - 2; \quad (-\infty, \infty)$$

3.2 Reduction of Order

- **Reduction of order** can be used to reduce a linear *second-order* DE with known solution y_1 into a linear *first-order* DE, which can be solved for a second solution y_2
- Applying reduction of order $y_2 = u(x)y_1$ to the standard form of a second-order linear homogeneous DE

$$y'' + P(x)y' + Q(x)y = 0$$

gives

$$\begin{aligned} y_1 u'' + (2y_1' + P(x)y_1)\, u' &= 0 \\ \Downarrow \; \times y_1, \; u' &= w \\ (y_1^2 w)' &= -(y_1^2 w)P(x) \\ \Downarrow & \\ y_2(x) &= y_1(x) \int \frac{\exp\left(-\int P(x)\,dx\right)}{y_1^2(x)}\,dx \end{aligned}$$

Example The indicated function $y_1(x)$ is a solution of the given equation. Use the reduction of order to find a second solution $y_2(x)$

- $y'' - 4y' + 4y = 0; \quad y_1 = e^{2x}$
- $y'' + 16y = 0; \quad y_1 = \cos 4x$
- $y'' - y = 0; \quad y_1 = \cosh x$

Example The indicated function $y_1(x)$ is a solution of the associated homogeneous equation. Use the reduction of order to find a second solution $y_2(x)$ of homogeneous equation and a particular solution $y_p(x)$ of the given nonhomogeneous equation

$$y'' - 4y = 2; \quad y_1 = e^{-2x}$$

3.3 Homogeneous Linear Equations with Constant Coefficients

The general solution of $ay'' + by' + cy = 0$ is found by substituting $y = e^{px}$ and solving the resulting **characteristic equation** $ap^2 + bp + c = 0$ for roots p_1 and p_2

- **Case I** p_1 and p_2 are real and distinct

$$y = c_1 e^{p_1 x} + c_2 e^{p_2 x}$$

- **Case II** p_1 and p_2 are real and equal

$$y = c_1 e^{p_1} x + c_2 x e^{p_1 x}$$

- **Case III** p_1 and p_2 are complex conjugate: $p_1, p_2 = \alpha \pm i\beta$

$$y = e^{\alpha} x \left(c_1 \cos \beta x + c_2 \sin \beta x\right)$$

Example Solve $y'' + \omega^2 y = 0$ and $y'' - \omega^2 y = 0$

Example Solve $3y''' + 5y'' + 10y' - 4y = 0$

```
import numpy as np
from scipy.integrate import solve_ivp

import matplotlib.pyplot as plt
```

```python
from sympy import Symbol, init_printing
from sympy.solvers import solve
init_printing()

x = Symbol('x')
solve(3*x**3 +5*x**2 +10*x -4, x)
```

$$\left[\frac{1}{3},\ -1-\sqrt{3}i,\ -1+\sqrt{3}i\right]$$

Example Find the general solution of the given second-order differential equation

- $4y'' + y' = 0$
- $y'' - y' - 6y = 0$
- $y'' + 8y' + 16y = 0$

Example Find the general solution of the given higher-order differential equation

- $y''' - 4y'' - 5y' = 0$
- $y''' - 5y'' + 3y' + 9y = 0$

Example Solve the initial-value problem

$y''' + 12y'' + 36y' = 0,\ \ y(0) = 0,\ y'(0) = 1,\ y''(0) = -7$

3.4 Undetermined Coefficients

Method of undetermined coefficients can be used to obtain a particular solution y_p

- The underlying idea is a conjecture about the form of y_p based on the kinds of functions making up the input function $g(x)$
- Limited to nonhomogeneous linear DEs where
  - Coefficients a_i, $i = 1, \cdots, n$, are constants

- $g(x)$ is a constant, a polynominal function, $e^{\alpha x}$, $\sin \beta x$ or $\cos \beta x$, or finite sums and products of these functions

- There are models of y_p for various functions

$g(x)$	Form of y_p
1. 1 (any constant)	A
2. $5x + 7$	$Ax + B$
3. $3x^2 - 2$	$Ax^2 + Bx + C$
4. $x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + E$
5. $\sin 4x$	$A \cos 4x + B \sin 4x$
6. $\cos 4x$	$A \cos 4x + B \sin 4x$
7. e^{5x}	Ae^{5x}
8. $(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9. x^2e^{5x}	$(Ax^2 + Bx + C)e^{5x}$
10. $e^{3x} \sin 4x$	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
11. $5x^2 \sin 4x$	$(Ax^2 + Bx + C) \cos 4x + (Ex^2 + Fx + G) \sin 4x$
12. $xe^{3x} \cos 4x$	$(Ax + B)e^{3x} \cos 4x + (Cx + E)e^{3x} \sin 4x$

- Finally, the general solution is obtained from the **superposition** of y_h and y_p

Example Solve $y'' + 4y = x \cos x$

- If $g(x)$ consists of a sum of, say, m terms of the kind listed in the table, then the assumption for a particular solution y_p consists of the sum of the trial forms
- If y_p contains terms that duplicate terms in y_h, then that y_p must be multiplied by x^n, where n is the smallest positive integer that eliminate that duplication

Example Solve $y'' - 6y' + 9y = 6x^2 + 2 - 12e^{3x}$

Example Solve $y^{(4)} + y''' = 1 - x^2e^{-x}$

Example Solve the given differential equation by undetermined coefficients

- $y'' + 2y' + y = \sin x + 3 \cos 2x$
- $y'' + 2y' - 24y = 16 - (x + 2)e^{4x}$
- $y''' - 6y'' = 3 - \cos x$

Example Solve the given initial-value problem

- $y'' + 4y = -2, \;\; y(\pi/8) = 1/2, \;\; y'(\pi/8) = 2$
- $5y'' + y' = -6x, \;\; y(0) = 0, \; y'(0) = -10$
- $\dfrac{d^2x}{dt^2} + \omega^2 x = F_0 \sin \omega t, \;\; x(0) = 0, \; x'(0) = 0$

3.5 Variation of Parameters

The **method of variation of parameters** can be used with linear higher-order DEs

- This method *always* yields a y_p provided the homogeneous equation can be solved
- This method is not limited to the types of input functions constraining the method of undetermined coefficients

To adapt the method of variation of parameters to a linear second-order DE

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = g(x),$$

we put the above equation in the standard form

$$y'' + P(x)y' + Q(x)y = f(x)$$

To solve,

- Find the homogeneous solutions y_1, y_2
- Seek a particular solution of the form

$$y_p = u_1(x)y_1 + u_2(x)y_2$$

$$y_p'' + P(x)y_p' + Q(x)y_p = f(x)$$

$$\Downarrow\ y_p = u_1y_1 + u_2y_2$$

$$\frac{d}{dx}[y_1u_1' + y_2u_2'] + P(x)[y_1u_1' + y_2u_2'] + y_1'u_1' + y_2'u_2' = f(x)$$

$$\Downarrow \text{ let } y_1u_1' + y_2u_2' = 0, \text{ then } y_1'u_1' + y_2'u_2' = f(x)$$

$$\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ f(x) \end{bmatrix}$$

$$\Downarrow$$

$$u_1 = \int \frac{W_1}{W}\,dx = -\int \frac{y_2}{W}\,f(x)\,dx, \quad u_2 = \int \frac{W_2}{W}\,dx = \int \frac{y_1}{W}\,f(x)\,dx$$

$$\text{where } W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}, \ W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}, \ W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}$$

Example Solve $y'' - y = \dfrac{1}{x}$

The method can be generalized to the standard form of nth order linear DE. A particular solution is

$$y_p = u_1(x)y_1 + u_2(x)y_2 + \cdots + u_n(x)y_n$$

where

$$u_k = \int \frac{W_k}{W}\,dx, \ k = 1, 2, \cdots, n$$

Example Solve each differential equation by variation of parameters

- $y'' + y = \sec x$
- $y'' + y = \sin x$
- $y'' + y = \cos^2 x$
- $3y'' - 6y' + 6y = e^x \sec x$

Example Solve each differential equation by variation of parameters subject to the initial conditions $y(0) = 1,\ y'(0) = 0$

- $4y'' - y = xe^{x/2}$
- $y'' - 2y' + y = e^x \sec^2 x$
- $y'' + 2y' - 8y = 2e^{-2x} - e^{-x}$

Example Solve each differential equation by variation of parameters subject to the initial conditions $y(0) = 1,\ y'(0) = 0$

- $y'' - 4y = \dfrac{e^{2x}}{x}$
- $2y'' + 2y' + y = 4\sqrt{x}$

3.6 Cauchy-Euler Equation

The **Cauchy-Euler equation** is a linear DE of the form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 x \frac{dy}{dx} + a_0 y = g(x)$$

where $a_n, a_{n-1}, \cdots, a_0$ are constants and the exponent of the coefficient matches the order of differentiation

$y = x^p$ is a solution of second order Cauchy-Euler equation whenever p is a solution of the **auxiliary equation**

$$a_2 p^2 + (a_1 - a_2)p + a_0 = 0$$

- **Case I:** Distinct Real Roots, p_1, p_2

$$y = c_1 x^{p_1} + c_2 x^{p_2}$$

- **Case II:** Repeated Real Roots, $p_1 = p_2$

$$y = c_1 x^{p_1} + c_2 x^{p_1} \ln x$$

- **Case III:** Complex Conjugate Roots, $p_1, p_2 = \alpha \pm i\beta$

$$y = x^{\alpha}\left[c_1\cos(\beta \ln x) + c_2\sin(\beta \ln x)\right]$$

Example Solve $x^2y'' - 3xy' + 3y = 2x^4e^x$

Example Solve the given differential equation

- $x^2y'' - 2y = 0$
- $xy'' + y' = 0$
- $x^2y'' + xy' + 4y = 0$

Example Solve the given differential equation by variation of parameters

- $xy'' - 4y' = x^4$
- $x^2y'' + xy' - y = \ln x$

3.7 Nonlinear Equations

- Nonlinear equations do not possess superposability

Example Verify that y_1 and y_2 are solutions of the given DE but that $c_1y_1 + c_2y_2$ is, in general, not a solution

$$(y'')^2 = y^2,\ y_1 = e^x,\ y_2 = \cos x$$

- The major difference between linear and nonlinear DEs of order two or higher lies in the realm of solvability. Nonlinear higher-order DEs virtually defy solution. This means that there are no analytical methods whereby either an explicit or implicit solution can be found

- There are still things that can be done; we can always analyze a nonlinear DE qualitatively and numerically
- Nonlinear second-order DE $F(y, y', y'') = 0$ can be reduced to two first-order equations by means of the substitution $u = y'$ and can sometimes be solved using first-order methods

Example Solve $y'' = 2x(y')^2$ and $yy'' = (y')^2$

- In some instances, a solution of a nonlinear IVP can be approximated by a Taylor series centered at x_0

Example Solve $y'' = x + y - y^2,\ y(0) = -1,\ y'(0) = 1$ by using

$$y(x) = y(0) + \frac{y'(0)}{1!}x + \frac{y''(0)}{2!}x^2 + \frac{y'''(0)}{3!}x^3 + \cdots$$

Example The dependent variable y is missing in the given differential equation. Solve the equation by using the substitution $u = y'$

$$y'' + (y')^2 + 1 = 0$$

Example The independent variable x is missing in the given differential equation. Solve the equation by using the substitution $u = y'$

$$yy'' + (y')^2 + 1 = 0$$

Example Consider the initial-value problem

$$y'' + yy' = 0,\ \ y(0) = 1,\ y'(0) = -1$$

1. Use the DE and a numerical solver to graph the solution curve
2. Find an explicit solution of the IVP. Use a graphing utility to graph this solution
3. Find an interval of definition for the solution in part 2

Example Show that the substitution $u = y'$ leads to a Bernoulli equation. Solve this equation

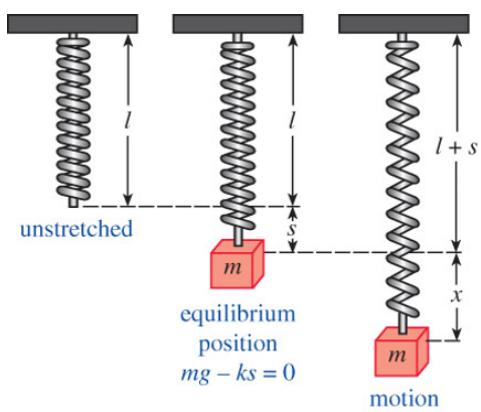
$$xy'' = y' + (y')^3$$

3.8 Linear Models: Initial-Value Problems

Several dynamical systems are modeled with linear 2nd-order DEs with constant coefficients and initial conditions at t_0

- Spring/Mass Systems
  - Free Undamped Motion
  - Free Damped Motion
  - Driven Motion

$$\begin{aligned} m\frac{d^2x}{dt^2} &= -kx - \beta\frac{dx}{dt} + f(t) \\ x(0) &= x_0 \\ \dot{x}(0) &= x_1 \end{aligned}$$



Example Show that the solution of the initial-value problem

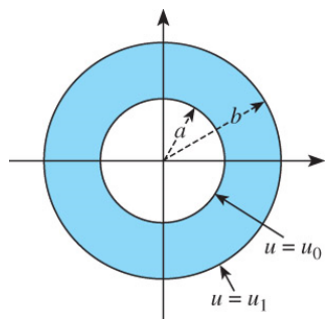
$$\frac{d^2x}{dt^2} + \omega^2 x = F_0 \cos\gamma t,\ x(0) = 0,\ x'(0) = 0$$

is $x(t) = \dfrac{F_0}{\omega^2 - \gamma^2}(\cos\gamma t - \cos\omega t)$

Example Evaluate $\lim\limits_{\gamma\to\omega} \dfrac{F_0}{\omega^2 - \gamma^2}(\cos\gamma t - \cos\omega t)$

3.9 Linear Models: Boundary-Value Problems

Example Temperature in a Ring



The temperature $u(r)$ in circular ring is determined from the boundary-value problem

$$r\frac{d^2u}{dr^2} + \frac{du}{dr} = 0,\ u(a) = u_0,\ u(b) = u_1$$

3.10 Green's Functions

3.10.1 Influence Function

- We shall consider the self-adjoint form

$$\frac{d}{dx}\left[p(x)\frac{du}{dx}\right] + q(x)u = -f(x) \tag{SA}$$

The function $p(x)$ is continuously differentiable and positive, and $q(x)$ and $f(x)$ are continuous for $\alpha < x < \beta$

- The homogeneous second-order differential equation

$$\frac{d}{dx}\left[p(x)\frac{dv}{dx}\right] + q(x)v = 0 \tag{SH}$$

has exactly two linearly independent solutions $v_1(x)$ and $v_2(x)$: any solution of (SH) can be written in the form

$$v(x) = c_1 v_1(x) + c_2 v_2(x)$$

where c_1 and c_2 are constants

- We now consider the function

$$w(x) = v_1(x)\int_\alpha^x v_2(\xi)f(\xi)\,d\xi - v_2(x)\int_\alpha^x v_1(\xi)f(\xi)\,d\xi$$

Differentiating w, we have

$$\begin{aligned}\frac{dw}{dx} &= v_1'(x)\int_\alpha^x v_2(\xi)f(\xi)\,d\xi - v_2'(x)\int_\alpha^x v_1(\xi)f(\xi)\,d\xi + \underbrace{[v_1(x)v_2(x) - v_2(x)v_1(x)]}_{=0} f(x)\\ &= v_1'(x)\int_\alpha^x v_2(\xi)f(\xi)\,d\xi - v_2'(x)\int_\alpha^x v_1(\xi)f(\xi)\,d\xi\end{aligned}$$

- Then

$$\begin{aligned}\frac{d}{dx}\left[p(x)\frac{dw}{dx}\right] &= \underbrace{\frac{d}{dx}\left[p(x)v_1'(x)\right]}_{-q(x)v_1}\int_\alpha^x v_2(\xi)f(\xi)\,d\xi - \underbrace{\frac{d}{dx}\left[p(x)v_2'(x)\right]}_{-q(x)v_2}\int_\alpha^x v_1(\xi)f(\xi)\,d\xi\\ &\quad + \underbrace{p(x)\left[v_1'(x)v_2(x) - v_2'(x)v_1(x)\right]}_{-K} f(x)\\ &= -q(x)w - Kf(x)\end{aligned}$$

where

$$
\begin{aligned}
&\frac{d}{dx}\left\{p(x)\left[v_1'(x)v_2(x)-v_2'(x)v_1(x)\right]\right\}\\
&\quad=\underbrace{\frac{d}{dx}\left[p(x)v_1'(x)\right]}_{-q(x)v_1(x)}v_2(x)-\underbrace{\frac{d}{dx}\left[p(x)v_2'(x)\right]}_{-q(x)v_2(x)}v_1(x)+p(x)v_1'(x)v_2'(x)-p(x)v_2'(x)v_1'(x)=0\\
&\quad\Downarrow\\
&\quad K \text{ is constant}
\end{aligned}
$$

- We have shown that w satisfies the equation:

$$\frac{d}{dx}\left[p(x)\frac{dw}{dx}\right]+q(x)w=-Kf(x)$$

Moreover, as $x\to\alpha$

$$w(\alpha)=w'(\alpha)=0$$

- Dividing by the constant K, we find that the function

$$u(x)=\int_\alpha^x R(x,\xi)\,f(\xi)\,d\xi \tag{IN}$$

where

$$R(x,\xi)=\frac{v_1(x)v_2(\xi)-v_2(x)v_1(\xi)}{K}=-\frac{v_1(x)v_2(\xi)-v_2(x)v_1(\xi)}{p(x)\left[v_1'(x)v_2(x)-v_2'(x)v_1(x)\right]}$$

is the solution of the initial value problem

$$\begin{aligned}\frac{d}{dx}\left[p(x)\frac{du}{dx}\right]+qu&=-f(x)\quad\text{for } x>\alpha\\ u(\alpha)=u'(\alpha)&=0\end{aligned}$$

- Since the denominator of $R(x,\xi)$ is a constant, the function $R(x,\xi)$ satisfies the homogeneous equation (SH) as either a function of x or ξ. In fact,

$$R(x,\xi)=-R(\xi,x)$$

- For a fixed value of ξ, $R(x,\xi)$ is completely characterized as the solution of the homogeneous initial value problem

$$\begin{aligned} \frac{d}{dx}\left[p(x)\frac{dR}{dx}\right] + q(x)R = 0 \quad \text{for } x > \xi \\ R\big|_{x=\xi} = 0 \\ R'\big|_{x=\xi} = -\frac{1}{p(\xi)} \end{aligned}$$

- The function $R(x,\xi)$ describes the influence on the value of u at x of a disturbance(impulse) concentrated at ξ. It is sometimes called the **influence function**, or the one-sided Green's function
- If the values of $u(\alpha)$ and $u'(\alpha)$ are prescribed to be other than zero, we must simply add a suitable solution $c_1v_1(x) + c_2v_2(x)$ to the expression (IN)

Example Consider the problem

$$\begin{aligned} u'' + u = -f(x) \quad \text{for } x > 0 \\ u(0) = 1 \\ u'(0) = -1 \end{aligned}$$

Solution $u(x) = w(x) + v(x)$

(1) $w(x)$

$$\begin{aligned} w'' + w = -f(x) \quad \text{for } x > 0 \\ w(0) = 0 \\ w'(0) = 0 \end{aligned}$$

For a fixed value of ξ, the influence function $R(x,\xi)$ satisfies

$$\begin{aligned} \frac{d^2R}{dx^2} + R = 0 \quad \text{for } x > \xi \\ R\big|_{x=\xi} = 0 \\ R'\big|_{x=\xi} = -1 \end{aligned}$$

Thus $R(x,\xi) = \sin(\xi - x)$ and the solution is

$$w(x) = \int_0^x \sin(\xi - x)\, f(\xi)\, d\xi$$

(2) $v(x)$

$$\begin{aligned} v'' + v &= 0 \quad \text{for } \ x > 0 \\ v(0) &= 1 \\ v'(0) &= -1 \end{aligned}$$

$$\Downarrow$$

$$v(x) = \cos x - \sin x$$

3.10.2 Green's Function

- Here, we research a two-point boundary value problem:

$$\begin{gathered} \frac{d}{dx}\left[p(x)\frac{du}{dx}\right] + q(x)u = -f(x) \quad \text{for } \alpha < x < \beta \\ u(\alpha) = u(\beta) = 0 \end{gathered} \tag{P1}$$

  Writing the general solution in the form

$$u(x) = \int_\alpha^x R(x,\xi)\, f(\xi)\, d\xi + c_1 v_1(x) + c_2 v_2(x)$$

  we obtain the two equations

$$\begin{aligned} c_1 v_1(\alpha) + c_2 v_2(\alpha) &= 0 \\ c_1 v_1(\beta) + c_2 v_2(\beta) &= -\int_\alpha^\beta R(\beta,\xi)\, f(\xi)\, d\xi \end{aligned}$$

  These two equations determine a unique pair of constants c_1 and c_2, provided the determinant of their coefficients is not zero; that is, provided

$$D \equiv v_1(\alpha)v_2(\beta) - v_2(\alpha)v_1(\beta) \neq 0$$

- We assume this for the moment. Then

$$
\begin{aligned}
c_1 &= \frac{v_2(\alpha)}{D}\int_\alpha^\beta R(\beta,\xi)\,f(\xi)\,d\xi \\
&= \frac{v_2(\alpha)}{D}\int_\alpha^x R(\beta,\xi)\,f(\xi)\,d\xi + \frac{v_2(\alpha)}{D}\int_x^\beta R(\beta,\xi)\,f(\xi)\,d\xi \\
c_2 &= -\frac{v_1(\alpha)}{D}\int_\alpha^\beta R(\beta,\xi)\,f(\xi)\,d\xi \\
&= -\frac{v_1(\alpha)}{D}\int_\alpha^x R(\beta,\xi)\,f(\xi)\,d\xi - \frac{v_1(\alpha)}{D}\int_x^\beta R(\beta,\xi)\,f(\xi)\,d\xi
\end{aligned}
$$

- The solution can then be written as

$$
\begin{aligned}
u(x) = &\int_\alpha^x \left[R(x,\xi) + \frac{v_2(\alpha)v_1(x) - v_1(\alpha)v_2(x)}{D}R(\beta,\xi)\right] f(\xi)\,d\xi \\
&+ \int_x^\beta \frac{v_2(\alpha)v_1(x) - v_1(\alpha)v_2(x)}{D}R(\beta,\xi)\,f(\xi)\,d\xi
\end{aligned}
$$

- We find after some algebraic manipulation that

$$
\begin{aligned}
G(x,\xi) &= R(x,\xi) + \frac{v_2(\alpha)v_1(x) - v_1(\alpha)v_2(x)}{D}R(\beta,\xi) \\
&= \frac{v_1(x)v_2(\xi) - v_2(x)v_1(\xi)}{K}\frac{D}{D} + \frac{v_2(\alpha)v_1(x) - v_1(\alpha)v_2(x)}{D}\frac{v_1(\beta)v_2(\xi) - v_2(\beta)v_1(\xi)}{K} \\
&= \frac{[v_1(\alpha)v_2(\xi) - v_2(\alpha)v_1(\xi)]\,[v_1(x)v_2(\beta) - v_2(x)v_1(\beta)]}{KD} \quad \text{for } \xi \le x
\end{aligned}
$$

$$
\begin{aligned}
G(x,\xi) &= \frac{v_2(\alpha)v_1(x) - v_1(\alpha)v_2(x)}{D}R(\beta,\xi) \\
&= \frac{[v_1(x)v_2(\alpha) - v_2(x)v_1(\alpha)]\,[v_1(\beta)v_2(\xi) - v_2(\beta)v_1(\xi)]}{KD} \quad \text{for } \xi \ge x
\end{aligned}
$$

- Then the solution of the two-point boundary value problem (P1) can be written in the form

$$
u(x) = \int_\alpha^\beta G(x,\xi)\,f(\xi)\,d\xi \tag{GR}
$$

- The function $G(x,\xi)$ is called the **Green's function** of the problem (P1). It is symmetric. That is

$$
G(x,\xi) = G(\xi,x)
$$

- To determine the Green's function, we note that for each ξ it satisfies the following boundary value problem

$$\begin{aligned}
\frac{d}{dx}\left[p(x)\frac{dG}{dx}\right] + q(x)G &= 0 \quad \text{for } x \neq \xi \\
G\big|_{x=\alpha} = G\big|_{x=\beta} &= 0 \\
G\big|_{x=\xi+0} - G\big|_{x=\xi-0} &= 0 \\
G'\big|_{x=\xi+0} - G'\big|_{x=\xi-0} &= -\frac{1}{p(\xi)}
\end{aligned}$$

Example Consider the problem

$$\begin{aligned}
u'' &= -f(x) \quad \text{for } 0 < x < 1 \\
u(0) &= 0 \\
u(1) &= 0
\end{aligned}$$

Solution

$$\begin{aligned}
G'' = 0 \;&\Rightarrow\; G = ax + b \\
&\Downarrow\; {\scriptstyle G|_{x=0}=G|_{x=1}=0} \\
G(x,\xi) &= \begin{cases} a_1(\xi)\,x & \text{for } x < \xi \\ a_2(\xi)(1-x) & \text{for } x > \xi \end{cases} \\
&\Downarrow\; {\scriptstyle G(x,\xi)=G(\xi,x)} \\
a_1(\xi) &= A(1-\xi) \\
a_2(\xi) &= A\xi \\
&\Downarrow\; {\scriptstyle G'|_{x=\xi+0}-G'|_{x=\xi-0}=-1} \\
-A\xi - A(1-\xi) &= -1 \to A = 1 \\
&\Downarrow \\
G(x,\xi) &= \begin{cases} (1-\xi)x & \text{for } x \leq \xi \\ \xi(1-x) & \text{for } x \geq \xi \end{cases}
\end{aligned}$$

The solution is

$$u(x) = \int_0^x \xi(1-x)\,f(\xi)\,d\xi + \int_x^1 (1-\xi)x\,f(\xi)\,d\xi$$

- Two-point boundary value problems with more general boundary conditions can be treated in the same manner. We consider the problem

$$\begin{aligned} \frac{d}{dx}\left[p(x)\frac{du}{dx}\right] + q(x)u &= -f(x) \quad \text{for } \alpha < x < \beta \\ -\mu_1 u'(\alpha) + \sigma_1 u(\alpha) &= 0 \\ \mu_2 u'(\beta) + \sigma_2 u(\beta) &= 0 \end{aligned} \tag{P2}$$

- The Green's function $G(x,\xi)$ is derived as before, provided the condition

$$\begin{aligned} D \equiv & \left[-\mu_1 v_1'(\alpha) + \sigma_1 v_1(\alpha)\right]\left[\mu_2 v_2'(\beta) + \sigma_2 v_2(\beta)\right] \\ & - \left[-\mu_1 v_2'(\alpha) + \sigma_1 v_2(\alpha)\right]\left[\mu_2 v_1'(\beta) + \sigma_2 v_1(\beta)\right] \neq 0 \end{aligned}$$

  is satisfied

- The Green's function $G(x,\xi)$ is the solution of the problem

$$\begin{aligned} \frac{d}{dx}\left[p(x)\frac{dG}{dx}\right] + q(x)G &= 0 \quad \text{for } x \neq \xi \\ -\mu_1\, G'\big|_{x=\alpha} + \sigma_1\, G\big|_{x=\alpha} = \mu_2\, G'\big|_{x=\beta} + \sigma_2\, G\big|_{x=\beta} &= 0 \\ G\big|_{x=\xi+0} - G\big|_{x=\xi-0} &= 0 \\ G'\big|_{x=\xi+0} - G'\big|_{x=\xi-0} &= -\frac{1}{p(\xi)} \end{aligned}$$

  It still satisfies the symmetry relation

$$G(x,\xi) = G(\xi,x)$$

Example Consider the problem

$$\begin{aligned} u'' &= -f(x) \quad \text{for } 0 < x < 1 \\ u(0) &= 0 \\ u'(1) + \sigma_2 u(1) &= 1 \end{aligned}$$

Solution $u(x) = w(x) + v(x)$

(1) $w(x)$

$$
\begin{aligned}
w'' &= -f(x) \quad \text{for } 0 < x < 1 \\
w(0) &= 0 \\
w'(1) + \sigma_2 w(1) &= 0
\end{aligned}
$$

$$
\begin{aligned}
G'' &= 0 \\
&\Downarrow \\
G &= ax + b \\
&\Downarrow \; G|_{x=0} = G'|_{x=1} + \sigma_2 G|_{x=1} = 0 \\
G(x,\xi) &= \begin{cases} a_1(\xi)\, x & \text{for } x < \xi \\ a_2(\xi)\,[1 + \sigma_2(1-x)] & \text{for } x > \xi \end{cases} \\
&\Downarrow \; G(x,\xi) = G(\xi,x) \\
a_1(\xi) &= A\,[1 + \sigma_2(1-\xi)] \\
a_2(\xi) &= A\xi \\
&\Downarrow \; G'|_{x=\xi+0} - G'|_{x=\xi-0} = -1 \\
A\xi \cdot -\sigma_2 - A\,[1 + \sigma_2(1-\xi)] &= -1 \to A = \frac{1}{1+\sigma_2} \\
&\Downarrow \\
G(x,\xi) &= \begin{cases} \frac{[1+\sigma_2(1-\xi)]\,x}{1+\sigma_2} & \text{for } x \le \xi \\ \frac{\xi\,[1+\sigma_2(1-x)]}{1+\sigma_2} & \text{for } x \ge \xi \end{cases}
\end{aligned}
$$

The solution is

$$
w(x) = \int_0^x \frac{\xi\,[1 + \sigma_2(1-x)]}{1+\sigma_2}\, f(\xi)\, d\xi + \int_x^1 \frac{[1 + \sigma_2(1-\xi)]\; x}{1+\sigma_2}\, f(\xi)\, d\xi
$$

(2) $v(x)$

$$
\begin{aligned}
v'' &= 0 \quad \text{for } \; 0 < x < 1 \\
v(0) &= 0 \\
v'(1) + \sigma_2 v(1) &= 1
\end{aligned}
$$

$$
\Downarrow
$$

$$
v(x) = \frac{x}{1+\sigma_2}
$$

Example Consider the problem

$$r^2u'' + 2ru' - n(n+1)u = -r^2F(r),\ u(R) = 0,\ u \sim \text{bounded}$$

Solution

$$\frac{d}{dr}\left[r^2\frac{dG}{dr}\right] - n(n+1)G = 0$$

$$\Downarrow$$

$$G = c_1r^n + c_2r^{-(n+1)}$$

$$\Downarrow G|_{r=0} = \text{bounded}\ ,\ G|_{r=R} = 0$$

$$G(r,\rho) = \begin{cases} a_1(\rho)r^n & \text{for } r < \rho \\ a_2(\rho)\left[\left(\frac{r}{R}\right)^{-(n+1)} - \left(\frac{r}{R}\right)^n\right] & \text{for } r > \rho \end{cases}$$

$$\Downarrow G(r,\rho) = G(\rho,r)$$

$$a_1(\rho) = A\left[\left(\frac{\rho}{R}\right)^{-(n+1)} - \left(\frac{\rho}{R}\right)^n\right]$$

$$a_2(\rho) = A\rho^n$$

$$\Downarrow G'|_{r=\rho+0} - G'|_{r=\rho-0} = -1/\rho^2$$

$$\frac{A\rho^n}{R}\left[-(n+1)\left(\frac{\rho}{R}\right)^{-(n+2)} - n\left(\frac{\rho}{R}\right)^{n-1}\right] - An\rho^{n-1}\left[\left(\frac{\rho}{R}\right)^{-(n+1)} - \left(\frac{\rho}{R}\right)^n\right] = -\frac{1}{\rho^2} \quad \rightarrow \quad A = \frac{1}{(2n+1)R^{n+1}}$$

$$\Downarrow$$

$$G(r,\rho) = \begin{cases} \frac{1}{(2n+1)R}\left(\frac{r}{R}\right)^n\left[\left(\frac{\rho}{R}\right)^{-(n+1)} - \left(\frac{\rho}{R}\right)^n\right] & \text{for } r \le \rho \\ \frac{1}{(2n+1)R}\left(\frac{\rho}{R}\right)^n\left[\left(\frac{r}{R}\right)^{-(n+1)} - \left(\frac{r}{R}\right)^n\right] & \text{for } r \ge \rho \end{cases}$$

The solution is

$$u(x) = \int_0^R G(r,\rho)\,F(\rho)\rho^2 d\rho$$

3.11 Solving Nonlinear Model

- In order to analyze an nth order IVP numerically, we express the nth order ODE as a system of n first-order equations

- For example,

$$\frac{d^2y}{dx^2} = f(x, y, y'), \;\; y(x_0) = y_0, \; y'(x_0) = y_1$$

$$\Downarrow y' = u$$

$$\text{Solve}: \begin{cases} y' = u \\ u' = f(x, y, u) \end{cases}$$

$$\text{subject to}: y(x_0) = y_0, \; u(x_0) = y_1$$

Example Use a numerical solver to obtain the solution curves satisfying the given initial conditions:

$$\frac{d^2x}{dt^2} + \frac{dx}{dt} + x + x^3 = 0$$

$$x(0) = -3, \; \dot{x}(0) = 4, \;\; \text{or} \;\; x(0) = 0, \; \dot{x}(0) = -8$$

Solution

$$\frac{d^2x}{dt^2} + \frac{dx}{dt} = -x - x^3$$

$$\Downarrow$$

$$\dot{x} = u$$

$$\dot{u} = -x - x^3 - u$$

```
import numpy as np
from scipy.integrate import solve_ivp

def func(t, y):
    return [y[1], -y[0] -y[0]**3 -y[1]]

tf = 14
t_eval = np.linspace(0, tf, 200)

sol1 = solve_ivp(func, [0, tf], [-3, 4], t_eval=t_eval)
sol2 = solve_ivp(func, [0, tf], [0, -8], t_eval=t_eval)
```

```
import matplotlib.pyplot as plt
plt.style.use('ggplot')

plt.figure(figsize=(6, 4))

plt.plot(sol1.t, sol1.y[0], 'b-', label=r'$x_0=-3,\; \dot{x}_0=4$')
plt.plot(sol2.t, sol2.y[0], 'r-', label=r'$x_0=0,\; \dot{x}_0=-8$')
plt.axis((0, tf, -3, 3))
plt.xlabel('t')
plt.ylabel('x')
plt.legend()

plt.show()
```

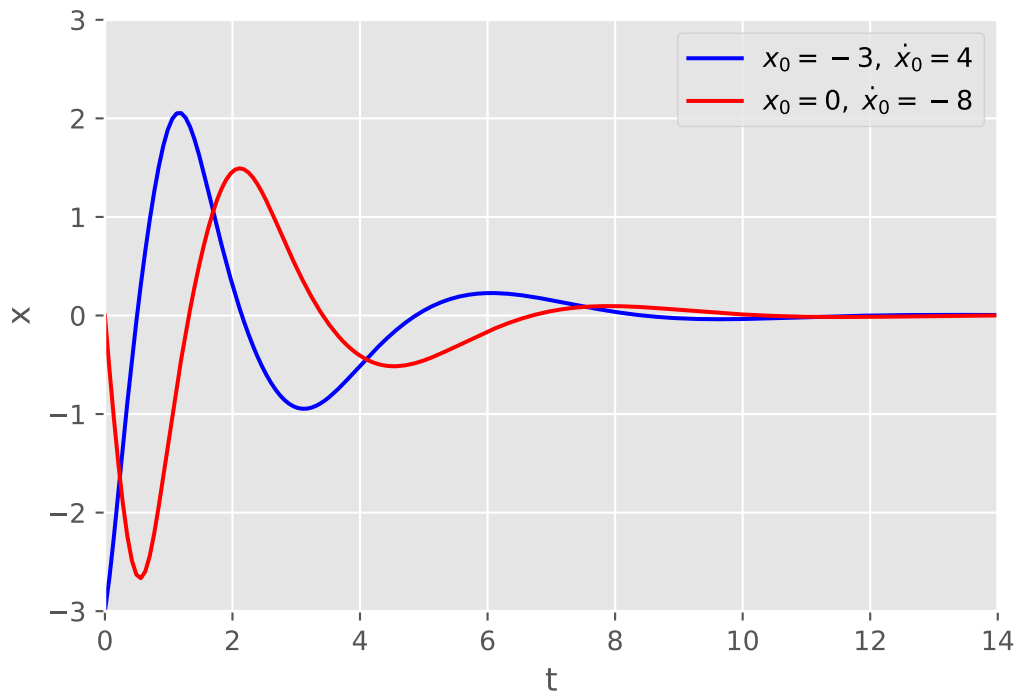
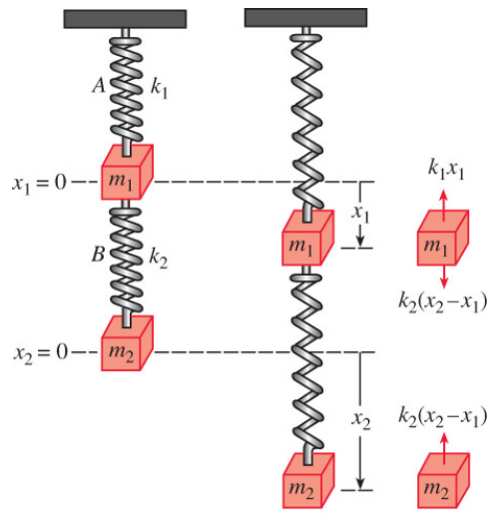


Figure 3.1: $\dfrac{d^2x}{dt^2} + \dfrac{dx}{dt} + x + x^3 = 0$

3.12 Solving System of Linear Equations

When physical systems are coupled, the mathematical model of the system usually consists of a set of coupled DEs



$$
\begin{aligned}
m_1\ddot{x}_1 &= -k_1x_1 + k_2(x_2 - x_1) \\
m_2\ddot{x}_2 &= -k_2(x_2 - x_1)
\end{aligned}
$$

Linear systems with constant coefficients can be solved by uncoupling the system into distinct linear ODEs in each dependent variable

Example Solve the above equation under the assumption that $k_1 = 6$, $k_2 = 4$, $m_1 = 1$, and $m_2 = 1$ subject to

$$x_1(0) = 0,\ x_1'(0) = 1,\ x_2(0) = 0,\ x_2'(0) = -1$$

```
k1, k2, m1, m2 = 6, 4, 1, 1

def func(t, y):
    return [y[1], -(k1 + k2)/m1 * y[0] + k2/m1 * y[2],
            y[3], k2/m2 * y[0] - k2/m2 * y[2]]

tf = 14
t_eval = np.linspace(0, tf, 200)
```

```
sol = solve_ivp(func, [0, tf], [0, 1, 0, -1], t_eval=t_eval)
```

```
plt.figure(figsize=(6, 4))

plt.plot(sol.t, sol.y[0], 'b-', label='$x_1$')
plt.plot(sol.t, sol.y[2], 'r-', label='$x_2$')
plt.axis((0, tf, -0.6, 0.6))
plt.legend()
plt.xlabel('t')
plt.ylabel('x')

plt.show()
```

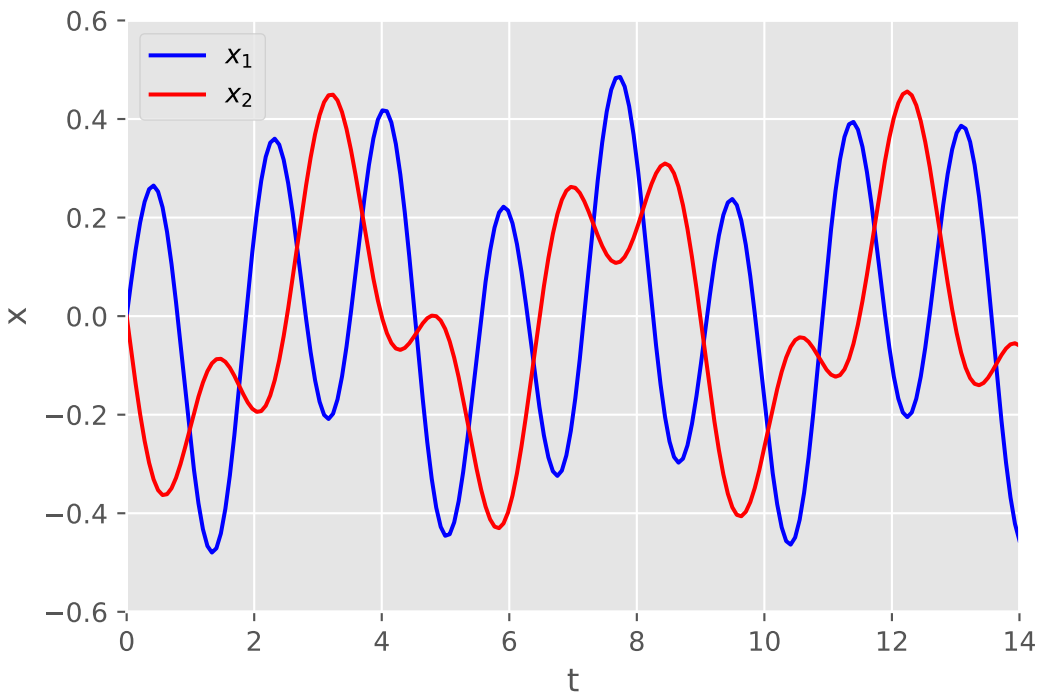


Figure 3.2: Coupled Spring/Mass System

Worked Exercises

4 The Laplace Transform

4.1 Definition of the Laplace Transform

- If f be a function defined for $t \geq 0$, then the *integral*

$$\mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st}\,dt = F(s)$$

  is the **Laplace Transform** of f provided the integral converges. The result is a function of s

Example Evaluate $\mathcal{L}\{1\}$, $\mathcal{L}\{t\}$, and $\mathcal{L}\{e^{-3t}\}$

Example Evaluate $\mathcal{L}\{f(t)\}$ for $f(t) = \begin{cases} 0, & 0 \leq t < 3 \\ 2, & t \geq 3 \end{cases}$

- $\mathcal{L}$ is a linear transform

$$\mathcal{L}\{\alpha f(t) + \beta g(t)\} = \alpha\mathcal{L}\{f(t)\} + \beta\mathcal{L}\{g(t)\} = \alpha F(s) + \beta G(s)$$

Example Find $\mathcal{L}\{f(t)\}$ by first using an appropriate trigonometric identity $f(t) = \sin^2 2t$

Example Find $\mathcal{L}\{f(t)\}$

- $f(t) = \begin{cases} -1, & 0 \leq t < 1 \\ 1, & t \geq 1 \end{cases}$
- $f(t) = \begin{cases} t, & 0 \leq t < 1 \\ 1, & t \geq 1 \end{cases}$

- $f(t) = \begin{cases} \sin t, & 0 \leq t < \pi \\ 0, & t \geq \pi \end{cases}$

Example Find $\mathcal{L}\{f(t)\}$

- $f(t) = e^{t+7}$
- $f(t) = e^t \cos t$
- $f(t) = t \cos t$
- $f(t) = \sin 3t \cos 3t$
- $f(t) = \sin^4 t$

4.2 The Inverse Transform and Transforms of Derivatives

- If $F(s)$ represents the Laplace transform of $f(t)$, then $f(t)$ is the **inverse Laplace transform** of $F(s)$

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

- $\mathcal{L}^{-1}$ is a linear transform

$$\mathcal{L}^{-1}\{\alpha F(s) + \beta G(s)\} = \alpha \mathcal{L}^{-1}\{F(s)\} + \beta \mathcal{L}^{-1}\{G(s)\} = \alpha f(t) + \beta g(t)$$

Example Find the given inverse transform

- $\mathcal{L}^{-1}\left\{\dfrac{1}{s^3}\right\}$
- $\mathcal{L}^{-1}\left\{\dfrac{1}{s^2} - \dfrac{48}{s^5}\right\}$
- $\mathcal{L}^{-1}\left\{\dfrac{(s+1)^3}{s^4}\right\}$

- $\mathcal{L}^{-1}\left\{\dfrac{1}{4s^2+1}\right\}$

- $\mathcal{L}^{-1}\left\{\dfrac{s+1}{s^2+2}\right\}$

Example Evaluate $\mathcal{L}^{-1}\left\{\dfrac{-2s+6}{s^2+4}\right\}$

- **Transforms of derivatives**

$$\begin{aligned}\mathcal{L}\{f'(t)\} &= sF(s) - f(0)\\ \mathcal{L}\{f''(t)\} &= s^2F(s) - sf(0) - f'(0)\\ &\vdots\end{aligned}$$

  $\mathcal{L}\left\{\dfrac{d^nf}{dt^n}\right\}$ depends on $F(s) = \mathcal{L}\{f(t)\}$ and the $n-1$ derivatives of $f(t)$ evaluated at $t = 0$

- If f is piecewise continuous on $[0, \infty]$ and of exponential order, then

$$\lim_{s\to\infty} \mathcal{L}\{f(t)\} = 0$$

- The Laplace transform of a linear DE with constant coefficients becomes an algebraic equation in $Y(s)$

Example Use the Laplace transform to solve the IVP

$$\frac{dy}{dt} + 3y = 13\sin 2t,\ y(0) = 6$$

Example Use the Laplace transform to solve the given initial-value problem

- $y'' + y = \sqrt{2}\sin\sqrt{2}t,\ \ y(0) = 10,\ y'(0) = 0$
- $2y''' + 3y'' - 3y' - 2y = e^{-t},\ \ y(0) = 0,\ y'(0) = 0,\ y''(0) = 1$

4.3 Translation Theorems

- **First Translation Theorem**

  If $\mathcal{L}\{f(t)\} = F(s)$ and a is any real number, then

$$\mathcal{L}\{e^{-at}f(t)\} = F(s+a)$$

Example Evaluate $\mathcal{L}\{e^{-2t}\cos 4t\}$ and $\mathcal{L}^{-1}\left\{\dfrac{2s+5}{(s+3)^2}\right\}$

- **Second Translation Theorem**

  If $\mathcal{L}\{f(t)\} = F(s)$ and $a > 0$, then

$$\mathcal{L}\{f(t-a)\mathcal{U}(t-a)\} = e^{-as}F(s)$$

- **Alternative Form**

$$\begin{aligned}\mathcal{L}\{g(t)\mathcal{U}(t-a)\} &= \int_a^\infty e^{-st}g(t)\,dt \\ &= \int_0^\infty e^{-s(t'+a)}g(t'+a)\,dt' = e^{-as}\mathcal{L}\{g(t+a)\}\end{aligned}$$

Example Find either $F(s)$ or $f(t)$

- $\mathcal{L}\{te^{10t}\}$
- $\mathcal{L}\{t^{10}e^{-7t}\}$
- $\mathcal{L}\{e^t\sin 3t\}$
- $\mathcal{L}^{-1}\left\{\dfrac{s}{(s+1)^2}\right\}$
- $\mathcal{L}^{-1}\left\{\dfrac{2s-1}{s^2(s+1)^3}\right\}$

Example Solve $y' + y = f(t),\ \ y(0) = 5$, where

$$f(t) = \begin{cases} 0, & 0 \le t < \pi \\ 3\cos t, & t \ge \pi \end{cases}$$

4.4 Additional Operational Properties

- **Derivatives of Transforms:** If $F(s) = \mathcal{L}\{f(t)\}$ and $n = 1, 2, \cdots$, then

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

- **Convolution Theorem:** $f * g = \displaystyle\int_0^t f(\tau) g(t-\tau)\, d\tau$

$$\mathcal{L}\{f * g\} = \mathcal{L}\{f(t)\}\mathcal{L}\{g(t)\} = F(s)G(s)$$

- **Transform of a Periodic Function:** $f(t+T) = f(t)$

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t)\, dt$$

Example Evaluate $\mathcal{L}\{t \sin \omega t\}$

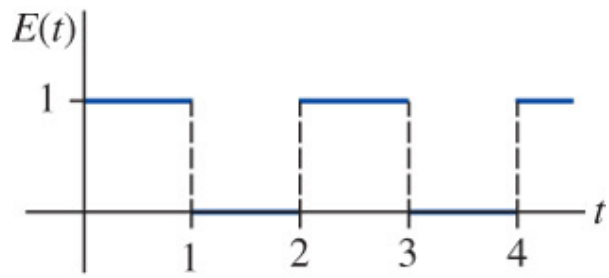
Example Solve $x'' + 16x = \cos 4t,\ x(0) = 1,\ x'(0) = 1$

Example Evaluate $\mathcal{L}^{-1}\left\{\dfrac{1}{(s^2 + \omega^2)^2}\right\}$

Example Evaluate $\mathcal{L}\left\{\displaystyle\int_0^t f(\tau)\, d\tau\right\}$

Example Solve $f(t) = 3t^2 - e^{-t} - \int_0^t f(\tau)\, e^{t-\tau}\, d\tau$ for $f(t)$

Example Find the Laplace transform of the periodic function



$$\mathcal{L}\{E(t)\} = \frac{1}{1-e^{-2s}} \int_0^2 e^{-st} E(t)\, dt = \frac{1}{s(1+e^{-s})}$$

Example Evaluate the given Laplace transform

- $\mathcal{L}\left\{te^{-10t}\right\}$
- $\mathcal{L}\left\{t\cos 2t\right\}$
- $\mathcal{L}\left\{te^{2t}\sin 6t\right\}$

Example Use the Laplace transform to solve the given initial-value problem

- $y' + y = t\sin t,\ y(0) = 0$
- $y'' + 9y = \cos 3t,\ y(0) = 2,\ y'(0) = 5$

Example Find the convolution $f*g$ of the given functions. After integrating find the Laplace transform $f*g$

- $f(t) = 4t,\ g(t) = 3t^2$
- $f(t) = e^{-t},\ g(t) = e^t$

Example Find the Laplace transform

- $\mathcal{L}\left\{e^{-t} * e^{t}\cos t\right\}$

Example Evaluate the given inverse transform

- $\mathcal{L}^{-1}\left\{\dfrac{1}{s^3(s-1)}\right\}$

Example Use the Laplace transform to solve the given integral or integrodifferential equation

- $f(t) + 2\displaystyle\int_0^t f(\tau)\cos(t-\tau)\,d\tau = 4e^{-t} + \sin t$
- $\dfrac{dy}{dt} = 10 - \displaystyle\int_0^t e^{-4t}y(t-\tau)\,d\tau,\ y(0) = 5$

Example The Laplace transform $\mathcal{L}\left\{e^{-t^2}\right\}$ exists, but without finding it solve the initial-value problem

- $y'' + 9y = 3e^{-t^2},\ \ y(0) = 0,\ y'(0) = 0$

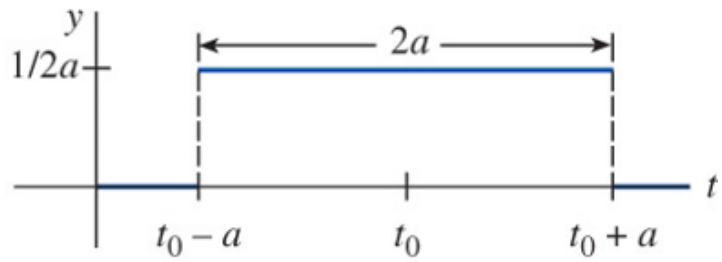
Example Solve the integral equation

- $f(t) = e^t + e^t\displaystyle\int_0^t e^{-\tau}f(\tau)\,d\tau$

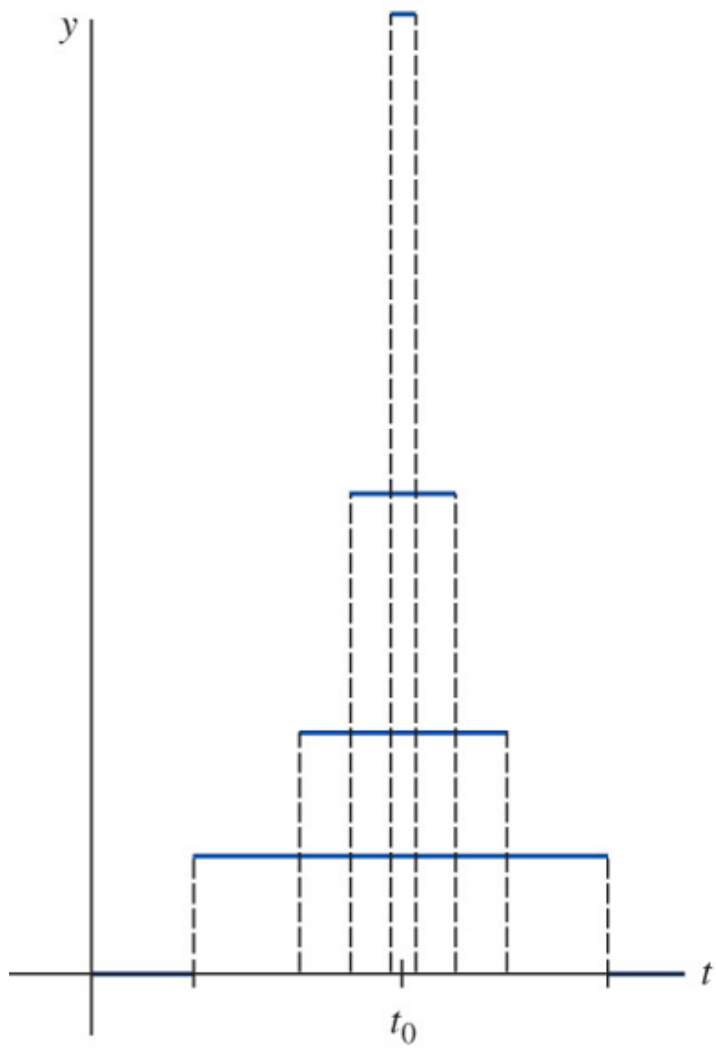
4.5 The Dirac Delta Function

Unit Pulse

$$\delta_a(t-t_0) = \begin{cases} 0, & 0 \le t < t_0 - a \\ \frac{1}{2a}, & t_0 - a \le t \le t_0 + a \\ 0, & t \ge t_0 + a \end{cases}$$



(a)



(b) Behavior of δ_a as $a \to 0$

- **The Dirac Delta Function**

$$\delta(t-t_0) = \lim_{a\to 0} \delta_a(t-t_0)$$
$$\Downarrow$$
$$\mathcal{L}\{\delta(t-t_0)\} = \lim_{a\to 0} \mathcal{L}\{\delta_a(t-t_0)\} = e^{-st_0} \lim_{a\to 0}\left(\frac{e^{sa}-e^{-sa}}{2sa}\right)$$
$$= e^{-st_0}$$

When $t_0 = 0,\ \mathcal{L}\{\delta(t)\} = 1$

Example Solve $y'' + y = 4\delta(t-2\pi)$ subject to $y(0) = 1,\ y'(0) = 0$

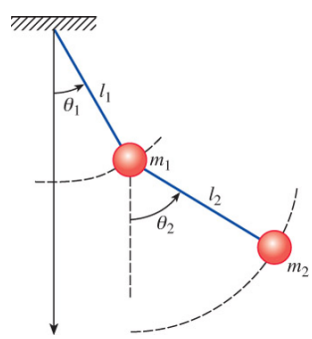
Example Use the Laplace transform to solve the given differential equation subject to the indicated initial conditions

- $y' - 3y = \delta(t-2),\ y(0) = 0$
- $y'' + y = \delta(t-2\pi),\ y(0) = 0,\ y'(0) = 1$
- $y'' + y = \delta(t-\pi/2) + \delta(t-3\pi/2),\ y(0) = 0,\ y'(0) = 0$

4.6 Systems of Linear Differential Equations

- When initial conditions are specified, the Laplace transform reduces a system of linear DEs to a set of simultaneous algebraic equations in the transformed functions

Example Double Pendulum



Linearization For small displacements θ_1 and θ_2,

$$\begin{aligned}(m_1+m_2)l_1\ddot{\theta}_1 + m_2 l_2\ddot{\theta}_2 + (m_1+m_2)g\,\theta_1 &= 0\\ l_2\ddot{\theta}_2 + l_1\ddot{\theta}_1 + g\,\theta_2 &= 0\end{aligned}$$

Solve the system when

$$m_1 = 3, m_2 = 1, l_1 = l_2 = 5, \text{ and } \ g = 10$$

$$\theta_1(0) = 1, \theta_2(0) = -1, \dot{\theta}_1(0) = 0, \dot{\theta}_2(0) = 0$$

Example Use the Laplace transform to solve the given system of differential equations

$$\begin{aligned}\dot{x} &= -x + y\\ \dot{y} &= 2x, \ \ x(0) = 0, \ y(0) = 1\end{aligned}$$

$$\begin{aligned}\dot{x} &= x - 2y\\ \dot{y} &= 5x - y, \ \ x(0) = -1, \ y(0) = 2\end{aligned}$$

$$\begin{aligned}2\dot{x} + \dot{y} - 2x &= 1\\ \dot{x} + \dot{y} - 3x - 3y &= 2, \ \ x(0) = 0, \ y(0) = 0\end{aligned}$$

Worked Exercises

5 Series Solutions of Linear Differential Equations

5.1 Solutions about Ordinary Points

- A power series in $x - a$ is an infinite series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots$$

  Such a series is also said to be a **power series centered at** a

- A power series is **convergent** if its sequence of partial sums converges
  - Convergence of power series can often be determined by the ratio test. Suppose that $c_n \neq 0$ for all n, and that

$$\lim_{n\to\infty}\left|\frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n}\right| = |x-a|\lim_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right| = L$$

    If $L < 1$, the series converges absolutely, if $L > 1$, the series diverges, and if $L = 1$, the test is inconclusive
  - Every power series has a **radius of convergence**, R. If $R > 0$, a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ converges for $|x-a| < R$
  - A function f ia **analytic** at point a if it can be represented by a power series in $x - a$ with a positive radius of convergence
  - Power series can be combined through the operations of addition, multiplication, and division

- Consider the linear second-order DE

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0, \quad a_2(x) \neq 0$$

- Divide by $a_2(x)$ to put into standard form

$$y'' + P(x)y' + Q(x)y = 0$$

- Point x_0 is an **ordinary point** of the DE if both $P(x)$ and $Q(x)$ are analytic at x_0. A point that is not an ordinary point is a **singular point** of the equation
- If $x = x_0$ is an ordinary point of the DE, we can always find **two linearly independent solutions** in the form of a power series centered at x_0
- A series solution converges at least on some interval defined by $|x - x_0| < R$, where R is the distance from x_0 to the closest singular point

Example: Solve $y'' + xy = 0$

- Since there are no finite singular points, two power series solutions are guaranteed, centered at 0, convergent for $|x| > \infty$
- Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ and the second derivative $y'' = \sum_{n=2}^{\infty} c_n n(n-1)x^{n-2}$ into the DE gives

$$\begin{aligned} y'' + xy &= \sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} + \sum_{n=0}^{\infty} c_n x^{n+1} \\ &= 2c_2 + \sum_{k=1}^{\infty} \left[(k+1)(k+2)c_{k+2} + c_{k-1}\right] x^k = 0 \end{aligned}$$

- The coefficient of each power of x be set equal to zero:

$$c_2 = 0 \text{ and } c_{k+2} = -\frac{c_{k-1}}{(k+1)(k+2)}, \; k = 1, 2, 3, \cdots$$

$$\begin{aligned} c_3 &= -\frac{c_0}{2 \cdot 3} \\ c_4 &= -\frac{c_1}{3 \cdot 4} \\ c_5 &= -\frac{c_2}{4 \cdot 5} = 0 \\ c_6 &= -\frac{c_3}{5 \cdot 6} = \frac{c_0}{2 \cdot 3 \cdot 5 \cdot 6} \\ c_7 &= -\frac{c_4}{6 \cdot 7} = \frac{c_1}{3 \cdot 4 \cdot 6 \cdot 7} \\ c_8 &= -\frac{c_5}{7 \cdot 8} = 0 \\ &\vdots \end{aligned}$$

- After grouping the terms containing c_0 and the terms containing c_1, we obtain $y = c_0 y_1(x) + c_1 y_2(x)$

$$y_1(x) = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{2 \cdot 3 \cdots (3n-1)(3n)} x^{3n}$$
$$y_2(x) = x + \sum_{n=1}^{\infty} \frac{(-1)^n}{3 \cdot 4 \cdots (3n)(3n+1)} x^{3n+1}$$

5.2 Solutions about Singular Points

- Consider the linear second-order DE

$$(x - x_0)^2 y'' + (x - x_0) p(x) y' + q(x) y = 0$$

  - Point x_0 is a **regular singular point** of the DE if both $p(x)$ and $q(x)$ are analytic at x_0
  - A singular point that is not regular is an **irregular singular point** of the equation

- To solve a DE about a regular singular point, we employ **Frobenius' Theorem**
  - If x_0 is a regular singular point of the standard DE, there exists **at least one nonzero solution** of the form

$$y = (x - x_0)^r \sum_{n=0}^{\infty} c_n (x - x_0)^n = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r}$$

    where r is a constant, and the series converges at least on some interval, $0 < x - x_0 < R$
  - After substituting $y = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r}$ into a DE and simplifying, the **indicial equation** is obtained, a quadratic equation in r that results from equating the total coefficient of the lowest power of x to zero
  - The **indicial roots** are the solutions to the quadratic equation and are then substituted into a recurrence relation

Suppose that $x = x_0$ ia a regular singular point of a DE and the indicial roots are r_1 and r_2: $r_1 \geq r_2$

- **Case I**

  r_1 and r_2 are distinct and do not differ by an integer,

$$y_1(x) = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r_1}$$
$$y_2(x) = \sum_{n=0}^{\infty} b_n (x - x_0)^{n+r_2}$$

- **Case II**

  $r_1 - r_2 = N$, where N is a positive integer,

$$y_1(x) = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r_1}, \ c_0 \neq 0$$
$$y_2(x) = C y_1(x) \ln(x - x_0) + \sum_{n=0}^{\infty} b_n (x - x_0)^{n+r_2}, \ b_0 \neq 0$$

- **Case III**

  $r_1 = r_2$,

$$y_1(x) = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r_1}, \ c_0 \neq 0$$
$$y_2(x) = y_1(x) \ln(x - x_0) + \sum_{n=0}^{\infty} b_n (x - x_0)^{n+r_1}$$

Example: Solve $2xy'' + (1 + x)y' + y = 0$

- Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ gives

$$\begin{aligned}
&2xy'' + (1+x)y' + y \\
&\quad = 2\sum_{n=0}^{\infty}(n+r)(n+r-1)c_n x^{n+r-1} + \sum_{n=0}^{\infty}(n+r)c_n x^{n+r-1} + \sum_{n=0}^{\infty}(n+r)c_n x^{n+r} + \sum_{n=0}^{\infty} c_n x^{n+r} \\
&\quad = x^r \left[r(2r-1)c_0 x^{-1} + \sum_{k=0}^{\infty}[(k+r+1)(2k+2r+1)c_{k+1} + (k+r+1)c_k]x^k \right] = 0
\end{aligned}$$

  which implies

$$r(2r-1)=0$$
$$(k+r+1)(2k+2r+1)c_{k+1}+(k+r+1)c_k=0, \quad k=0,1,\cdots$$

- We see that the indicial roots are $r_1=\frac{1}{2}$ and $r_2=0$

$$r_1=\frac{1}{2}, \ c_{k+1}=-\frac{c_k}{2(k+1)}, \ k=0,1,2,\cdots$$
$$r_2=0, \quad c_{k+1}=-\frac{c_k}{2k+1}, \ k=0,1,2,\cdots$$

- For $r_1=\frac{1}{2}$,

$$\begin{aligned}
c_1 &= -\frac{c_0}{2\cdot 1} \\
c_2 &= -\frac{c_1}{2\cdot 2}=\frac{c_0}{2^2\cdot 2!} \\
c_3 &= -\frac{c_2}{2\cdot 3}=\frac{-c_0}{2^3\cdot 3!} \\
c_4 &= -\frac{c_3}{2\cdot 4}=\frac{c_0}{2^4\cdot 4!} \\
&\vdots \\
c_n &= \frac{(-1)^n c_0}{2^n n!}
\end{aligned}$$

- For $r_2=0$,

$$\begin{aligned}
c_1 &= -\frac{c_0}{1} \\
c_2 &= -\frac{c_1}{3}=\frac{c_0}{1\cdot 3} \\
c_3 &= -\frac{c_2}{5}=\frac{-c_0}{1\cdot 3\cdot 5} \\
c_4 &= -\frac{c_3}{7}=\frac{c_0}{1\cdot 3\cdot 5\cdot 7} \\
&\vdots \\
c_n &= \frac{(-1)^n c_0}{1\cdot 3\cdot 5\cdot 7\cdots(2n-1)}
\end{aligned}$$

- The series solutions are

$$y_1(x)=x^{1/2}\left[1+\sum_{n=1}^{\infty}\frac{(-1)^n}{2^n n!}x^n\right]$$
$$y_2(x)=1+\sum_{n=1}^{\infty}\frac{(-1)^n}{1\cdot 3\cdot 5\cdot 7\cdots(2n-1)}x^n$$

5.3 Special Functions

The following DEs occur frequently in advanced studies in applied mathematics, physics, and engineering

- **Bessel's equation of order** ν, solutions are **Bessel functions**

$$x^2y'' + xy' + (x^2 - \nu^2)y = 0$$

- **Legendre's equation of order** n, solutions are **Legendre polynomials**

$$(1 - x^2)y'' - 2xy' + n(n+1)y = 0$$

5.3.1 Bessel Functions

- Because $x = 0$ is a regular singular point of Bessel's equation, there exists at least one solution of the form $y = \sum_{n=0}^{\infty} c_n x^{n+r}$

$$\begin{aligned}
&x^2y'' + xy' + (x^2 - \nu^2)y \\
&\quad = \sum_{n=0}^{\infty} c_n(n+r)(n+r-1)x^{n+r} + \sum_{n=0}^{\infty} c_n(n+r)x^{n+r} \\
&\qquad + \sum_{n=0}^{\infty} c_n x^{n+r+2} - \nu^2 \sum_{n=0}^{\infty} c_n x^{n+r} \\
&\quad = c_0(r^2 - \nu^2)x^r + x^r \sum_{n=1}^{\infty} c_n[(n+r)^2 - \nu^2]x^n + x^r \sum_{n=0}^{\infty} c_n x^{n+2} = 0
\end{aligned}$$

- The indicial equation is $r^2 - \nu^2 = 0$ so that the indicial roots are $r_1 = \nu$ and $r_2 = -\nu$. When $r_1 = \nu \geq 0$,

$$
\begin{aligned}
&x^\nu \sum_{n=1}^{\infty} c_n n(n+2\nu)x^n + x^\nu \sum_{n=0}^{\infty} c_n x^{n+2} \\
&\qquad = x^\nu \left[(1+2\nu)c_1 x + \sum_{k=0}^{\infty} \left[(k+2)(k+2+2\nu)c_{k+2} + c_k \right] x^{k+2} \right] = 0 \\
&\qquad \Downarrow \\
&\qquad c_1 = 0, \quad c_{k+2} = \frac{-c_k}{(k+2)(k+2+2\nu)}, \quad k = 0, 1, 2, \cdots \\
&\qquad \Downarrow \\
&\qquad c_3 = c_5 = c_7 = \cdots = 0, \\
&\qquad c_{2n} = -\frac{c_{2n-2}}{2^2 n(n+\nu)} \quad \leftarrow \quad k+2 = 2n
\end{aligned}
$$

- Thus

$$
\begin{aligned}
c_2 &= -\frac{c_0}{2^2 \cdot 1 \cdot (1+\nu)} \\
c_4 &= -\frac{c_2}{2^2 \cdot 2 \cdot (2+\nu)} \\
&= \frac{c_0}{2^4 \cdot 1 \cdot 2(1+\nu)(2+\nu)} \\
c_6 &= -\frac{c_4}{2^2 \cdot 3 \cdot (3+\nu)} \\
&= -\frac{c_0}{2^6 \cdot 1 \cdot 2 \cdot 3(1+\nu)(2+\nu)(3+\nu)} \\
&\vdots \\
c_{2n} &= \frac{(-1)^n c_0}{2^{2n} n!(1+\nu)(2+\nu)\cdots(n+\nu)}
\end{aligned}
$$

- It is standard practice to choose

$$
\begin{aligned}
c_0 &= \frac{1}{2^\nu \Gamma(1+\nu)} \\
&\text{thus,} \\
c_{2n} &= \frac{(-1)^n}{2^{2n+\nu} n!\, \Gamma(1+\nu+n)}
\end{aligned}
$$

Note $\Gamma(\nu) = \displaystyle\int_0^\infty x^{\nu-1} e^{-x}\, dx, \;\; \Gamma(\nu+1) = \nu\Gamma(\nu)$

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.special import gamma, factorial

x = np.linspace(-3.5, 5.5, 2251)
y = gamma(x)
```

```python
def gamma_plot():
    fig = plt.figure(figsize=(6, 4))

    plt.plot(x, y, 'b', alpha=0.6, label='$\Gamma(x)$')
    k = np.arange(1, 7)
    plt.plot(k, factorial(k -1), 'ko', alpha=0.6,
         label='$(x-1)!,\; x = 1, 2, 3\cdots$')
    plt.xlim(-3.5, 5.5)
    plt.ylim(-10, 25)
    plt.grid('true')
    plt.xlabel('x')
    plt.ylabel('$\Gamma(x)$')
    plt.legend(loc='lower right')

    plt.show()

gamma_plot()
```

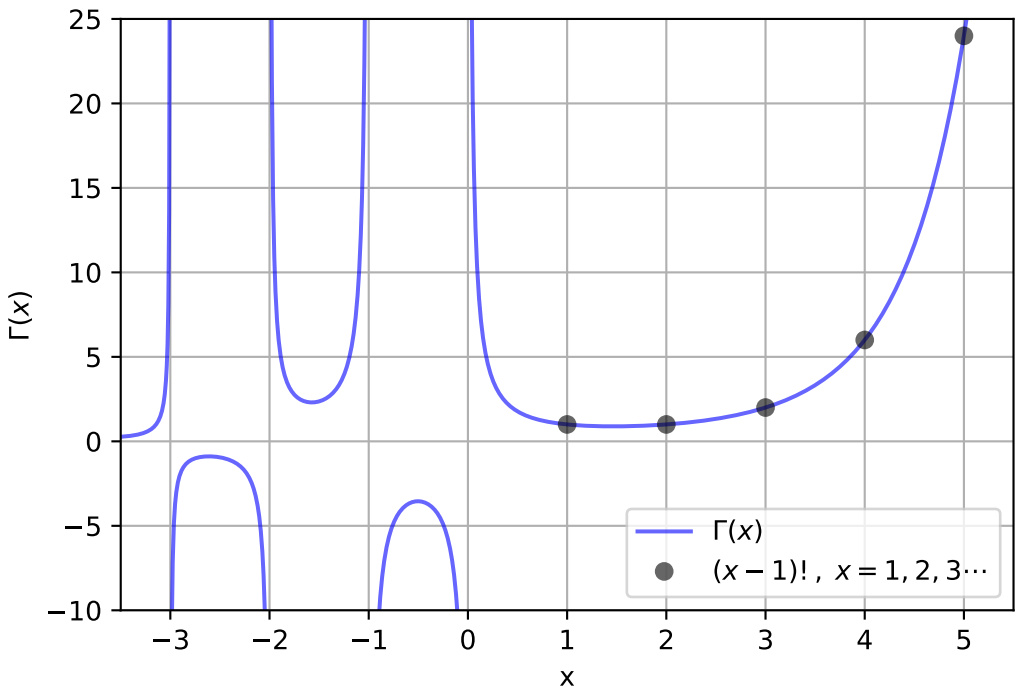


Figure 5.1: Gamma Function

- **Bessel Functions of the First Kind**

The series solution $y_1 = \sum_{n=0}^{\infty} c_{2n} x^{2n+\nu}$ is usually denoted by $J_\nu(x)$

$$J_\nu(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(1+\nu+n)} \left(\frac{x}{2}\right)^{2n+\nu}$$

Also, for the second exponent $r_2 = -\nu$

$$J_{-\nu}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(1-\nu+n)} \left(\frac{x}{2}\right)^{2n-\nu}$$

- When $\nu = 0$, $J_0(x)$
- When $r_1 - r_2 = 2\nu$ is not positive integer, $J_\nu(x)$ and $J_{-\nu}(x)$ are linearly independent
- When $r_1 - r_2 = 2\nu$ is positive integer, there are two possibilities
  - When $\nu = m =$ positive integer, $J_{-m}(x)$ is a constant multiple of $J_m(x)$: $J_m(x) = (-1)^m J_{-m}(x)$
  - When ν is half an odd positive integer, $J_\nu(x)$ and $J_{-\nu}(x)$ are linearly independent

Example Solve $x^2y'' + xy' + (x^2 - \frac{1}{4})y = 0$

- **Bessel Functions of the Second Kind**

  If $\nu \neq$ integer

$$Y_\nu(x) = \frac{\cos \nu\pi J_\nu(x) - J_{-\nu}(x)}{\sin \nu\pi}$$

  $J_\nu(x)$ and $Y_\nu(x)$ are linearly independent solutions of

$$x^2y'' + xy' + (x^2 - \nu^2)y = 0$$

  As $\nu \to m$ (an integer),

$$Y_m(x) = \lim_{\nu \to m} Y_\nu(x)$$

  $J_m(x)$ and $Y_m(x)$ are linearly independent solutions of

$$x^2y'' + xy' + (x^2 - m^2)y = 0$$

- Hence for *any* value of ν, the general solution of Bessel equation can be written as

$$y = c_1 J_\nu(x) + c_2 Y_\nu(x)$$

  $Y_\nu(x)$ is called the **Bessel function of the second kind** of order ν

Example Solve $x^2y'' + xy' + (x^2 - 9)y = 0$

5.3.1.1 Properties of Bessel Functions

- When m is integer
  - $J_m(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+m)!} \left(\frac{x}{2}\right)^{2n+m}$
  - $J_{-m}(x) = (-1)^m J_m(x)$
  - $J_m(-x) = (-1)^m J_m(x)$
  - $J_m(0) = \begin{cases} 0, & m > 0 \\ 1, & m = 0 \end{cases}$
  - $\lim_{x \to 0^+} Y_m(x) = -\infty$

```
import numpy as np
from scipy.special import jv, yv, spherical_jn
from scipy.special import eval_legendre

import matplotlib.pyplot as plt
plt.style.use('ggplot')

fig = plt.figure(figsize=(6, 8))

ax1 = fig.add_subplot(211)

x = np.linspace(0, 20, 200)
```

```
for m in range(5):
    y = jv(m, x)
    ax1.plot(x, y, label=f'$J_{m}(x)$')

ax1.axis((0, 20, -0.6, 1))

ax1.set_ylabel('$J_m(x)$')
ax1.legend()

ax2 = fig.add_subplot(212)

for m in range(5):
    y = yv(m, x)
    ax2.plot(x, y, label=f'$Y_{m}(x)$')

ax2.axis((0, 20, -3, 1))

ax2.set_xlabel('x')
ax2.set_ylabel('$Y_m(x)$')
ax2.legend()

plt.show()
```

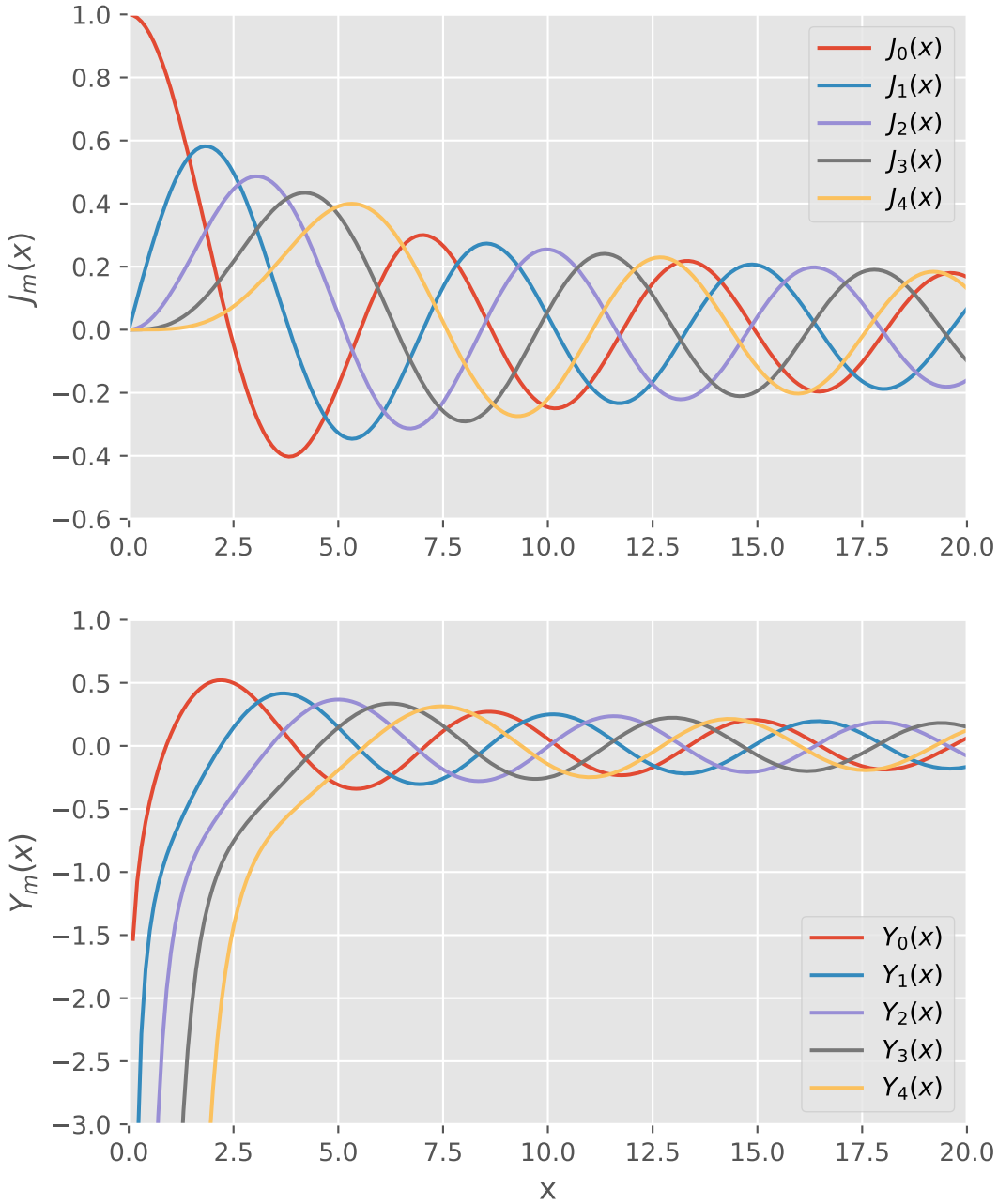


Figure 5.2: Bessel functions

- **Differential Recurence Relation**

$$
\begin{aligned}
xJ_\nu'(x) &= \sum_{n=0}^{\infty} \frac{(-1)^n(2n+\nu)}{n!\,\Gamma(1+\nu+n)} \left(\frac{x}{2}\right)^{2n+\nu} \\
&= \nu \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(1+\nu+n)} \left(\frac{x}{2}\right)^{2n+\nu} + x \sum_{n=1}^{\infty} \frac{(-1)^n}{(n-1)!\,\Gamma(1+\nu+n)} \left(\frac{x}{2}\right)^{2n+\nu-1} \\
&= \nu J_\nu(x) - x \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(2+\nu+k)} \left(\frac{x}{2}\right)^{2k+\nu+1} \\
&= \nu J_\nu(x) - x J_{\nu+1}(x) \\
&\Big\Downarrow \times x^{-\nu-1} \\
\frac{d}{dx}[x^{-\nu} J_\nu(x)] &= -x^{-\nu} J_{\nu+1}(x)
\end{aligned}
$$

and

$$
\begin{aligned}
xJ_\nu'(x) &= -\nu J_\nu(x) + xJ_{\nu-1}(x) \\
&\Big\Downarrow \times x^{\nu-1} \\
\frac{d}{dx}[x^{\nu} J_\nu(x)] &= x^{\nu} J_{\nu-1}(x)
\end{aligned}
$$

Example $J_0'(x) = -J_1(x)$

5.3.1.2 DEs Solvable in Terms of Bessel Functions

- Parametric Bessel equation of order ν

$$
\begin{aligned}
x^2y'' + xy' + (\alpha^2x^2 - \nu^2)y &= 0 \\
&\Big\Downarrow t = \alpha x,\ \alpha > 0 \\
t^2\frac{d^2y}{dt^2} + t\frac{dy}{dt} + (t^2 - \nu^2)y &= 0 \\
&\Downarrow \\
y = c_1J_\nu(\alpha x) + c_2Y_\nu(\alpha x)
\end{aligned}
$$

- Modified Bessel equation of order ν

$$
\begin{aligned}
&x^2y'' + xy' - (x^2+\nu^2)y = 0 \\
&\qquad\qquad \Big\Downarrow\, t = ix \\
&t^2\frac{d^2y}{dt^2} + t\frac{dy}{dt} + (t^2-\nu^2)y = 0 \\
&\qquad\qquad \Big\Downarrow\, I_\nu(x) = i^{-\nu}J_\nu(ix),\ K_\nu(x) = \frac{\pi}{2}\frac{I_{-\nu}(x)-I_\nu(x)}{\sin\nu\pi},\ K_m(x) = \lim_{\nu\to m} K_\nu(x) \\
&y = c_1 I_\nu(x) + c_2 K_\nu(x)
\end{aligned}
$$

- Yet another equation

$$
\begin{aligned}
&y'' + \frac{1-2a}{x}y' + \left(b^2c^2x^{2c-2} + \frac{a^2-p^2c^2}{x^2}\right)y = 0,\ p \geq 0 \\
&\qquad\qquad \Big\Downarrow\, z = bx^c,\ y(x) = \left(\frac{z}{b}\right)^{a/c} w(z) \\
&y = x^a\left[c_1J_p(bx^c) + c_2Y_p(bx^c)\right]
\end{aligned}
$$

- The aging spring

$$
\begin{aligned}
&m\ddot{x} + ke^{-\alpha t}x = 0,\ \alpha > 0 \\
&\qquad\qquad \Big\Downarrow\, s = \frac{2}{\alpha}\sqrt{\frac{k}{m}}e^{-\alpha t/2} \\
&s^2\frac{d^2x}{ds^2} + s\frac{dx}{ds} + s^2x = 0
\end{aligned}
$$

- **Spherical Bessel Functions:**

  When solving the Helmholtz equation in spherical coordinates by separation of variables, the radial equation has the form

$$
x^2y'' + 2xy' + \left(x^2 - n(n+1)\right)y = 0
$$

  The two linearly independent solutions to this equation are called the spherical Bessel functions j_n and y_n

$$
j_n(x) = \sqrt{\frac{\pi}{2x}}J_{n+\frac{1}{2}}(x)
$$

$$y_n(x) = \sqrt{\frac{\pi}{2x}} Y_{n+\frac{1}{2}}(x) = (-1)^{n+1} \sqrt{\frac{\pi}{2x}} J_{-n-\frac{1}{2}}(x)$$

when $\nu = n + \frac{1}{2}$ is half an odd integer, that is, $\pm\frac{1}{2}, \pm\frac{3}{2}, \pm\frac{5}{2}, \cdots$

Let's consider the case when $\nu = \frac{1}{2}$,

$$J_{1/2}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(1+\frac{1}{2}+n)} \left(\frac{x}{2}\right)^{2n+1/2}$$

where

$$\begin{aligned}
\Gamma\left(\frac{1}{2}\right) &= \sqrt{\pi} \\
\Gamma\left(\frac{3}{2}\right) &= \Gamma\left(1+\frac{1}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{1}{2}\sqrt{\pi} \\
\Gamma\left(\frac{5}{2}\right) &= \Gamma\left(1+\frac{3}{2}\right) = \frac{3}{2}\Gamma\left(\frac{3}{2}\right) = \frac{3}{2^2}\sqrt{\pi} = \frac{3\cdot 2}{2^3}\sqrt{\pi} = \frac{3!}{2^3}\sqrt{\pi} \\
\Gamma\left(\frac{7}{2}\right) &= \Gamma\left(1+\frac{5}{2}\right) = \frac{5}{2}\Gamma\left(\frac{5}{2}\right) = \frac{5\cdot 3}{2^3}\sqrt{\pi} = \frac{5\cdot 4\cdot 3\cdot 2}{2^3\cdot 4\cdot 2}\sqrt{\pi} = \frac{5!}{2^5 2!}\sqrt{\pi} \\
&\vdots \\
\Gamma\left(1+\frac{1}{2}+n\right) &= \frac{(2n+1)!}{2^{2n+1}n!}\sqrt{\pi}
\end{aligned}$$

Hence

$$J_{1/2}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\frac{(2n+1)!}{2^{2n+1}n!}\sqrt{\pi}} \left(\frac{x}{2}\right)^{2n+1/2} = \sqrt{\frac{2}{\pi x}} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = \sqrt{\frac{2}{\pi x}} \sin x$$

Example Show that $J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$

```
plt.style.use('ggplot')

fig = plt.figure(figsize=(6, 4))
ax = fig.add_subplot(111)

x = np.linspace(np.finfo(np.float32).eps, 10, 200)
for n in range(5):
    y = spherical_jn(n, x)
    ax.plot(np.append(0, x), np.append(0, y),
        label=f'$j_{n}(x)$')

ax.axis((0, 10, -0.4, 1.0))
```

```
ax.set_title(r'$j_{n}(x), n=0,1,2,3,4$')
ax.set_xlabel('$x$')
ax.set_ylabel('$j_n(x)$')
ax.legend()

plt.show()
```

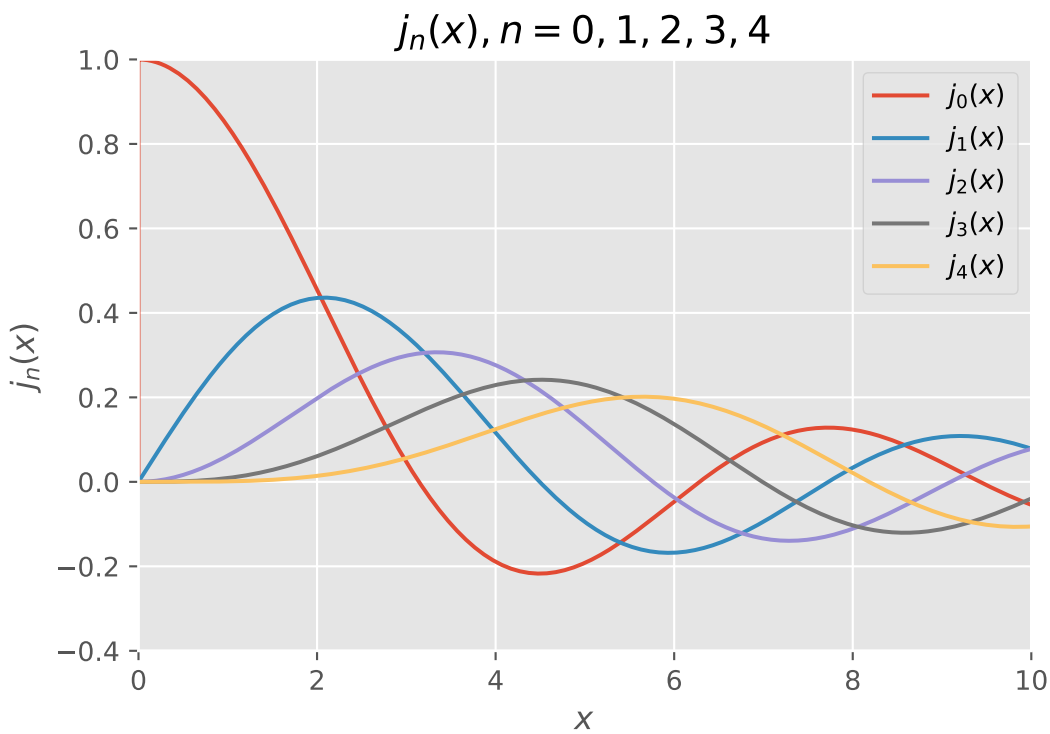


Figure 5.3: Spherical Bessel Function

Example Find the general solution of the given differential equation on $(0, \infty)$

- $x^2y'' + xy' + (x^2 - \frac{1}{9})y = 0$
- $xy'' + y' + xy = 0$
- $x^2y'' + xy' + (9x^2 - 4)y = 0$
- $x^2y'' + xy' - (16x^2 + \frac{4}{9})y = 0$

Example Find the general solution of the given differential equation on $(0, \infty)$

- $x^2y'' + 2xy' + \alpha^2x^2y = 0;\ y = x^{-1/2}u(x)$
- $xy'' + 2y' + 4y = 0$

Example Verify that $I_\nu(x) = i^{-\nu}J_\nu(ix)$ is a real function

Example Express the general solution of the given differential equation in terms of the modified Bessel functions

$$xy'' + y' - 7x^3y = 0$$

Example First express the general solution of the given differential equation in terms of Bessel functions and then express the general solution in terms of elementary functions

$$x^2y'' + 4xy' + (x^2 + 2)y = 0$$

Example Derive the following result

$$\int_0^x rJ_0(r)\,dr = xJ_1(x)$$

$$J_0'(x) = J_{-1}(x) = -J_1(x)$$

Example Use the solution of the aging spring equation $mx'' + ke^{-at}x = 0$ to discuss the behavior of $x(t)$ as $t \to \infty$ in the three cases

- $c_1 \neq 0,\ c_2 = 0$
- $c_1 = 0,\ c_2 \neq 0$
- $c_1 \neq 0,\ c_2 \neq 0$

5.3.2 Legendre Polynomials

- Since $x = 0$ is an ordinary point of Legendre's equation, we substitute the series $y = \sum_{j=0}^{\infty} c_j x^j$ to get

$$
\begin{aligned}
&(1-x^2)y'' - 2xy' + n(n+1)y = \\
&[n(n+1)c_0 + 2c_2] + [(n-1)(n+2)c_1 + 6c_3]x \\
&\qquad + \sum_{j=2}^{\infty}[(j+2)(j+1)c_{j+2} + (n-j)(n+j+1)c_j]x^j = 0 \\
&\qquad\qquad \Downarrow \\
&\qquad\qquad c_2 = -\frac{n(n+1)}{2}c_0 \\
&\qquad\qquad c_3 = -\frac{(n-1)(n+2)}{6}c_1 \\
&\qquad\qquad \vdots \\
&\qquad\qquad c_{j+2} = -\frac{(n-j)(n+j+1)}{(j+2)(j+1)}c_j, \quad j = 2, 3, 4, \cdots
\end{aligned}
$$

- The recurrence relation yields

$$
\begin{aligned}
c_2 &= -\frac{n(n+1)}{2!}c_0 \\
c_4 &= -\frac{(n-2)(n+3)}{4\cdot 3}c_2 = \frac{(n-2)n(n+1)(n+3)}{4!}c_0 \\
c_6 &= -\frac{(n-4)(n+5)}{6\cdot 5}c_4 = -\frac{(n-4)(n-2)n(n+1)(n+3)(n+5)}{6!}c_0 \\
&\vdots \\
c_3 &= -\frac{(n-1)(n+2)}{3!}c_1 \\
c_5 &= -\frac{(n-3)(n+4)}{5\cdot 4}c_3 = \frac{(n-3)(n-1)(n+2)(n+4)}{5!}c_1 \\
c_7 &= -\frac{(n-5)(n+6)}{7\cdot 6}c_5 = -\frac{(n-5)(n-3)(n-1)(n+2)(n+4)(n+6)}{7!}c_1 \\
&\vdots
\end{aligned}
$$

- Thus for at least $|x| < 1$, we obtain two linearly independent power series solutions:

$$
\begin{aligned}
y_1(x) &= c_0\left[1 - \frac{n(n+1)}{2!}x^2 + \frac{(n-2)n(n+1)(n+3)}{4!}x^4 - \frac{(n-4)(n-2)n(n+1)(n+3)(n+5)}{6!}x^6 \cdots\right] \\
y_2(x) &= c_1\left[x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}x^5 - \frac{(n-5)(n-3)(n-1)(n+2)(n+4)(n+6)}{7!}x^7 \cdots\right]
\end{aligned}
$$

  - If n is an even integer, the series $y_1(x)$ reduces to a polynomial of degree n with only even powers of x and the series $y_2(x)$ diverges

- If n is an odd integer, the series $y_2(x)$ reduces to a polynomial of degree n with only odd powers of x and the series $y_1(x)$ diverges

- The general solution for an integer n is then given by the **polynomials**

  For example, if $n = 4$, then

$$y_1(x) = c_0 \left[1 - \frac{4 \cdot 5}{2!}x^2 + \frac{2 \cdot 4 \cdot 5 \cdot 7}{4!}x^4\right] = c_0 \left[1 - 10x^2 + \frac{35}{3}x^4\right]$$

- It is traditional to choose specific values for c_0 or c_1, depending on whether n is an even or odd positive integer, respectively

  - For $n = 0$, $c_0 = 1$, and for $n = 2, 4, 6, \cdots$,

    $c_0 = (-1)^{n/2} \dfrac{1 \cdot 3 \cdots (n-1)}{2 \cdot 4 \cdots n}$

  - For $n = 1$, $c_1 = 1$, and for $n = 3, 5, 7, \cdots$,

    $c_1 = (-1)^{(n-1)/2} \dfrac{1 \cdot 3 \cdots n}{2 \cdot 4 \cdots (n-1)}$

- For example, when $n = 4$, we have

$$y_1(x) = (-1)^{4/2} \frac{1 \cdot 3}{2 \cdot 4} \left[1 - 10x^2 + \frac{35}{3}x^4\right] = \frac{1}{8}(35x^4 - 30x^2 + 3)$$

- **Legendre Polynomials**

  There specific n-th degree polynomials are called **Legendre polynomials** and denoted by $P_n(x)$

$$\begin{aligned}
P_0(x) &= 1 \\
P_1(x) &= x \\
P_2(x) &= \frac{1}{2}(3x^2 - 1) \\
P_3(x) &= \frac{1}{2}(5x^3 - 3x) \\
P_4(x) &= \frac{1}{8}(35x^4 - 30x^2 + 3) \\
P_5(x) &= \frac{1}{8}(63x^5 - 70x^3 + 15x)
\end{aligned}$$

```
fig = plt.figure(figsize=(5, 5))
ax = fig.add_subplot(111)

x = np.linspace(-1, 1, 100)
for n in range(6):
    y = eval_legendre(n, x)
    ax.plot(x, y, label=f'$P_{n}(x)$')

ax.axis((-1.05, 1.05, -1.05, 1.05))
ax.set_xlabel('x')
ax.set_ylabel('$P_n(x)$')
ax.legend(loc='lower left', bbox_to_anchor=(1, 0))

plt.show()
```

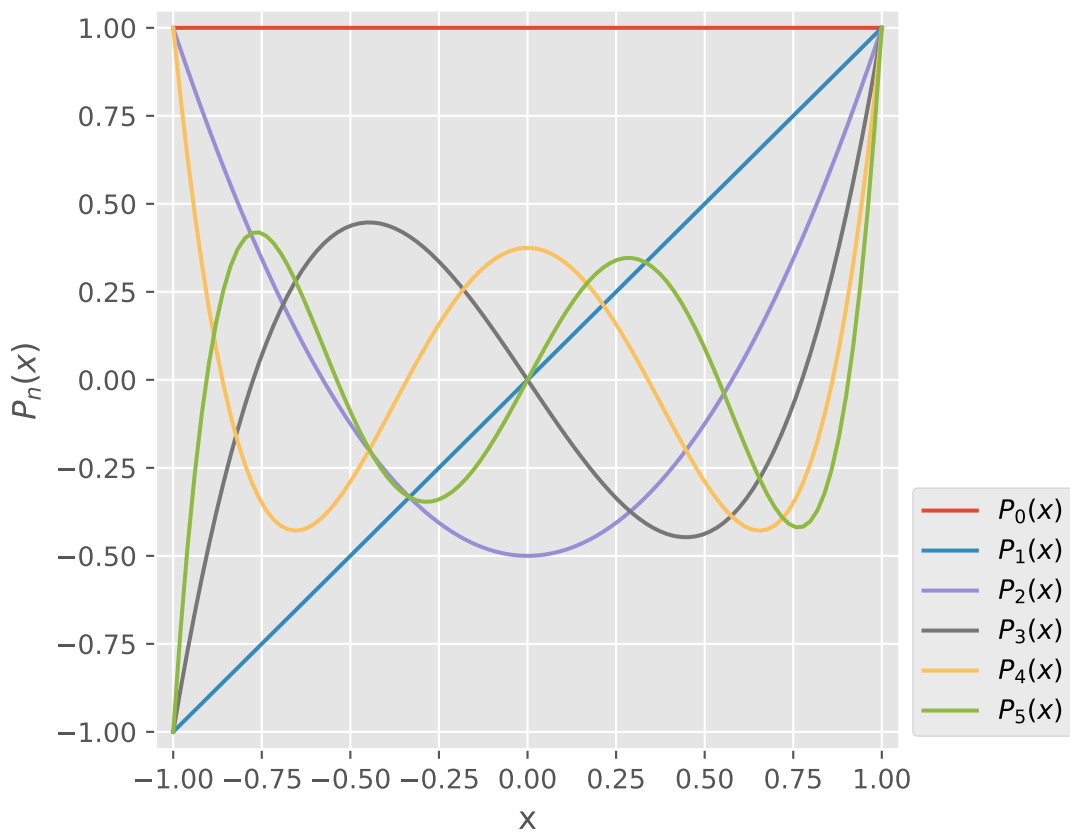


Figure 5.4: Legendre polynomials

- **Properties of Legendre Polynomials**
  - $P_n(-x) = (-1)^n P_n(x)$
  - $P_n(1) = 1$
  - $P_n(-1) = (-1)^n$
  - $P_n(0) = 0, \;\; n$ odd
  - $P_n'(0) = 0, \;\; n$ even

- **Rodrigues' Formula**

  The Legendre polynomials can also be represented using **Rodrigues' Formula**,

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} \left(x^2 - 1\right)^n \tag{RF}$$

  This can be demonstrated through the following observations

  - The right hand side of (RF) is a n-th order polynomial
  - Treating $\left(x^2 - 1\right)^n = (x-1)^n (x+1)^n$ as a product and using Leibnitz' rule to differentiate n times, we have

$$\frac{d^n}{dx^n}(x-1)^n(x+1)^n = n!(x+1)^n + \text{terms with } (x-1) \text{ as a factor}$$

    so that $P_n(1) = \dfrac{n! 2^n}{2^n n!} = 1$
  - If $h(x) = \left(1 - x^2\right)^n$, then $h'(x) = -2nx(1-x^2)^{n-1}$, so that

$$\left(1 - x^2\right) h' + 2nx\, h = 0$$

    Now differentiate $n+1$ times, using Leibnitz, to get

$$\begin{aligned} &\left(1 - x^2\right) h^{n+2} - 2(n+1)xh^{n+1} - 2\frac{(n+1)n}{2} h^n + 2nxh^{n+1} + 2n(n+1)h^n = 0 \\ &\qquad\qquad \Downarrow \\ &\left(1 - x^2\right) h^{n+2} - 2xh^{n+1} + n(n+1)h^n = 0 \end{aligned}$$

  - As the equation is linear and $P_n(x) \propto h^n(x)$, $P_n(x)$ satisfies the Legendre equation of order n

$$\left(1 - x^2\right) P_n''(x) - 2xP_n'(x) + n(n+1)P_n(x) = 0$$

- **Integral Relations**

$$\int_{-1}^{1} P_k(x)P_l(x)\,dx = \delta_{k,l}\frac{2}{2l+1}, \quad k \le l$$

$$\int_{-1}^{1} xP_s(x)P_r(x)\,dx = \delta_{s+1,r}\frac{2r}{(2r-1)(2r+1)}, \; s \le r$$

- **Recurrence Relation**

  The Legendre polynomials satisfy the following recurrence relation

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$$

  Consider the polynomial $xP_n(x)$. It has degree $n+1$ and is thus in the linear span of $P_0, \cdots, P_{n+1}$. We can hence write $xP_n(x)$ as a linear combination of the first $n+2$ Legendre polynomials:

$$xP_n(x) = c_0P_0(x) + c_1P_1(x) + \cdots + c_{n+1}P_{n+1}(x)$$

  - Thus

$$\begin{aligned}\int_{-1}^{1} xP_n(x)P_k(x)\,dx &= c_k\int_{-1}^{1} P_k^2(x)\,dx \\ \rightarrow c_k &= \frac{2k+1}{2}\int_{-1}^{1} xP_n(x)P_k(x)\,dx\end{aligned}$$

  These integrals vanish unless $k = n \pm 1$ and for this case, we can use

$$c_{n-1} = \frac{2n-1}{2}\int_{-1}^{1} xP_n(x)P_{n-1}(x)\,dx = \frac{n}{2n+1}$$

$$c_{n+1} = \frac{2n+3}{2}\int_{-1}^{1} xP_n(x)P_{n+1}(x)\,dx = \frac{n+1}{2n+1}$$

  - Hence

$$\begin{aligned}xP_n(x) &= c_{n-1}P_{n-1}(x) + c_{n+1}P_{n+1}(x) \\ &= \frac{n}{2n+1}P_{n-1}(x) + \frac{n+1}{2n+1}P_{n+1}(x)\end{aligned}$$

    This is what we wanted to prove

Example Use the recurrence relation and $P_0(x) = 1,\ P_1(x) = x$, to generate the next six Legendre polynomials

Example Show that the differential equation

$$\sin\theta \frac{d^2y}{d\theta^2} + \cos\theta \frac{dy}{d\theta} + n(n+1)\,\sin\theta\, y = 0$$

can be transformed into Legendre's equation by means of the substitution $x = \cos\theta$

Example Find the first three positive values of λ for which the problem

$$(1-x^2)y'' - 2xy' + \lambda y = 0$$

$$y(0) = 0,\ y(x),\ y'(x) \ \text{ bounded on } [-1, 1]$$

has nontrivial solutions

- The differential equation

$$(1-x^2)y'' - 2xy' + \left[n(n+1) - \frac{m^2}{1-x^2}\right] y = 0$$

  is known as the **associated Legendre equation**. When $m = 0$, this equation reduces to Legendre's equation. A solution of the associated equation is

$$P_n^m(x) = (1-x^2)^{m/2} \frac{d^m}{dx^m} P_n(x)$$

  where $P_n(x),\ n = 0, 1, 2, \cdots$ are the Legendre polynomials. The solutions $P_n^m(x)$ for $m = 0, 1, 2, \cdots$, are called **associated Legendre functions**

Example Answer the following questions

- Find the associated Legendre functions $P_0^0(x)$, $P_1^0(x)$, $P_1^1(x)$, $P_2^1(x)$, $P_2^2(x)$, $P_3^1(x)$, $P_3^2(x)$, and $P_3^3(x)$

- What can you say about $P_n^m(x)$ when m is an even non-negative integer?
- What can you say about $P_n^m(x)$ when m is a non-negative integer and $m > n$?
- Verify that $y = P_1^1(x)$ satisfies the associated Legendre equation when $n = 1$ and $m = 1$

Worked Exercises

6 Matrices

6.1 Matrix Algebra

- A **matrix** is any rectangular array of numbers or functions

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \ddots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

  - The numbers or functions in the array are **entries** or **elements**
  - An $n \times n$ matrix is a **square** matrix of **order** n

- **Column** and **row vectors** are $n \times 1$ and $1 \times n$ matrices

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}, \begin{pmatrix} a_1 & a_2 & \cdots & a_n \end{pmatrix}$$

- **Equality of Matrices**

$$\mathbf{A} = \left(a_{ij}\right)_{m\times n} \text{ and } \mathbf{B} = \left(b_{ij}\right)_{m\times n} \text{ are equal}$$
$$\text{if } a_{ij} = b_{ij} \text{ for each } i \text{ and } j$$

- **Matrix Addition**

$$\mathbf{A} + \mathbf{B} = \left(a_{ij} + b_{ij}\right)_{m\times n}$$

- **Scalar Multiplication**

$$k\mathbf{A} = \left(ka_{ij}\right)_{m\times n}$$

- **Properties of Matrix Addition and Scalar Multiplication**

  Suppose $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are $m \times n$ matrices and k_1 and k_2 are scalars. Then

$$\begin{aligned}
\mathbf{A} + \mathbf{B} &= \mathbf{B} + \mathbf{A} \\
\mathbf{A} + (\mathbf{B} + \mathbf{C}) &= (\mathbf{A} + \mathbf{B}) + \mathbf{C} \\
(k_1 k_2)\,\mathbf{A} &= k_1\,(k_2 \mathbf{A}) \\
k_1\,(\mathbf{A} + \mathbf{B}) &= k_1 \mathbf{A} + k_1 \mathbf{B} \\
(k_1 + k_2)\,\mathbf{A} &= k_1 \mathbf{A} + k_2 \mathbf{A}
\end{aligned}$$

- **Matrix multiplication**

$$\mathbf{AB} = \left(\sum_{k=1}^{p} a_{ik} b_{kj} \right)_{m \times n}$$

  where $\mathbf{A}$ is an $m \times p$ matrix, $\mathbf{B}$ is a $p \times n$ matrix, and $\mathbf{AB}$ is the $m \times n$ matrix

  - In general, $\mathbf{AB} \neq \mathbf{BA}$
  - **Associative Law:** $\mathbf{A}\,(\mathbf{BC}) = (\mathbf{AB})\,\mathbf{C}$
  - **Distributive Law:** $\mathbf{A}\,(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$

- **Transpose of a Matrix**

$$\mathbf{A}^T = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \ddots & a_{m2} \\ \vdots & \ddots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}$$

- **Properties of Transpose**

$$\begin{aligned}
\left(\mathbf{A}^T\right)^T &= \mathbf{A} \\
(\mathbf{A} + \mathbf{B})^T &= \mathbf{A}^T + \mathbf{B}^T \\
(\mathbf{AB})^T &= \mathbf{B}^T \mathbf{A}^T \\
(k\mathbf{A})^T &= k\mathbf{A}^T
\end{aligned}$$

- **Special Matrices**

  - In a **zero matrix**, all entries are zeros
  - In a **triangular matrix**, all entries above or below the main diagonal are zeros (lower triangular or upper triangular)

- In a **diagonal matrix**, all entries not on the main diagonal are zeros
- A **scalar matrix** is a diagonal one where all entries on the main diagonal are equal. If those entries are 1's, it is an **identity matrix**, $\mathbf{I}$ (or $\mathbf{I}_n$ when there is a need to emphasize the order of the matrix)
- An $n \times n$ matrix $\mathbf{A}$ is **symmetric** if $\mathbf{A}^T = \mathbf{A}$

Example If $\mathbf{A} = \begin{pmatrix} 2 & -3 \\ -5 & 4 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} -1 & 6 \\ 3 & 2 \end{pmatrix}$,

find (a) $\mathbf{AB}$, (b) $\mathbf{BA}$, (c) $\mathbf{A}^2$, (d) $\mathbf{B}^2$

Example Show that if $\mathbf{A}$ is an $m \times n$ matrix, then $\mathbf{AA}^T$ is symmetric

Example In matrix theory, many of the familiar properties of the real number system are not valid. If a and b are real numbers, then $ab = 0$ implies that $a = 0$ or $b = 0$. Find two matrices such that $\mathbf{AB} = \mathbf{0}$ but $\mathbf{A} \neq \mathbf{0}$ and $\mathbf{B} \neq \mathbf{0}$

Example Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. Explain why, in general, the given formula is not valid

$$(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + 2\mathbf{AB} + \mathbf{B}^2$$

Example Find the resulting vector $\mathbf{b}$ if the given vector $\mathbf{a} = \langle 1, 1 \rangle$ is rotated through the indicated angle $\theta = \pi/2$

Example Verify that the quadratic form $ax^2 + bxy + cy^2$ is the same as

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a & \frac{1}{2}b \\ \frac{1}{2}b & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

6.2 Systems of Linear Algebraic Equations

- **General Form**

  A system of m linear equations in n unknowns has the general form

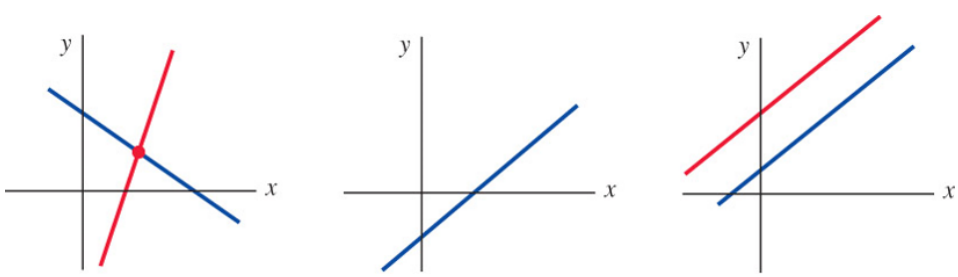
$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

- The **coefficients** of the unknowns can be abbreviated as a_{ij}. The numbers $b_1, b_2, \cdots, b_m$ are called the **constants** of the system. If all the constants are zero, the system is said to be **homogeneous**, otherwise it is **nonhomogeneous**

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & \ddots & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

$$\mathbf{Ax} = \mathbf{b}$$

- A linear system of equations is said to be **consistent** if it has at least one solution and **inconsistent** if it has no solutions. If a linear system is consistent, it has either
  - a unique solution (that is, precisely one solution), or
  - infinitely many solutions



(a) Consistent (b) Consistent (c) Inconsistent

- **Augmented Matrix**

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \ddots & a_{2n} & b_2 \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right)$$

- A system can be solved with **elementary operations** (**row reduction** for matrices) on an augmented matrix

ElementaryOperations	Meaning
R_{ij}	Interchange rows i and j
cR_i	Multiply the row i by the nonzero constant c
$cR_i + R_j$	Multiply the row i by c and add to the row j

- In the **Gaussian elimination**, we row-reduce the augmented matrix until we arrive at a row-equivalent augmented matrix in **row-echelon form**
  - The first nonzero entry in a nonzero row is a 1
  - In consecutive nonzero rows, the first entry 1 in the lower row appears to the right of the 1 in the higher row
  - Rows consisting of all zeros are at the bottom of the matrix

Example Solve

$$\begin{pmatrix} 2 & 6 & 1 \\ 1 & 2 & -1 \\ 5 & 7 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \\ 9 \end{pmatrix}$$

Using row operations on the augmented matrix, we obtain

$$\left(\begin{array}{ccc|c} 2 & 6 & 1 & 7 \\ 1 & 2 & -1 & -1 \\ 5 & 7 & -4 & 9 \end{array}\right) \xrightarrow{R_{12}} \left(\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 2 & 6 & 1 & 7 \\ 5 & 7 & -4 & 9 \end{array}\right) \xrightarrow[-5R_1 + R_3]{-2R_1 + R_2} \left(\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & 2 & 3 & 9 \\ 0 & -3 & 1 & 14 \end{array}\right)$$

$$\overset{\frac{1}{2}R_2}{\Longrightarrow}\left(\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & 1 & \frac{3}{2} & \frac{9}{2} \\ 0 & -3 & 1 & 14 \end{array}\right) \overset{3R_2+R_3}{\Longrightarrow}\left(\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & 1 & \frac{3}{2} & \frac{9}{2} \\ 0 & 0 & \frac{11}{2} & \frac{55}{2} \end{array}\right)$$

$$\overset{\frac{2}{11}R_3}{\Longrightarrow}\left(\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & 1 & \frac{3}{2} & \frac{9}{2} \\ 0 & 0 & 1 & 5 \end{array}\right)$$

The last matrix is in row-echelon form. We can make the last matrix above to be in reduced row-echelon form

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & 1 & \frac{3}{2} & \frac{9}{2} \\ 0 & 0 & 1 & 5 \end{array}\right) \overset{-2R_2+R_1}{\Longrightarrow}\left(\begin{array}{ccc|c} 1 & 0 & -4 & -10 \\ 0 & 1 & \frac{3}{2} & \frac{9}{2} \\ 0 & 0 & 1 & 5 \end{array}\right) \overset{\substack{-4R_3+R_1 \\ -\frac{3}{2}R_3+R_2}}{\Longrightarrow}\left(\begin{array}{ccc|c} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 5 \end{array}\right)$$

We see that the solution is $x_1 = 10,\ \ x_2 = -3,\ \ x_3 = 5$

Example Solve

$$\begin{pmatrix} 1 & 3 & -2 \\ 4 & 1 & 3 \\ 2 & -5 & 7 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -7 \\ 5 \\ 19 \end{pmatrix}$$

Using row operations on the augmented matrix, we obtain

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & -7 \\ 4 & 1 & 3 & 5 \\ 2 & -5 & 7 & 19 \end{array}\right) \overset{\substack{-4R_1+R_2 \\ -2R_1+R_3}}{\Longrightarrow}\left(\begin{array}{ccc|c} 1 & 3 & -2 & -7 \\ 0 & -11 & 11 & 33 \\ 0 & -11 & 11 & 33 \end{array}\right) \overset{\substack{-R_2+R_3 \\ -\frac{1}{11}R_2}}{\Longrightarrow}\left(\begin{array}{cc|c|c} 1 & 3 & -2 & -7 \\ 0 & 1 & -1 & -3 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$$

In this case, the last matrix implies that the original system of three equations is really equivalent to two equations

$$\overset{-3R_2+R_1}{\Longrightarrow}\left(\begin{array}{cc|c|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & -3 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$$

If we let $x_3 = t$, $x_1 = -t + 2$ and $x_2 = t - 3$, then we see that the system has infinitely many solutions

Example Solve

$$\begin{pmatrix} 1 & 1 \\ 4 & -1 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -6 \\ 8 \end{pmatrix}$$

Using row operations on the augmented matrix, we obtain

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 4 & -1 & -6 \\ 2 & -3 & 8 \end{array}\right) \overset{\substack{-4R_1+R_2 \\ -2R_1+R_3}}{\Longrightarrow} \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & -5 & -10 \\ 0 & -5 & 6 \end{array}\right) \overset{\substack{-R_2+R_3 \\ -\frac{1}{5}R_2}}{\Longrightarrow} \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & 2 \\ \hline 0 & 0 & 16 \end{array}\right)$$

The system has no solution

- A **homogeneous system** of linear equations is **always consistent**. The solution consisting of all zeros is called the **trivial solution**. A homogeneous system either possesses only the trivial solution or possesses the trivial solution along with infinitely many nontrivial solutions
- **A homogeneous system possesses nontrivial solutions if the number m of equations is less than the number n of unknowns** $(m < n)$

Example Find the positive integers x_1, x_2, x_3, and x_4 so that

$$x_1 C_2H_6 + x_2 O_2 \rightarrow x_3 CO_2 + x_4 H_2O$$

Because the number of atoms of each element must be the same on each side of the last equation, we have:

Atom	
C	$2x_1 = x_3$
H	$6x_1 = 2x_4$
O	$2x_2 = 2x_3 + x_4$

$$\left(\begin{array}{cccc|c} 2 & 0 & -1 & 0 & 0 \\ 6 & 0 & 0 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{array}\right) \overset{\substack{R_{12} \\ R_{23}}}{\Longrightarrow} \left(\begin{array}{cccc|c} 6 & 0 & 0 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \\ 2 & 0 & -1 & 0 & 0 \end{array}\right) \overset{\text{row operations}}{\Longrightarrow} \left(\begin{array}{ccc|c|c} 1 & 0 & 0 & -\frac{1}{3} & 0 \\ 0 & 1 & 0 & -\frac{7}{6} & 0 \\ 0 & 0 & 1 & -\frac{2}{3} & 0 \end{array}\right)$$

Then when we let $x_4 = t,\ \ x_1 = \frac{1}{3}t,\ x_2 = \frac{7}{6}t,\ x_3 = \frac{2}{3}t.$ If we pick $t = 6,\ x_1 = 2,\ x_2 = 7,$ $x_3 = 4,\ x_4 = 6$

Example Use either Gaussian elimination or Gauss-Jordan elimination to solve the given system or show that no solution exists

$$\begin{aligned} x_1 - x_2 &= 11 \\ 4x_1 + 3x_2 &= -5 \end{aligned}$$

$$\begin{aligned} 9x_1 + 3x_2 &= -5 \\ 2x_1 + x_2 &= -1 \end{aligned}$$

$$\begin{aligned} x_1 - x_2 - x_3 &= -3 \\ 2x_1 + 3x_2 + 5x_3 &= 7 \\ x_1 - 2x_2 + 3x_3 &= -11 \end{aligned}$$

$$\begin{aligned} x_1 + x_2 + x_3 &= 0 \\ x_1 + x_2 + 3x_3 &= 0 \end{aligned}$$

$$\begin{aligned} x_1 - x_2 - x_3 &= 8 \\ x_1 - x_2 + x_3 &= 3 \\ -x_1 + x_2 + x_3 &= 4 \end{aligned}$$

Example Balance the given chemical equation:

$$\mathrm{C_5H_8 + O_2 \rightarrow CO_2 + H_2O}$$

$$\mathrm{Cu + HNO_3 \rightarrow Cu(NO_3)_2 + H_2O + NO}$$

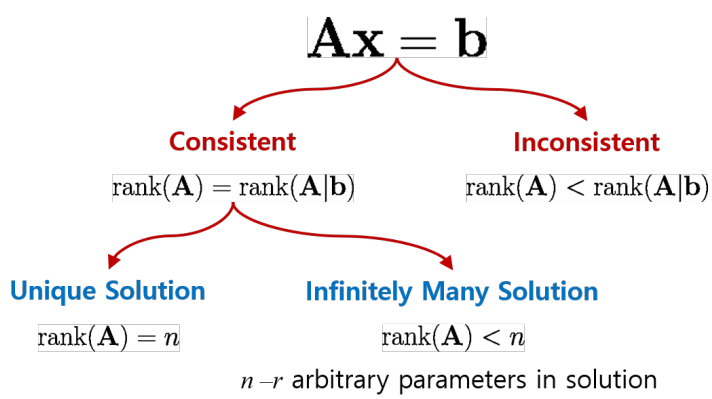
Example Compute the given product for an arbitrary 3×3 matrix $\mathbf{A}$

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & c & 1 \end{pmatrix} \mathbf{A}$$

6.3 Rank of a Matrix

- The **rank** of an $m \times n$ matrix $\mathbf{A}$, $\operatorname{rank}(\mathbf{A})$, is **the maximum number of linearly independent row vectors**. If a matrix $\mathbf{A}$ is now equivalent to a row-echelon form $\mathbf{B}$, then
  - the row space of $\mathbf{A}$ = the row space of $\mathbf{B}$
  - the nonezero rows of $\mathbf{B}$ form a basis for the row space of $\mathbf{A}$, and
  - $\operatorname{rank}(\mathbf{A})$ = the number of nonzero rows in $\mathbf{B}$
- **Consistency of $\mathbf{Ax} = \mathbf{b}$**
  - A linear system of equations $\mathbf{Ax} = \mathbf{b}$ is consistent if and only if $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}|\mathbf{b})$
  - Suppose a linear system $\mathbf{Ax} = \mathbf{b}$ with m equations and n unknowns is consistent. If $\operatorname{rank}(\mathbf{A}) = r \leq n$, then the solution of the system contains $n - r$ parameters. This means that we have the unique solution when $r = n$

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \ddots & a_{2n} & b_2 \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right) \xrightarrow{\text{row operations}} \left(\begin{array}{cccc|ccc|c} 1 & a'_{12} & \cdots & a'_{1r} & a'_{1r+1} & \cdots & a'_{1n} & b'_1 \\ 0 & 1 & \ddots & \vdots & a'_{2r+1} & \ddots & a'_{2n} & b'_2 \\ \vdots & \ddots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 1 & a'_{rr+1} & \cdots & a'_{rn} & b'_r \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$



Example Find the rank of the given matrix

$$\begin{pmatrix} 2 & 1 & 3 \\ 6 & 3 & 9 \\ -1 & -\frac{1}{2} & -\frac{3}{2} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 4 \\ 1 & 4 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 4 & 1 & 0 & 5 & 1 \\ 2 & 1 & \frac{2}{3} & 3 & \frac{1}{3} \\ 6 & 6 & 6 & 12 & 0 \end{pmatrix}$$

Example Determine whether the given set of vectors is linearly dependent or linearly independent

$$\mathbf{u}_1 = \langle 1, 2, 3 \rangle,\ \mathbf{u}_2 = \langle 1, 0, 1 \rangle,\ \mathbf{u}_3 = \langle 1, -1, 5 \rangle$$

$$\mathbf{u}_1 = \langle 1, -1, 3, -1 \rangle,\ \mathbf{u}_2 = \langle 1, -1, 4, 2 \rangle,\ \mathbf{u}_3 = \langle 1, -1, 5, 7 \rangle$$

Example Suppose the system $\mathbf{Ax} = \mathbf{b}$ is consistent and $\mathbf{A}$ is a 5×8 matrix and $\operatorname{rank} \mathbf{A} = 3$. How many parameters does the solution of the system have?

Example Let $\mathbf{A}$ be a nonzero 4×6 matrix

1. What is the maximum rank that $\mathbf{A}$ can have?

2. If $\operatorname{rank}(\mathbf{A}|\mathbf{b}) = 2$, then for what value(s) of $\operatorname{rank}(\mathbf{A})$ is the system $\mathbf{Ax} = \mathbf{b}$, $\mathbf{b} \neq \mathbf{0}$, inconsistent? Consistent?

3. If $\operatorname{rank}(\mathbf{A}) = 3$, then how many parameters does the solution of the system $\mathbf{Ax} = \mathbf{0}$ have?

Example Let $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ be the first, second, and third column vectors, respectively, of the matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 7 \\ 1 & 0 & 2 \\ -1 & 5 & 13 \end{pmatrix}$$

What can we conclude about rank($\mathbf{A}$) from the observation $2\mathbf{v}_1 + 3\mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}$?

6.4 Determinants

- **Determinant of a 2×2 Matrix**

$$\det(\mathbf{A}) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

- **Determinant of a 3×3 Matrix**

$$\det(\mathbf{A}) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}(-1)^{1+1}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + a_{12}(-1)^{1+2}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}(-1)^{1+3}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

- **Determinant of a $n \times n$ Matrix**

$$\det \mathbf{A} = \sum (-1)^h a_{1l_1} a_{2l_2} \cdots a_{nl_n}$$

  where the summation is over all permutations l_1, l_2, $\cdots$, l_n of 1, 2, $\cdots$, n and the sign accords with the parity of the permutation

- **Cofactor and Minor**

  The **cofactor of** a_{ij} is the determinant

$$C_{ij} = (-1)^{i+j} M_{ij}$$

  where M_{ij} is the determinant of the submatrix obtained by deleting the i-th row and the j-th column of $\mathbf{A}$. The determinant M_{ij} is called a **minor determinant**

- **Cofactor Expansion of a Determinant: Laplace Development**

  Let $\mathbf{A} = \left(a_{ij}\right)_{n\times n}$ be an $n \times n$ matrix. For each $1 \le i \le n$, **the cofactor expansion of** $\det(\mathbf{A})$ **along the i-th row** is

$$\det(\mathbf{A}) = \sum_{k=1}^{n} a_{ik} C_{ik}$$

  For each $1 \le j \le n$, **the cofactor expansion of** $\det(\mathbf{A})$ **along the j-th column** is

$$\det(\mathbf{A}) = \sum_{k=1}^{n} a_{kj} C_{kj}$$

Example Suppose

$$\mathbf{A} = \begin{pmatrix} 2 & 3 & 4 \\ 1 & -1 & 2 \\ -2 & 3 & 5 \end{pmatrix}$$

Evaluate the indicated minor determinant or cofactor

1. M_{12} **2.** M_{32} **3.** C_{13} **4.** C_{22}

Example Evaluate the determinant of the matrix

$$\begin{pmatrix} 3 & 5 \\ -1 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 1-\lambda & 3 \\ 2 & 2-\lambda \end{pmatrix}$$

Example Evaluate the determinant of the given matrix by cofactor expansion

$$\begin{pmatrix} -2 & -1 & 4 \\ -3 & 6 & 1 \\ -3 & 4 & 8 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ x & y & z \\ 2 & 3 & 4 \end{pmatrix}$$

6.5 Properties of Determinants

- If $\mathbf{A}^T$ is the transpose of the $n \times n$ matrix $\mathbf{A}$, then $\det(\mathbf{A}^T) = \det(\mathbf{A})$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{vmatrix}$$

- If any two rows (columns) of an $n \times n$ matrix $\mathbf{A}$ are the same, then $\det(\mathbf{A}) = 0$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 0$$

- If all the entries in a row (column) of an $n \times n$ matrix $\mathbf{A}$ are zero, then $\det(\mathbf{A}) = 0$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 0$$

- If $\mathbf{B}$ is the matrix obtained by interchanging any two rows (columns) of an $n \times n$ matrix $\mathbf{A}$, then $\det(\mathbf{B}) = -\det(\mathbf{A})$

$$\begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = -\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

- If $\mathbf{B}$ is the matrix obtained by multiplying a row (column) by a nonzero real number k, then $\det(\mathbf{B}) = k\det(\mathbf{A})$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ ka_{21} & ka_{22} & ka_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = k\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

- If $\mathbf{A}$ and $\mathbf{B}$ are both $n \times n$ matrices, then $\det(\mathbf{AB}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$

$$\left|\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}\begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix}\right| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}\begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix}$$

- Suppose $\mathbf{B}$ is the matrix obtained from an $n \times n$ matrix $\mathbf{A}$ by multiplying a row(column) by a nonzero k and adding the result to another row(column). Then $\det(\mathbf{B}) = \det(\mathbf{A})$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ ka_{21} + a_{31} & ka_{22} + a_{32} & ka_{23} + a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

- If $\mathbf{A}$ is an $n \times n$ triangular matrix, then $\det(\mathbf{A}) = a_{11}a_{22}\cdots a_{nn}$

$$\begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33}$$

Alien Cofactors

- Suppose $\mathbf{A}$ is an $n \times n$ matrix. If $a_{i1}, a_{i2}, \cdots, a_{in}$ are the entries in the i-th row and $C_{p1}, C_{p2}, \cdots, C_{pn}$ are the cofactors of the entries in the p-th row, then

$$\sum_{k=1}^{n} a_{ik}C_{pk} = 0 \text{ for } i \neq p$$

- If $a_{1j}, a_{2j}, \cdots, a_{nj}$ are the entries in the j-th column and $C_{1p}, C_{2p}, \cdots, C_{np}$ are the cofactors of the entries in the p-th column, then

$$\sum_{k=1}^{n} a_{kj}C_{kp} = 0 \text{ for } j \neq p$$

Example State the appropriate theorem(s) in this section that justifies the given equality

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -\begin{vmatrix} 3 & 4 \\ 1 & 2 \end{vmatrix}$$

$$\begin{vmatrix} -5 & 6 \\ 2 & -8 \end{vmatrix} = \begin{vmatrix} 1 & 6 \\ -6 & -8 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 2 & 18 \\ 5 & 9 & -12 \end{vmatrix} = 6\begin{vmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 5 & 9 & -4 \end{vmatrix}$$

Example Evaluate the determinant of the given matrix using the result

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 5$$

1. $\mathbf{A} = \begin{pmatrix} a_3 & a_2 & a_1 \\ b_3 & b_2 & b_1 \\ c_3 & c_2 & c_1 \end{pmatrix}$

2. $\mathbf{A} = \begin{pmatrix} 2a_1 & a_2 & a_3 \\ 6b_1 & 3b_2 & 3b_3 \\ 2c_1 & c_2 & c_3 \end{pmatrix}$

3. $\mathbf{A} = \begin{pmatrix} 4a_1 - 2a_3 & a_2 & a_3 \\ 4b_1 - 2b_3 & b_2 & b_3 \\ 2c_1 - c_3 & \frac{1}{2}c_2 & \frac{1}{2}c_3 \end{pmatrix}$

Example Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ x & y & z \\ y+z & x+z & x+y \end{pmatrix}$$

Without expanding, show that $\det \mathbf{A} = 0$

Example Evaluate

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{vmatrix}$$

6.6 Inverse of a Matrix

- If $\mathbf{A}$ is an $n \times n$ matrix and there exists an $n \times n$ matrix $\mathbf{B}$ such that

  $$\mathbf{AB} = \mathbf{BA} = \mathbf{I}$$

  then $\mathbf{A}$ is said to be **nonsingular** or **invertible** and $\mathbf{B}$ is the **inverse** of $\mathbf{A}$

- An $n \times n$ matrix that has no inverse is called **singular**. If $\mathbf{A}$ is nonsingular, its inverse is denoted by $\mathbf{B} = \mathbf{A}^{-1}$

- **Properties of the Inverse**

  Let $\mathbf{A}$ and $\mathbf{B}$ be nonsingular matrices. Then

  $$\begin{aligned} \left(\mathbf{A}^{-1}\right)^{-1} &= \mathbf{A} \\ (\mathbf{AB})^{-1} &= \mathbf{B}^{-1}\mathbf{A}^{-1} \\ \left(\mathbf{A}^{T}\right)^{-1} &= \left(\mathbf{A}^{-1}\right)^{T} \end{aligned}$$

- **Adjoint Matrix**

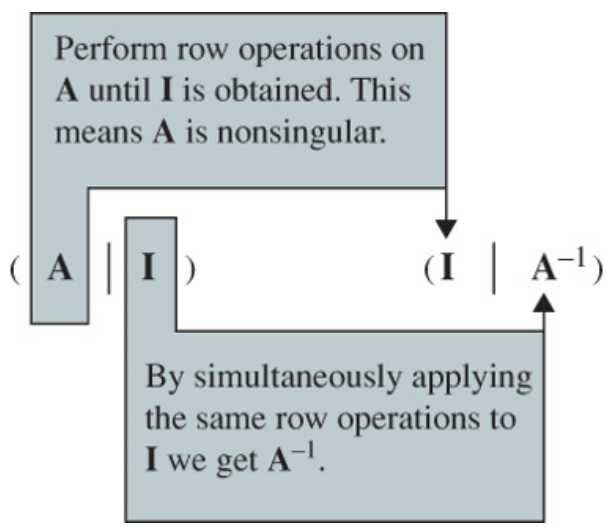
$$\operatorname{adj}(\mathbf{A}) = \begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & & & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{pmatrix}^T$$

- **Finding the Inverse**

  Let $\mathbf{A}$ be an $n \times n$ matrix. If $\det(\mathbf{A}) \neq 0$ (**nonsingular**), then

$$\mathbf{A}^{-1} = \frac{\operatorname{adj}(\mathbf{A})}{\det(\mathbf{A})}$$

  or



- **Using the Inverse to Solve Systems**

  The coefficient matrix $\mathbf{A}$ is $n \times n$. In particular, if $\mathbf{A}$ is nonsingular, the system $\mathbf{Ax} = \mathbf{b}$ can be solved by

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

  A homogeneous system of n linear equations in n unknowns $\mathbf{Ax} = \mathbf{0}$ has

  - only the trivial solution if and only if $\mathbf{A}$ is nonsingular
  - a nontrivial solution if and only if $\mathbf{A}$ is singular

Example Verify that the matrix $\mathbf{B}$ is the inverse of the matrix $\mathbf{A}$

$$\mathbf{A} = \begin{pmatrix} 1 & \frac{1}{2} \\ 2 & \frac{3}{2} \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix}$$

Example Determine whether the given matrix is singular or nonsingular. If it is nonsingular, find the inverse using $\mathbf{A}^{-1} = \frac{\operatorname{adj} \mathbf{A}}{\det \mathbf{A}}$

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \quad \begin{pmatrix} 0 & -1 & 1 & 4 \\ 3 & 2 & -2 & 1 \\ 0 & 4 & 0 & 1 \\ 1 & 0 & -1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 0 & -3 & 0 & 1 \end{pmatrix}$$

Example Find the inverse of the given matrix or show that no inverse exists

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}, \quad \begin{pmatrix} 4 & 2 & 3 \\ 2 & 1 & 0 \\ -1 & -2 & 0 \end{pmatrix}$$

Example If $\mathbf{A}$ is nonsingular, then $\left(\mathbf{A}^T\right)^{-1} = \left(\mathbf{A}^{-1}\right)^T$, verify this for

$$\mathbf{A} = \begin{pmatrix} 1 & 4 \\ 2 & 10 \end{pmatrix}$$

Example Find the inverse of the rotation matrix

$$\mathbf{M} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

What does $\mathbf{A} = \mathbf{M}^{-1}\mathbf{B}$ represents?

Example A nonsingular matrix $\mathbf{A}$ is said to be **orthogonal** if $\mathbf{A}^{-1} = \mathbf{A}^T$

1. Verify tha the rotation matrix is orthogonal

2. Verify that

$$\mathbf{A} = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & 0 & -2 \\ \sqrt{2} & \sqrt{3} & 1 \\ \sqrt{2} & -\sqrt{3} & 1 \end{pmatrix}$$

is an orthogonal matrix

Example Answer the questions based on the supposition

- Show that if $\mathbf{A}$ is an orthogonal matrix, then $\det \mathbf{A} = \pm 1$
- Suppose $\mathbf{A}$ and $\mathbf{B}$ are nonsingular $n \times n$ matrices. Then show that $\mathbf{AB}$ is nonsingular
- Suppose $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ matrices and that either $\mathbf{A}$ or $\mathbf{B}$ is singular. Then show that $\mathbf{AB}$ is singular
- Suppose $\mathbf{A}$ is a nonsingular matrix. Then show that $\det \mathbf{A}^{-1} = \dfrac{1}{\det \mathbf{A}}$
- Suppose $\mathbf{A}^2 = \mathbf{A}$. Then show that either $\mathbf{A} = \mathbf{I}$ or $\mathbf{A}$ is singular
- Suppose $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ matrices. $\mathbf{A}$ is nonsingular, and $\mathbf{AB} = \mathbf{0}$. Then show that $\mathbf{B} = \mathbf{0}$
- Suppose $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ matrices. $\mathbf{A}$ is nonsingular, and $\mathbf{AB} = \mathbf{AC}$. Then show that $\mathbf{B} = \mathbf{C}$
- Suppose $\mathbf{A}$ and $\mathbf{B}$ are nonsingular $n \times n$ matrices. Is $\mathbf{A} + \mathbf{B}$ necessarily nonsingular?
- Suppose $\mathbf{A}$ is a nonsingular matrix. Then show that $\mathbf{A}^T$ is nonsingular
- Suppose $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ nonzero matrices and $\mathbf{AB} = \mathbf{0}$. Then show that both $\mathbf{A}$ and $\mathbf{B}$ are singular

6.7 Cramer's Rule

If $\det(\mathbf{A}) \neq 0$, the solution of the system is given by

$$x_k = \frac{\det(\mathbf{A}_k)}{\det(\mathbf{A})}, \quad k = 1, 2, \cdots, n$$

where

$$\mathbf{A}_k = \begin{pmatrix} a_{11} & \cdots & a_{1k-1} & b_1 & a_{1k+1} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2k-1} & b_2 & a_{2k+1} & \cdots & a_{2n} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & \cdots & a_{nk-1} & b_n & a_{nk+1} & \cdots & a_{nn} \end{pmatrix}$$

Example Solve the given system of equations by Cramer's rule

$$\begin{aligned} -3x_1 + x_2 &= 3 \\ 2x_1 - 4x_2 &= -6 \end{aligned}$$

$$\begin{aligned} 0.1x_1 - 0.4x_2 &= 0.13 \\ x_1 - x_2 &= 0.4 \end{aligned}$$

$$\begin{aligned} 2x + y &= 1 \\ 3x + 2y &= -2 \end{aligned}$$

Example Consider the system

$$\begin{aligned} x_1 + x_2 &= 1 \\ x_1 + \epsilon x_2 &= 2 \end{aligned}$$

When ϵ is close to 1, the lines that make up the system are almost parallel

1. Use Cramer's rule to show that a solution of the system is

$$x_1 = 1 - \frac{1}{\epsilon - 1}, \quad x_2 = \frac{1}{\epsilon - 1}$$

2. The system is said to be ill-conditioned since small changes in the input data(for example, the coefficients) causes a significant or large change in the output or solution. Verify this by finding the solution of the system for $\epsilon = 1.01$ and then for $\epsilon = 0.99$

6.8 The Eigenvalue Problem

- Let $\mathbf{A}$ be an $n \times n$ matrix. A number λ is said to be an **eigenvalue** of $\mathbf{A}$ if there exists a nonzero solution vector $\mathbf{k}$ of the linear system

  $$\mathbf{Ak} = \lambda \mathbf{k}$$

  and the solution vector $\mathbf{k}$ is said to be an **eigenvector** corresponding to the eigenvalue λ

- The problem of solving $\mathbf{A}\mathbf{k} = \lambda\mathbf{k}$ for nonzero vectors $\mathbf{k}$ is called to be the **eigenvalue problem** for $\mathbf{A}$
- We must solve the **characteristic equation** $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$ to find an eigenvalue λ
- To find an eigenvector $\mathbf{k}$ corresponding to an eigenvalue λ, we solve $(\mathbf{A} - \lambda\mathbf{I})\mathbf{k} = \mathbf{0}$ by applying Gauss elimination to $(\mathbf{A} - \lambda\mathbf{I}|\mathbf{0})$

Example Find the eigenvalues and eigenvectors of

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix}$$

To find the eigenvalues, we solve

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 1-\lambda & 2 & 1 \\ 6 & -1-\lambda & 0 \\ -1 & -2 & -1-\lambda \end{vmatrix} = 0$$

It follows that the characteristic equation is

$$-\lambda^3 - \lambda^2 + 12\lambda = -\lambda(\lambda+4)(\lambda-3) = 0$$

Hence the eigenvalues are $\lambda_1 = -4,\ \lambda_2 = 0,\ \lambda_3 = 3$

For $\lambda_1 = -4$, we have

$$(\mathbf{A} + 4\mathbf{I}|\mathbf{0}) = \left(\begin{array}{ccc|c} 5 & 2 & 1 & 0 \\ 6 & 3 & 0 & 0 \\ -1 & -2 & 3 & 0 \end{array}\right) \overset{\text{row operations}}{\Longrightarrow} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$$

Thus $k_1 = -k_3,\ k_2 = 2k_3$. Choosing $k_3 = 1$ gives the eigenvector

$$\mathbf{k}_1 = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$$

For $\lambda_2 = 0$, we have

$$(\mathbf{A} - 0\mathbf{I}|\mathbf{0}) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 6 & -1 & 0 & 0 \\ -1 & -2 & -1 & 0 \end{array}\right) \overset{\text{row operations}}{\Longrightarrow} \left(\begin{array}{ccc|c} 1 & 0 & \frac{1}{13} & 0 \\ 0 & 1 & \frac{6}{13} & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$$

Thus $k_1 = -\frac{1}{13}k_3$, $k_2 = -\frac{6}{13}k_3$. Choosing $k_3 = 1$ gives the eigenvector

$$\mathbf{k}_2 = \begin{pmatrix} -\frac{1}{13} \\ -\frac{6}{13} \\ 1 \end{pmatrix}$$

For $\lambda_3 = 3$, we have

$$(\mathbf{A} - 3\mathbf{I}|\mathbf{0}) = \left(\begin{array}{ccc|c} -2 & 2 & 1 & 0 \\ 6 & -4 & 0 & 0 \\ -1 & -2 & -4 & 0 \end{array}\right) \overset{\text{row operations}}{\Longrightarrow} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$$

Thus $k_1 = -k_3$, $k_2 = -\frac{3}{2}k_3$. Choosing $k_3 = 1$ gives the eigenvector

$$\mathbf{k}_3 = \begin{pmatrix} -1 \\ -\frac{3}{2} \\ 1 \end{pmatrix}$$

```
import pprint

import numpy as np
from scipy.linalg import lu

import sympy
from sympy import plot_implicit, symbols, Eq
sympy.init_printing()

A = sympy.Matrix([[1, 2, 1], [6, -1, 0], [-1, -2, -1]])

A.eigenvects()
```

$$\left[\left(-4,\ 1,\ \left[\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \right] \right),\ \left(0,\ 1,\ \left[\begin{bmatrix} -\frac{1}{13} \\ -\frac{6}{13} \\ 1 \end{bmatrix} \right] \right),\ \left(3,\ 1,\ \left[\begin{bmatrix} -1 \\ -\frac{3}{2} \\ 1 \end{bmatrix} \right] \right) \right]$$

Example Find the eigenvalues and eigenvectors of

$$\mathbf{A} = \begin{pmatrix} 9 & 1 & 1 \\ 1 & 9 & 1 \\ 1 & 1 & 9 \end{pmatrix}$$

The characteristic equation

$$\det(\mathbf{A}-\lambda\mathbf{I}) = \begin{vmatrix} 9-\lambda & 1 & 1 \\ 1 & 9-\lambda & 1 \\ 1 & 1 & 9-\lambda \end{vmatrix} = -(\lambda-11)(\lambda-8)^2 = 0$$

shows that $\lambda_1 = 11$ and that $\lambda_2 = \lambda_3 = 8$ is an eigenvalue of multiplicity 2

For $\lambda_1 = 11$, we have

$$(\mathbf{A}-11\mathbf{I}|\mathbf{0}) = \left(\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -2 & 0 \end{array}\right) \overset{\text{row operations}}{\Longrightarrow} \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right)$$

Thus $k_1 = k_3$, $k_2 = k_3$. Choosing $k_3 = 1$ gives the eigenvector

$$\mathbf{k}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

For $\lambda_2 = 8$, we have

$$(\mathbf{A}-8\mathbf{I}|\mathbf{0}) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right) \overset{\text{row operations}}{\Longrightarrow} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Here $k_1 + k_2 + k_3 = 0$, we are free to select two of the variables arbitrarily

Choosing, on the one hand, $k_2 = 1$, $k_3 = 0$, and on the other, $k_2 = 0$, $k_3 = 1$, we obtain two linearly independent eigenvectors

$$\mathbf{k}_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \text{ and } \mathbf{k}_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

corresponding to a single eigenvalue

If instead we choose $k_2 = 1$, $k_3 = 1$ and then $k_2 = 1$, $k_3 = -1$, we obtain, respectively, two entirely different but orthogonal eigenvectors

$$\mathbf{k}_2 = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \text{ and } \mathbf{k}_3 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

```
A = sympy.Matrix([[9, 1, 1], [1, 9, 1], [1, 1, 9]])
A.eigenvects()
```

$$\left[\left(8,\ 2,\ \left[\begin{bmatrix}-1\\1\\0\end{bmatrix},\ \begin{bmatrix}-1\\0\\1\end{bmatrix}\right]\right),\ \left(11,\ 1,\ \left[\begin{bmatrix}1\\1\\1\end{bmatrix}\right]\right)\right]$$

Example Find the eigenvalues and eigenvectors of

$$\mathbf{A} = \begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$$

From the characteristic equation

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 3-\lambda & 4 \\ -1 & 7-\lambda \end{vmatrix} = (\lambda - 5)^2 = 0$$

we see $\lambda_1 = \lambda_2 = 5$ is an eigenvalue of algebraic multiplicity 2

To find the eigenvector(s) corresponding to $\lambda_1 = 5$, we resort to the system $(\mathbf{A} - 5\mathbf{I}|\mathbf{0})$

$$(\mathbf{A} - 5\mathbf{I}|\mathbf{0}) = \left(\begin{array}{cc|c} -2 & 4 & 0 \\ -1 & 2 & 0 \end{array}\right) \overset{\text{row operations}}{\Longrightarrow} \left(\begin{array}{cc|c} 1 & -2 & 0 \\ \hline 0 & 0 & 0 \end{array}\right)$$

Thus $k_1 = 2k_2$. If we choose $k_2 = 1$, we find the single eigenvector

$$\mathbf{k}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

We define the geometric multiplicity of an eigenvalue to be the number of linearly independent eigenvectors for the eigenvalue

When the geometric multiplicity of an eigenvalue is less than the algebraic multiplicity, we say the matrix is *defective*. In the case of defective matrices, we must search for additional system

$$(\mathbf{A} - 5\mathbf{I}|\mathbf{k}_1) = \left(\begin{array}{cc|c} -2 & 4 & 2 \\ -1 & 2 & 1 \end{array}\right) \overset{\text{row operations}}{\Longrightarrow} \left(\begin{array}{cc|c} 1 & -2 & -1 \\ \hline 0 & 0 & 0 \end{array}\right)$$

Thus $k_1 - 2k_2 = -1$. If we choose $k_2 = 0$, we find the generalized eigenvector

$$\mathbf{k}_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

```
A = sympy.Matrix([[3, 4], [-1, 7]])
A.eigenvects()
```

$$\left[\left(5,\ 2,\ \left[\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right]\right)\right]$$

- Let $\mathbf{A}$ be a square matrix with real entries. If $\lambda = \alpha + i\beta,\ \beta \neq 0,$ is a complex eigenvalue of $\mathbf{A}$,

$$\mathbf{A}\bar{\mathbf{k}} = \bar{\lambda}\bar{\mathbf{k}}$$

- $\lambda = 0$ is an eigenvalue of $\mathbf{A}$ if and only if $\mathbf{A}$ is singular
- If λ is an eigenvalue of nonsingular $\mathbf{A}$ with eigenvector $\mathbf{k}$, $1/\lambda$ is an eigenvalue of $\mathbf{A}^{-1}$ with the same eigenvector $\mathbf{k}$
- The eigenvalues of an upper triangular, lower triangular, and diagonal matrix are the main diagonal entries

Example Find the eigenvalues and eigenvectors of the given matrix. State whether the matrix is singular or nonsingular

$$\begin{pmatrix} -1 & 2 \\ -7 & 8 \end{pmatrix}, \quad \begin{pmatrix} 4 & 8 \\ 0 & -5 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 1 & 1 & -1 \end{pmatrix}$$

Example Find the eigenvalues and eigenvectors of the given nonsingular matrix $\mathbf{A}$. Then without finding $\mathbf{A}^{-1}$, find its eigenvalues and corresponding eigenvectors

$$\begin{pmatrix} 5 & 1 \\ 1 & 5 \end{pmatrix}, \quad \begin{pmatrix} 4 & 2 & -1 \\ 0 & 3 & -2 \\ 0 & 0 & 5 \end{pmatrix}$$

Example True or False: If λ is an eigenvalue of an $n \times n$ matrix $\mathbf{A}$, then the matrix $\mathbf{A} - \lambda\mathbf{I}$ is singular. Justify your answer

Example Suppose λ is an eigenvalue with corresponding eigenvector $\mathbf{k}$ of an $n \times n$ matrix $\mathbf{A}$

1. If $\mathbf{A}^2 = \mathbf{AA}$, then show that $\mathbf{A}^2\mathbf{k} = \lambda^2\mathbf{k}$. Explain the meaning of the last equation

2. Verify the result obtained in part 1 for the matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 3 \\ 5 & 4 \end{pmatrix}$$

3. Generalize the result in part 1

Example Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. The matrix $\mathbf{B}$ is said to be **similar** to the matrix $\mathbf{A}$ if there exists a nonsingular matrix $\mathbf{S}$ such that $\mathbf{B} = \mathbf{S}^{-1}\mathbf{AS}$. If $\mathbf{B}$ is similar to $\mathbf{A}$, then show that $\mathbf{A}$ is similar to $\mathbf{B}$

Example Suppose $\mathbf{A}$ and $\mathbf{B}$ are similar matrices. Show that $\mathbf{A}$ and $\mathbf{B}$ have the same eigenvalues

6.9 Powers of Matrices

- **Cayley-Hamilton Theorem**

  If $(-1)^n\lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0 = 0$ is the characteristic equation of $n \times n$ matrix $\mathbf{A}$, then

  $$(-1)^n\mathbf{A}^n + c_{n-1}\mathbf{A}^{n-1} + \cdots + c_1\mathbf{A} + c_0\mathbf{I} = \mathbf{0}$$

  And we can write

  $$\mathbf{A}^m = a_0\mathbf{I} + a_1\mathbf{A} + \cdots + a_{n-1}\mathbf{A}^{n-1}$$

  and the equation for the eigenvalues

$$\lambda^m = a_0 + a_1\lambda + \cdots + a_{n-1}\lambda^{n-1}$$

hold for the same constants

Example Verify that the given matrix satisfies its own characteristic equation

$$\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 4 & 5 \end{pmatrix}$$

Example Compute $\mathbf{A}^m$

$$\mathbf{A} = \begin{pmatrix} 8 & 5 \\ 4 & 0 \end{pmatrix}; \quad m = 5, \quad \mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 0 \end{pmatrix}; \quad m = 10$$

Example Show that the given matrix has an eigenvalue λ_1 of multiplicity two. As a consequence, the equation $\lambda^m = c_0 + c_1\lambda$ does not yield enough independent equations to form a system for determining the coefficients c_i. Use the derivative (with respect to λ) of this equation evaluated at λ_1 as the extra needed equation to form a system. Compute $\mathbf{A}^m$ and use this result to compute the indicated power of the matrix $\mathbf{A}$

$$\mathbf{A} = \begin{pmatrix} 7 & 3 \\ -3 & 1 \end{pmatrix}; \quad m = 6$$

Example $\lambda = 0$ is an eigenvalue of each matrix. In this case, show that the coefficient c_0 in the characteristic equation

$$(-1)^n\mathbf{A}^n + c_{n-1}\mathbf{A}^{n-1} + \cdots + c_1\mathbf{A} + c_0\mathbf{I} = \mathbf{0}$$

is 0. Compute $\mathbf{A}^m$ in each case. In part (a), explain why we do not have to solve any system for the coefficients c_1 in determining $\mathbf{A}^m$

$$(a) \;\; \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}, \quad (b) \;\; \mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & -2 \\ 1 & 1 & 3 \end{pmatrix}$$

Example A non-zero $n \times n$ matrix $\mathbf{A}$ is said to be **nilpotent of index** m if m is the smallest positive integer for which $\mathbf{A}^m = \mathbf{0}$.

1. Explain why any nilpotent matrix $\mathbf{A}$ is singular

2. Show that all the eigenvalues of a nilpotent matrix $\mathbf{A}$ are 0

6.10 Orthogonal Matrices

- Let $\mathbf{A}$ be a *symmetric* matrix ($\mathbf{A} = \mathbf{A}^T$) with *real* entries. Then the eigenvalues of $\mathbf{A}$ are *real*
- Let $\mathbf{A}$ be a *symmetric* matrix. Then eigenvectors corresponding to distinct(different) eigenvalues are *orthogonal*
- An $n \times n$ matrix $\mathbf{A}$ is *orthogonal* ($\mathbf{A}^{-1} = \mathbf{A}^T$) if and only if its columns $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n$ form an orthonormal set

$$\mathbf{x}_i \cdot \mathbf{x}_j = 0,\ i \neq j \text{ and } \mathbf{x}_i \cdot \mathbf{x}_i = 1$$

- It may not be possible to find n linearly independent eigenvectors for an $n \times n$ matrix $\mathbf{A}$ when some of eigenvalues are repeated (defective matrix)
- But a symmetric matrix is an exception. It can be proved that a set of n linearly independent eigenvectors can always be found for an $n \times n$ symmetric matrix $\mathbf{A}$ even there is some repetition of the eigenvalues
- However, this does not mean that all eigenvectors are mutually orthogonal for an $n \times n$ symmetric matrix $\mathbf{A}$. The set of eigenvectors corresponding to distinct eigenvalues are orthogonal; but different eigenvectors corresponding to a repeated eigenvalue may not be orthogonal
- But it is always possible to *find* or *construct* a set of n mutually orthogonal eigenvectors by using Gram-Schmidt orthogonalization

Example Construct an orthogonal matrix from the eigenvectors of

$$\mathbf{A} = \begin{pmatrix} 7 & 4 & -4 \\ 4 & -8 & -1 \\ -4 & -1 & -8 \end{pmatrix}$$

```
A = sympy.Matrix([[7, 4, -4], [4, -8, -1], [-4, -1, -8]])
A.eigenvects()
```

$$\left[\left(-9, \ 2, \ \left[\begin{bmatrix} -\frac{1}{4} \\ 1 \\ 0 \end{bmatrix}, \ \begin{bmatrix} \frac{1}{4} \\ 0 \\ 1 \end{bmatrix} \right] \right), \ \left(9, \ 1, \ \left[\begin{bmatrix} -4 \\ -1 \\ 1 \end{bmatrix} \right] \right) \right]$$

```
v1 = A.eigenvects()[0][2][0].T
v2 = A.eigenvects()[0][2][1].T
v3 = A.eigenvects()[1][2][0].T

B = sympy.Matrix(sympy.GramSchmidt([v1, v2, v3], True)); B
```

$$\begin{bmatrix} -\frac{\sqrt{17}}{17} & \frac{4\sqrt{17}}{17} & 0 \\ \frac{2\sqrt{34}}{51} & \frac{\sqrt{34}}{102} & \frac{\sqrt{34}}{6} \\ -\frac{2\sqrt{2}}{3} & -\frac{\sqrt{2}}{6} & \frac{\sqrt{2}}{6} \end{bmatrix}$$

```
B.T * B
```

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example Determine whether the given matrix is orthogonal

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 1 \\ -\frac{12}{13} & \frac{5}{13} & 0 \\ \frac{5}{13} & \frac{12}{13} & 0 \end{pmatrix}$$

Example (a) Verify that the indicated column vectors are eigenvectors of the given symmetric matrix and (b) identify the corresponding eigenvalues. (c) Use Gram-Schmidt process to construct an orthogonal matrix $\mathbf{P}$ from the eigenvectors

$$\mathbf{A} = \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix}; \ \mathbf{k}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \ \mathbf{k}_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \ \mathbf{k}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Example Answer the questions based on the supposition

- Suppose $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ orthogonal matrices. Then show that $\mathbf{AB}$ is orthogonal
- Suppose $\mathbf{A}$ is an orthogonal matrix. Is $\mathbf{A}^2$ is orthogonal
- Suppose $\mathbf{A}$ is an orthogonal matrix. Then show that $\mathbf{A}^{-1}$ is orthogonal
- Suppose $\mathbf{A}$ is an orthogonal matrix such that $\mathbf{A}^2 = \mathbf{I}$. Then show that $\mathbf{A}^T = \mathbf{A}$
- Show that the rotation matrix is orthogonal

6.11 Approximation of Eigenvalues

- Let λ_1, λ_2, $\cdots$, λ_k, $\cdots$, λ_n denote the eigenvalues of an $n \times n$ matrix $\mathbf{A}$. The eigenvalue λ_k is said to be the **dominant eigenvalue** of $\mathbf{A}$ if $|\lambda_k| > |\lambda_i|$, $i = 1, 2, \cdots, n$, but $i \neq k$
- An eigenvector corresponding to λ_k is called the **dominant eigenvector** of $\mathbf{A}$
- **Power Method**

  Let us assume that the eigenvalues of $\mathbf{A}$ are such that $|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \cdots \geq |\lambda_n|$ and that the corresponding n eigenvectors $\mathbf{k}_1$, $\mathbf{k}_2$, $\cdots$, $\mathbf{k}_n$ are linearly independent. Because of this last assumption, n eigenvectors can serve as a basis for $\mathbb{R}^n$. For any nonzero $n \times 1$ vector $\mathbf{x}_0$,

$$\mathbf{x}_0 = c_1\mathbf{k}_1 + c_2\mathbf{k}_2 + c_3\mathbf{k}_3 + \cdots + c_n\mathbf{k}_n$$

  We shall also assume $\mathbf{x}_0$ is chosen so that $c_1 \neq 0$. We do the following procedure

$$\begin{aligned}
\mathbf{A}\mathbf{x}_0 &= c_1\mathbf{A}\mathbf{k}_1 + c_2\mathbf{A}\mathbf{k}_2 + c_3\mathbf{A}\mathbf{k}_3 + \cdots + c_n\mathbf{A}\mathbf{k}_n \\
&\Downarrow \ \mathbf{x}_i = \mathbf{A}\mathbf{x}_{i-1},\ \mathbf{A}\mathbf{k}_j = \lambda_j\mathbf{k}_j \\
\mathbf{x}_1 &= c_1\lambda_1\mathbf{k}_1 + c_2\lambda_2\mathbf{k}_2 + c_3\lambda_3\mathbf{k}_3 + \cdots + c_n\lambda_n\mathbf{k}_n \\
&\Downarrow \\
\mathbf{A}\mathbf{x}_1 &= c_1\lambda_1\mathbf{A}\mathbf{k}_1 + c_2\lambda_2\mathbf{A}\mathbf{k}_2 + c_3\lambda_3\mathbf{A}\mathbf{k}_3 + \cdots + c_n\lambda_n\mathbf{A}\mathbf{k}_n \\
&\Downarrow \\
\mathbf{x}_2 &= c_1\lambda_1^2\mathbf{k}_1 + c_2\lambda_2^2\mathbf{k}_2 + c_3\lambda_3^2\mathbf{k}_3 + \cdots + c_n\lambda_n^2\mathbf{k}_n \\
&\Downarrow \\
\mathbf{x}_m &= c_1\lambda_1^m\mathbf{k}_1 + c_2\lambda_2^m\mathbf{k}_2 + c_3\lambda_3^m\mathbf{k}_3 + \cdots + c_n\lambda_n^m\mathbf{k}_n \\
&= \lambda_1^m\left[c_1\mathbf{k}_1 + c_2\left(\frac{\lambda_2}{\lambda_1}\right)^m\mathbf{k}_2 + c_3\left(\frac{\lambda_3}{\lambda_1}\right)^m\mathbf{k}_3 + \cdots + c_n\left(\frac{\lambda_n}{\lambda_1}\right)^m\mathbf{k}_n\right] \\
&\Downarrow \ m \to \infty \\
\mathbf{x}_m &\simeq \lambda_1^m c_1\mathbf{k}_1
\end{aligned}$$

- For large values of m and under all the assumptions that were made, the $n \times 1$ vector $\mathbf{x}_m$ is an approximation to a dominant eigenvector associated with the dominant eigenvalue λ_1
- If $\mathbf{x}_m$ is an approximation to a dominant eigenvector, then the dominant eigenvalue λ_1 can be approximated by the **Rayleigh quotient**

$$\lambda_1 = \frac{\mathbf{A}\mathbf{x}_m \cdot \mathbf{x}_m}{\mathbf{x}_m \cdot \mathbf{x}_m}$$

- Iteration often results in vectors whose entries become very large. Large numbers can cause a problem in computation. One way around this difficulty is to use a **scaled-down** normalized vector

$$\mathbf{x}_m \leftarrow \frac{\mathbf{x}_m}{\|\mathbf{x}_m\|}$$

```
def power_method(A, n_iteration):
    # Ideally choose a random vector
    # To decrease the chance that
    #   our vector is orthogonal to the eigenvector
    k = np.random.rand(A.shape[1])

    for _ in range(n_iteration):
        # calculate the matrix-by-vector product Ak
        k_1 = np.dot(A, k)

        # calculate the norm
        k_1_norm = np.linalg.norm(k_1)

        # re normalize the vector
        k = k_1 /k_1_norm

    return k

def Rayleigh_quotient(A, k):
    # calculate Rayleigh quotient
    lambda_ = np.dot(np.dot(A, k), k) /np.dot(k, k)

    return lambda_

A = np.array([[4, 2], [3, -1]])
k_1 = power_method(A, 10)
```

```
lambda_1 = Rayleigh_quotient(A, k_1)

print('A =')
pprint.pprint(A)
print('\nDominant Eigenvector:', np.around(k_1, 3))
print('Dominant Eigenvalue:', np.around(lambda_1, 3))
```

```
A =
array([[ 4,  2],
       [ 3, -1]])

Dominant Eigenvector: [0.894 0.447]
Dominant Eigenvalue: 5.0
```

- **Method of Deflation**
  - After we have found the dominant eigenvalue λ_1 of a matrix $\mathbf{A}$, it may still be necessary to find nondominant eigenvalues. We will limit the discussion to the case where $\mathbf{A}$ is a *symmetric* matrix
  - Suppose λ_1 and $\mathbf{k}_1$ are, respectively, the dominant eigenvalue and a corresponding *normalized* eigenvector of a symmetric matrix $\mathbf{A}$. Furthermore, suppose the eigenvalues of $\mathbf{A}$ are such that

    $$|\lambda_1| > |\lambda_2| > |\lambda_3| > \cdots > |\lambda_n|$$

    In this case, the matrix

    $$\mathbf{A}_1 = \mathbf{A} - \lambda_1 \mathbf{k}_1 \mathbf{k}_1^T$$

    has eigenvalues 0, $|\lambda_2|$, $|\lambda_3|$, $\cdots$, $|\lambda_n|$ and that eigenvectors of $\mathbf{A}_1$ are also eigenvectors of $\mathbf{A}$. Note that λ_2 is now the dominant eigenvalue of $\mathbf{A}_1$

```
def deflation_method(A, n_iteration):

    n = A.shape[1]

    K = np.zeros((n, n))
    Lambda = np.zeros(n)

    k0 = np.zeros(n)
```

```
    L0 = 0
    for i in range(n):
        A = A -L0 *np.outer(k0, k0)
        k0 = power_method(A, n_iteration)
        L0 = Rayleigh_quotient(A, k0)
        K[i, :] = k0
        Lambda[i] = L0

    return K, Lambda

# symmetric matrix
A = np.array([[1, 2, -1], [2, 1, 1], [-1, 1, 0]])

K, Lambda = deflation_method(A, 20)

print('A =')
pprint.pprint(A)

print('\nk_1 =', np.around(K[0, :], 3))
print('k_2 =', np.around(K[1, :], 3))
print('k_3 =', np.around(K[2, :], 3))

print('\nlambda_1 =', np.around(Lambda[0], 3))
print('lambda_2 =', np.around(Lambda[1], 3))
print('lambda_3 =', np.around(Lambda[2], 3))
```

```
A =
array([[ 1,  2, -1],
       [ 2,  1,  1],
       [-1,  1,  0]])

k_1 = [0.707 0.707 0.   ]
k_2 = [ 0.578 -0.577  0.577]
k_3 = [-0.408  0.408  0.816]

lambda_1 = 3.0
lambda_2 = -2.0
lambda_3 = 1.0
```

- **Inverse Power Method**

  If we want to find the smallest eigenvalue instead of the largest one, then we perform

power iteration for $\mathbf{A}^{-1}$ (since the eigenvalues of $\mathbf{A}^{-1}$ are the reciprocals of the eigenvalues of $\mathbf{A}$). Of course, we do not want to compute $\mathbf{A}^{-1}$

```
def inverse_power_method(A, n_iteration):
    # Ideally choose a random vector
    # To decrease the chance that
    #   our vector is orthogonal to the eigenvector
    k = np.random.rand(A.shape[1])

    for _ in range(n_iteration):
        # calculate the matrix-by-vector product Ak
        w = np.linalg.solve(A, k)

        # calculate the norm
        w_norm = np.linalg.norm(w)

        # re normalize the vector
        k = w /w_norm

    return k

A = np.array([[4, 2], [3, -1]])
k_2 = inverse_power_method(A, 10)
lambda_2 = Rayleigh_quotient(A, k_2)

print('A =')
pprint.pprint(A)
print('\nEigenvector =', np.around(k_2, 3))
print('Eigenvalue of Least Magnitude =', np.around(lambda_2, 3))
```

```
A =
array([[ 4,  2],
       [ 3, -1]])

Eigenvector = [-0.316  0.949]
Eigenvalue of Least Magnitude = -2.0
```

6.12 Diagonalization

- For an $n \times n$ matrix $\mathbf{A}$, can we find an $n \times n$ nonsingular matrix $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$ is a diagonal matrix?

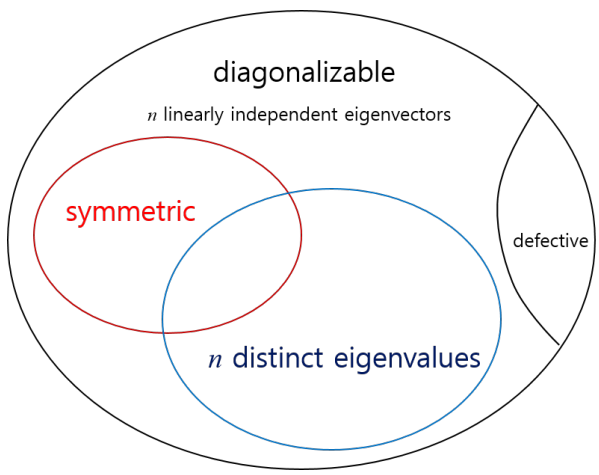
- An $n \times n$ matrix $\mathbf{A}$ is diagonalizable if and only if $\mathbf{A}$ has n linearly independent eigenvectors
- Let $\mathbf{k}_1, \mathbf{k}_2, \cdots, \mathbf{k}_n$ be linearly independent eigenvectors corresponding to eigenvalues $\lambda_1, \lambda_2, \cdots, \lambda_n$. Next form the matrix $\mathbf{P}$ with column vectors $\mathbf{k}_1, \mathbf{k}_2, \cdots, \mathbf{k}_n$

$$\mathbf{P} = \begin{pmatrix} \mathbf{k}_1 & \mathbf{k}_2 & \cdots & \mathbf{k}_n \end{pmatrix}$$

We can wrtie the product $\mathbf{AP}$ as

$$\begin{aligned} \mathbf{AP} &= \begin{pmatrix} \mathbf{Ak}_1 & \mathbf{Ak}_2 & \cdots & \mathbf{Ak}_n \end{pmatrix} = \begin{pmatrix} \lambda_1\mathbf{k}_1 & \lambda_2\mathbf{k}_2 & \cdots & \lambda_n\mathbf{k}_n \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{k}_1 & \mathbf{k}_2 & \cdots & \mathbf{k}_n \end{pmatrix} \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix} = \mathbf{PD} \end{aligned}$$

Multiplying by $\mathbf{P}^{-1}$ on the left then gives $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$



- If an $n \times n$ matrix $\mathbf{A}$ has n distinct eigenvalues, it is diagonalizable
- An $n \times n$ matrix $\mathbf{A}$ can be *orthogonally* diagonalized if and only if $\mathbf{A}$ is symmetric

Example Diagonalize

$$\mathbf{A} = \begin{pmatrix} 9 & 1 & 1 \\ 1 & 9 & 1 \\ 1 & 1 & 9 \end{pmatrix}$$

- The eigenvalues and corresponding orthogonal eigenvectors are $\lambda_1 = 11$, $\lambda_2 = \lambda_3 = 8$

$$\mathbf{k}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{k}_2 = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \text{ and } \mathbf{k}_3 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

- Multiplying these vectors, in turn, by the reciprocals of the norms $\|\mathbf{k}_1\| = \sqrt{3}$, $\|\mathbf{k}_2\| = \sqrt{6}$ and $\|\mathbf{k}_3\| = \sqrt{2}$, we obtain an orthonormal set

$$\mathbf{k}_1 = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}, \ \mathbf{k}_2 = \begin{pmatrix} -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix} \text{ and } \mathbf{k}_3 = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

  We then use these vectors as columns to construct an orthogonal matrix that diagonalizes $\mathbf{A}$

$$\mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

- This transforms $\mathbf{A}$ to $\mathbf{D}$

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{P}^T\mathbf{A}\mathbf{P} = \begin{pmatrix} 11 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{pmatrix} = \mathbf{D}$$

  The entries in $\mathbf{D}$ are the eigenvalues of $\mathbf{A}$ and the order in which these numbers appear on the diagonal corresponds to the order in which the eigenvectors are used as columns in the matrix $\mathbf{P}$

Example - Quadratic Forms: Identify the conic section whose equation $2x^2 + 4xy - y^2 = 1$

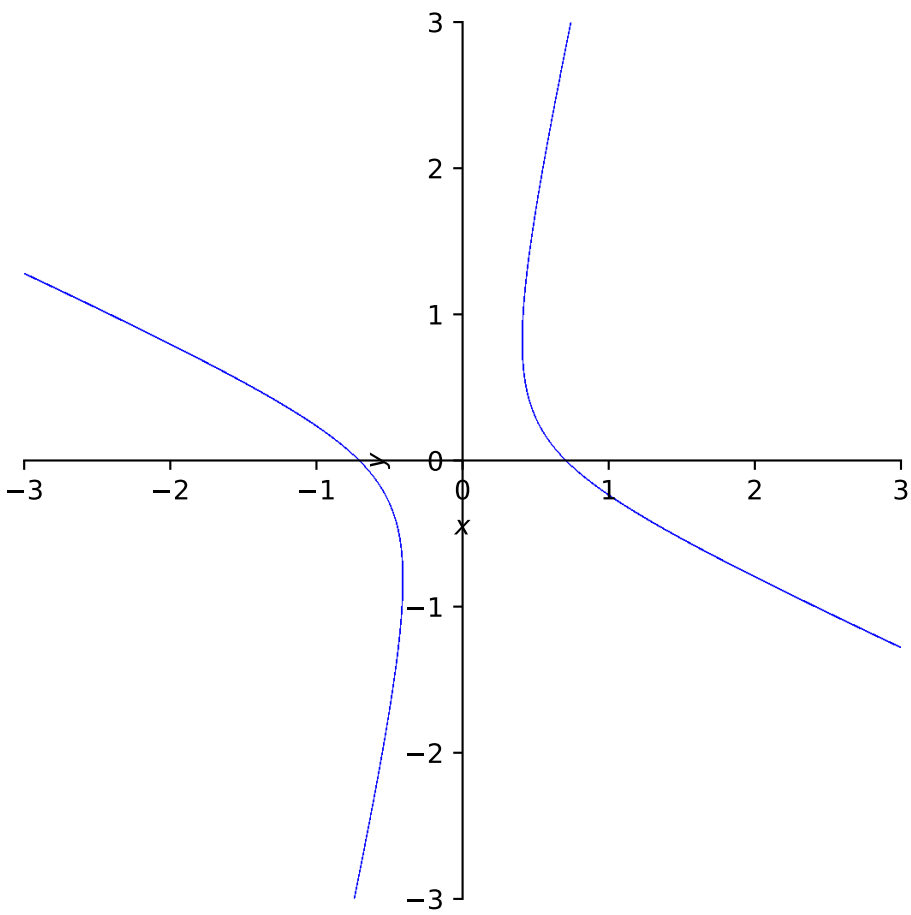
- We can write the given equation as

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 1 \text{ or } \mathbf{x}^T\mathbf{A}\mathbf{x} = 1$$

- The eigenvalues and corresponding eigenvectors of $\mathbf{A}$ are found to be

$$\lambda_1 = -2,\ \lambda_2 = 3,\ \mathbf{k}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix},\ \mathbf{k}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

```
x, y = symbols('x y')

p1 = plot_implicit(Eq(2*x**2 + 4*x*y - y**2, 1),
                   (x, -3, 3), (y, -3, 3),
                   aspect_ratio=(1, 1), size=(5, 5))
```



- Observe that $\mathbf{k}_1$ and $\mathbf{k}_2$ are orthogonal. Moreover, $\|\mathbf{k}_1\| = \|\mathbf{k}_2\| = \sqrt{5}$, and so the vectors

$$\mathbf{k}_1 = \begin{pmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{pmatrix} \text{ and } \mathbf{k}_2 = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix}$$

are orthogonal. Hence, the matrix

$$\mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{pmatrix}$$

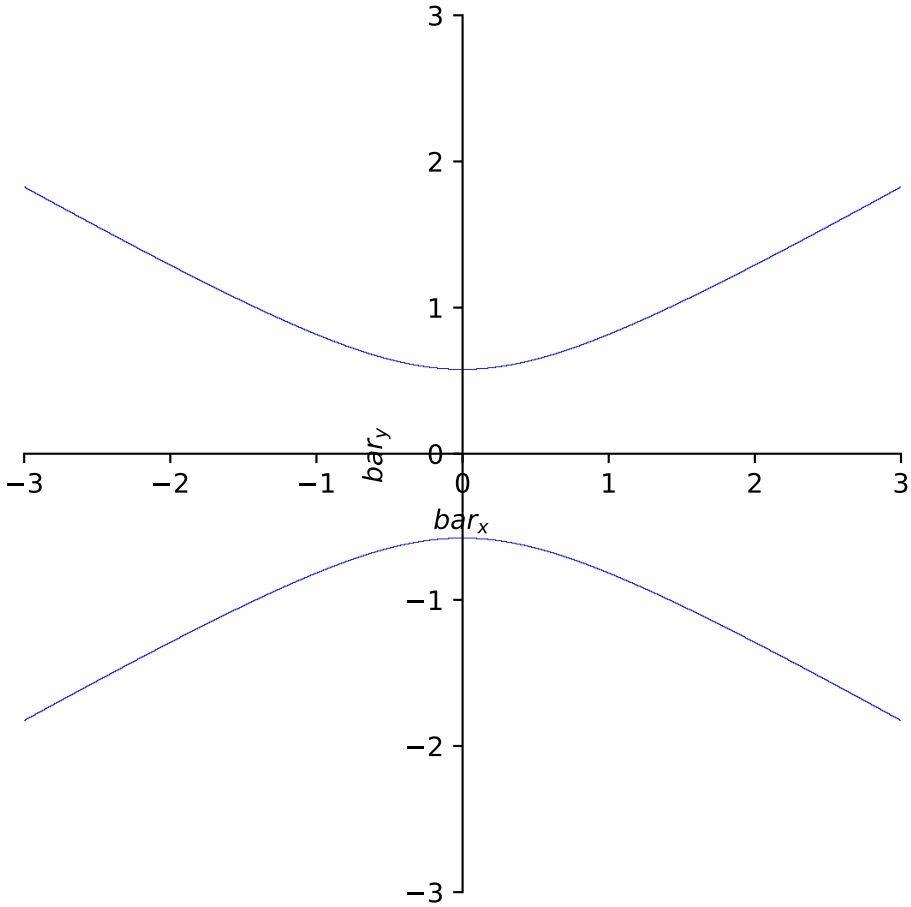
is orthogonal

- If we define the change of variables $\mathbf{x} = \mathbf{P}\bar{\mathbf{x}}$ where $\bar{\mathbf{x}} = \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$, then the quadratic form can be written

$$\begin{aligned} \mathbf{x}^T\mathbf{A}\mathbf{x} = \bar{\mathbf{x}}^T\mathbf{P}^T\mathbf{A}\mathbf{P}\bar{\mathbf{x}} = \bar{\mathbf{x}}^T\mathbf{D}\bar{\mathbf{x}} \\ = \begin{pmatrix} \bar{x} & \bar{y} \end{pmatrix} \begin{pmatrix} -2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix} = 1 \\ \text{or} \\ -2\bar{x}^2 + 3\bar{y}^2 = 1 \end{aligned}$$

This last equation is recognized as the standard form of a hyperbola

```
bar_x, bar_y = symbols('bar_x bar_y')

p1 = plot_implicit(Eq(-bar_x**2 + 3*bar_y**2, 1),
                   (bar_x, -3, 3), (bar_y, -3, 3),
                   aspect_ratio=(1, 1), size=(5, 5))
```



Example Determine whether the given matrix $\mathbf{A}$ is diagonalizable. If so, find the matrix $\mathbf{P}$ that diagonalizes $\mathbf{A}$ and the disgonal matrix $\mathbf{D}$ such that $\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$

$$\begin{pmatrix} -9 & 13 \\ -2 & 6 \end{pmatrix}, \quad \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Example The given matrix $\mathbf{A}$ is symmetric. Find an orthogonal matrix $\mathbf{P}$ that diagonalizes $\mathbf{A}$ and the diagonal matrix $\mathbf{D}$ such that $\mathbf{D} = \mathbf{P}^T\mathbf{A}\mathbf{P}$

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 7 \\ 0 & 1 & 0 \\ 7 & 0 & 1 \end{pmatrix}$$

Example Identify the given conic section

$$5x^2 - 2xy + 5y^2 = 24$$

$$-3x^2 + 8xy + 3y^2 = 20$$

Example Find 3×3 symmetric matrix that has eigenvalues $\lambda_1 = 1$, $\lambda_2 = 3$, and $\lambda_3 = 5$ and corresponding eigenvectors

$$\mathbf{k}_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{k}_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \mathbf{k}_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

Example If $\mathbf{A}$ is an $n \times n$ diagonalizable matrix, then $\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$, where $\mathbf{D}$ is a diagonal matrix. Show that if m is a positive integer, then $\mathbf{A}^m = \mathbf{P}\mathbf{D}^m\mathbf{P}^{-1}$

Example Find the indicated power of the given matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}, \quad \mathbf{A}^5$$

$$\mathbf{A} = \begin{pmatrix} 6 & -10 \\ 3 & -5 \end{pmatrix}, \quad \mathbf{A}^{10}$$

Example Suppose $\mathbf{A}$ is a nonsingular diagonalizable matrix. Then show that $\mathbf{A}^{-1}$ is diagonalizable

Example Suppose $\mathbf{A}$ is a diagonalizable matrix. Is the matrix $\mathbf{P}$ unique?

6.13 LU Factorization

- Let $\mathbf{A}$ be a square matrix. An *LU factorization* refers to the factorization of $\mathbf{A}$ into two factors – a lower triangular matrix $\mathbf{L}$ and an upper triangular matrix $\mathbf{U}$:

$$\mathbf{A} = \mathbf{LU}$$

- Without a proper ordering or permutations in the matrix, the factorization may fail to materialize. This is a procedural problem. It can be removed by simply reordering the rows of $\mathbf{A}$. It turns out that a proper permutation in rows (or columns) is sufficient for LU factorization. LU factorization with partial pivoting (LUP) refers often to LU factorization with row permutations only

$$\mathbf{PA} = \mathbf{LU}$$

  where $\mathbf{P}$ is a permutation matrix, which, when left-multiplied to $\mathbf{A}$, reorders the rows of $\mathbf{A}$. It turns out that all square matrices can be factorized in this form

- If $\mathbf{A}$ is invertible, then it admits an LU factorization if and only if all its leading principal minors are nonzero. If $\mathbf{A}$ is a singular matrix of rank k, then it admits an LU factorization if the first k leading principal minors are nonzero

- LU decomposition is basically a modified form of Gaussian elimination. We transform the matrix $\mathbf{A}$ into an upper triangular matrix $\mathbf{U}$ by eliminating the entries below the main diagonal. The **Doolittle algorithm** does the column-by-column elimination, starting from the left, by multiplying $\mathbf{A}$ to the left with atomic lower triangular matrices. It results in a *unit* lower triangular matrix and an upper triangular matrix

- **Doolittle Algorithm**
  - We define

$$\mathbf{A}^{(0)} = \mathbf{A}$$

  - We eliminate the matrix elements below the main diagonal in the k-th column of $\mathbf{A}^{(k-1)}$ by adding to the i-th row of this matrix the k-th row multiplied by

$$l_{i,k} = \frac{a_{i,k}^{(k-1)}}{a_{k,k}^{(k-1)}} \quad \text{for} \quad i = k+1, \cdots, n$$

This can be done by multiplying $\mathbf{A}^{(k-1)}$ to the left with the lower triangular matrix

$$\mathbf{L}_k = \begin{pmatrix} 1 & 0 & & \cdots & & 0 \\ 0 & \ddots & \ddots & & & \\ & & 1 & & & \\ \vdots & & -l_{k+1,k} & & & \vdots \\ & & \vdots & & \ddots & 0 \\ 0 & & -l_{n,k} & & 0 & 1 \end{pmatrix}$$

– We set

$$\mathbf{A}^{(k)} = \mathbf{L}_k \mathbf{A}^{(k-1)},\ k = 1, \cdots, n-1$$

After $n-1$ steps, we eliminated all the matrix elements below the main diagonal, so we obtain an upper triangular matrix $\mathbf{A}^{(n-1)}$. We find the decomposition

$$\begin{aligned} \mathbf{A} &= \mathbf{L}_1^{-1}\mathbf{L}_1\mathbf{A}^{(0)} = \mathbf{L}_1^{-1}\mathbf{A}^{(1)} = \mathbf{L}_1^{-1}\mathbf{L}_2^{-1}\mathbf{L}_2\mathbf{A}^{(1)} = \mathbf{L}_1^{-1}\mathbf{L}_2^{-1}\mathbf{A}^{(2)} \\ &\vdots \\ &= \mathbf{L}_1^{-1}\cdots\mathbf{L}_{n-1}^{-1}\mathbf{A}^{(n-1)} \end{aligned}$$

Denote the upper triangular matrix $\mathbf{A}^{(n-1)}$ by $\mathbf{U}$, and $\mathbf{L} = \mathbf{L}_1^{-1}\cdots\mathbf{L}_{n-1}^{-1}$

– Because the inverse of a lower triangular matrix $\mathbf{L}_k$ is again a lower triangular matrix, and the multiplication of two lower triangular matrices is again a lower triangular matrix, it follows that $\mathbf{L}$ is a lower triangular matrix:

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & & \cdots & & 0 \\ l_{2,1} & \ddots & \ddots & & & \\ & & 1 & & & \\ \vdots & & l_{k+1,k} & & & \vdots \\ & & \vdots & & 1 & 0 \\ l_{n,1} & \cdots & l_{n,k} & \cdots & l_{n,n-1} & 1 \end{pmatrix}$$

We obtain $\mathbf{A} = \mathbf{LU}$

NOTE It is clear that in order for this algorithm to work, one needs to have $a_{k,k}^{(k-1)}$ at each step (see the definition of $l_{i,k}$). If this assumption fails at some point, one needs to interchange k-th row with another row below it before continuing. This is why an LU decomposition in general looks like $\mathbf{PA} = \mathbf{LU}$

```python
def lu_factor(A):
    """
        LU factorization with partial pivoting

        PA = LU
        P(permutation),
        L(unit Lower triangular) and
        U(upper triangular)

        Return P, L, U
    """
    n = A.shape[0]
    U = A.copy()
    P = np.identity(n)
    L = np.identity(n)

    for k in range(n -1):

        # Partial Pivoting
        max_row_index = np.argmax(abs(U[k:n,k])) +k
        P[[k,max_row_index]] = P[[max_row_index,k]]
        U[[k,max_row_index]] = U[[max_row_index,k]]

        # LU
        L[k+1:,k] = U[k+1:,k] /U[k,k]
        U[k+1:,k] = 0.0
        U[k+1:,k+1:] -= np.tensordot(L[k+1:,k],
                            U[k,k+1:], axes=0)

    return P, L, U

A = np.array([[7, 3, -1, 2], [3, 8, 1, -4], [-1, 1, 4, -1],
              [2, -4, -1, 6]], dtype='float64')
P, L, U = lu_factor(A)

print('A ='); pprint.pprint(A)
print('\nP ='); pprint.pprint(P)
print('\nL ='); pprint.pprint(L)
print('\nU ='); pprint.pprint(U)
```

```
A =
array([[ 7.,  3., -1.,  2.],
```

```
       [ 3.,  8.,  1., -4.],
       [-1.,  1.,  4., -1.],
       [ 2., -4., -1.,  6.]])

P =
array([[1., 0., 0., 0.],
       [0., 1., 0., 0.],
       [0., 0., 1., 0.],
       [0., 0., 0., 1.]])

L =
array([[ 1.        ,  0.        ,  0.        ,  0.        ],
       [ 0.42857143,  1.        ,  0.        ,  0.        ],
       [-0.14285714,  0.21276596,  1.        ,  0.        ],
       [ 0.28571429, -0.72340426,  0.08982036,  1.        ]])

U =
array([[ 7.        ,  3.        , -1.        ,  2.        ],
       [ 0.        ,  6.71428571,  1.42857143, -4.85714286],
       [ 0.        ,  0.        ,  3.55319149,  0.31914894],
       [ 0.        ,  0.        ,  0.        ,  1.88622754]])
```

```python
A = np.array([[7, 3, -1, 2], [3, 8, 1, -4], [-1, 1, 4, -1],
              [2, -4, -1, 6]], dtype='float64')
P, L, U = lu(A)

print('P =')
pprint.pprint(P)
print('\nL =')
pprint.pprint(L)
print('\nU =')
pprint.pprint(U)
```

```
P =
array([[1., 0., 0., 0.],
       [0., 1., 0., 0.],
       [0., 0., 1., 0.],
       [0., 0., 0., 1.]])

L =
```

```
array([[ 1.        ,  0.        ,  0.        ,  0.        ],
       [ 0.42857143,  1.        ,  0.        ,  0.        ],
       [-0.14285714,  0.21276596,  1.        ,  0.        ],
       [ 0.28571429, -0.72340426,  0.08982036,  1.        ]])

U =
array([[ 7.        ,  3.        , -1.        ,  2.        ],
       [ 0.        ,  6.71428571,  1.42857143, -4.85714286],
       [ 0.        ,  0.        ,  3.55319149,  0.31914894],
       [ 0.        ,  0.        ,  0.        ,  1.88622754]])
```

- **Solving Linear Equations**

  Given a system of linear equations in matrix form

  $$\mathbf{Ax} = \mathbf{b}$$

  Suppose we have already obtained the LUP decomposition of $\mathbf{A}$ such that $\mathbf{PA} = \mathbf{LU}$, so

  $$\mathbf{LUx} = \mathbf{Pb}$$

  In this case the solution is done in two logical steps:

  - First, we solve the equation $\mathbf{Ly} = \mathbf{Pb}$ for $\mathbf{y}$
  - Second, we solve the equation $\mathbf{Ux} = \mathbf{y}$ for $\mathbf{x}$
  - Note that in both cases we are dealing with triangular matrices $\mathbf{L}$ and $\mathbf{U}$, which can be solved directly by forward and backward substitution without using the Gaussian elimination process (however we do need this process or equivalent to compute the LU decomposition itself)
  - The cost of solving a system of linear equations is approximately $\frac{2}{3}n^3$ floating-point operations
  - The above procedure can be repeatedly applied to solve the equation multiple times for different $\mathbf{b}$. In this case it is faster (and more convenient) to do an LU decomposition of the matrix $\mathbf{A}$ once and then solve the triangular matrices for the different $\mathbf{b}$, rather than using Gaussian elimination each time
  - The matrices $\mathbf{L}$ and $\mathbf{U}$ could be thought to have *encoded* the Gaussian elimination process

```python
def ufsub(L, b):
    """ Unit row oriented forward substitution """
    y = b.copy()
    for i in range(1, L.shape[0]):
        y[i] -= np.dot(L[i,:i], y[:i])
    return y

def bsub(U, y):
    """ Row oriented backward substitution """
    x = y.copy()
    x[-1] /= U[-1,-1]
    for i in range(U.shape[0] -2, -1, -1):
        x[i] -= np.dot(U[i,i+1:], x[i+1:])
        x[i] /= U[i,i]
    return x

A = np.array([[7, 3, -1, 2], [3, 8, 1, -4],
              [-1, 1, 4, -1], [2, -4, -1, 6]], dtype='float64')
b = np.array([1, 2, 3, 4], dtype='float64')

P, L, U = lu_factor(A)
Pb = np.matmul(P, b)

y = ufsub(L, Pb)
x = bsub(U, y)

print('A ='); pprint.pprint(A)
print('\nb =', end=' '); pprint.pprint(b)
print('\nx =', end=' '); pprint.pprint(x)
```

```
A =
array([[ 7.,  3., -1.,  2.],
       [ 3.,  8.,  1., -4.],
       [-1.,  1.,  4., -1.],
       [ 2., -4., -1.,  6.]])

b = array([1., 2., 3., 4.])

x = array([-1.27619048,  1.87619048,  0.57142857,  2.43809524])
```

- **Inverting a Matrix**

  In matrix inversion, instead of vector $\mathbf{b}$, we have matrix $\mathbf{I}_n$ so that we are trying to find a matrix $\mathbf{X}$

$$\mathbf{LUX} = \mathbf{I}_n$$

  We can use the same algorithm presented earlier to solve for each column of matrix $\mathbf{X}$

- **Computing the Determinant**

  Given the LUP decomposition $\mathbf{A} = \mathbf{P}^{-1}\mathbf{LU}$ of a square matrix $\mathbf{A}$, the determinant of $\mathbf{A}$ can be computed straightforwardly as

$$\det \mathbf{A} = \det \mathbf{P}^{-1} \det \mathbf{L} \det \mathbf{U} = (-1)^s \prod_{i=1}^{n} l_{ii} \prod_{i=1}^{n} u_{ii}$$

  where s is the number of row exchanges in the permutation matrix

Example Find the LU factorization of the given matrix

$$\begin{pmatrix} 2 & -2 \\ 1 & 2 \end{pmatrix}, \quad \begin{pmatrix} -1 & 4 \\ 2 & 2 \end{pmatrix}, \quad \begin{pmatrix} 4 & -2 & 1 \\ -4 & 1 & 2 \\ 12 & 1 & 3 \end{pmatrix}$$

Example Use the LU factorization to solve the given linear system of equations

$$\begin{pmatrix} 2 & -2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\begin{pmatrix} 4 & -2 & 1 \\ -4 & 1 & 2 \\ 12 & 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 7 \\ 7 \\ 28 \end{pmatrix}$$

6.14 Applications